Defining Language

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Defining Language: A local grammar of definition sentences
by Geoff Barnbrook

Defining Language

A local grammar of definition sentences

Geoff Barnbrook
University of Birmingham

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For John Sinclair — teacher, colleague, friend

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Brief description

This book describes the analysis of the main features of the language used in English definition sentences, using as a corpus the definitions contained in the *Collins Cobuild Student's Dictionary*. It examines the usefulness of the information provided by dictionaries in natural language processing work and the nature of the language used in dictionary definitions in general and in the Cobuild range in particular. It provides a general survey of monolingual English dictionaries, including a brief history of their development, and a detailed investigation of the nature of learners' dictionaries and their special features. The concept of sublanguages is examined, together with the justification for regarding definition sentences as a sublanguage and for the application to them of a local grammar of definition. Grammars and parsers are considered in general terms, and in their relevance to the creation of a model for the language of definitions.

The methodology adopted for the development of the language model is described, together with a detailed account of the taxonomy, local grammar and associated parser developed for definition sentences. The implications of the results of the analysis and future possible applications of the taxonomy, grammar and parser are described and assessed.

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Chapter 1

Language, definitions and dictionaries

Definitions set out to explain the meanings of certain words in terms of certain other words. The processes by which they do this, and the forms that definitions take, are by no means straightforward. The study of these processes and forms is rewarding in more than one way. In itself, it constitutes the investigation of a significant function of language, and of a significant number of the utterances that make up language. The grammar of definition sentences presented in this book describes a major aspect of the English language in general, and the parser which implements it facilitates the proper analysis of some of the most basic metalinguistic statements in common use. Beyond the boundaries of language description, the contents of these definition sentences also provide valuable resources in the search for information about words for use in automatic language processing. This book, then, describes the structure of definition sentences in English, based on the characteristics of a sample represented by a typically rich source of definitions: a monolingual English dictionary. It also describes a parser developed for definition sentences taken from the sample dictionary which both implements the grammar and allows the definitions to be analysed to yield information for automatic language processing.

1.1 Dictionary entries and definitions

Definitions form one of the basic functions of language, and a description of their structure and operation describes a major aspect of language use. While most forms of text include examples of definition sentences, some form a richer source for them than others. In particular, dictionary entries, as explained in more detail in Chapter 2, set out to describe the linguistic properties and behaviour patterns of their headwords. To do this they use definitions, among other things, to describe the meanings of these headwords. When these definitions take the form of normal English sentences they can be used as a

sample of definitions in general to enable us to examine the grammar of this area of language use. This is the basis of this study.

If the contents of these entries can be made available to a computer in a readily accessible form, they can also provide extremely valuable data for use in natural language processing systems. Meijs (1994, p. 69) describes the usefulness of machine-readable dictionaries (MRDs) in enabling the construction of 'large-scale lexicons with a realistic level of coverage instead of the customary purpose-built 'toy' lexicons containing just a few sample items'. In a survey of the ASCOT, *Natural primitives?*, LINKS and ACQUILEX projects he describes the extraction of syntactic and semantic information and the construction of semantic taxonomies and lexical knowledge bases. These projects used a variety of information from their source MRDs, including both items that were separately encoded in the dictionary entries, such as box codes and subject field codes, and elements of the definition text itself.

Going beyond dictionary entries to definition sentences in other texts, the approach described in this study could allow definitions to be extracted automatically from their environment and analysed to provide information about the words that they define, a particularly powerful tool for texts which introduce specialised terminology.

In this investigation, the definition text of the dictionary is the sole focus of study, and it is analysed both to gain a general understanding of the components and operations of the definition process, and to unlock the rich and varied information that it can provide about the word being defined. To achieve both of these aims the structure of a sample of definition sentences provided by the selected dictionary has been examined in detail. From this examination, a grammar of their structures has been produced, and a parser based on that grammar has been developed. This parser allows the information contained in the definitions to be accessed more readily. The reasons for doing this are considered next.

1.2 Why parse dictionary entries?

The information provided by dictionary entries is often presented in a complex, highly structured form of language, relying heavily on a densely encoded system of typographical formats and abbreviations. The exact method, and the nature and extent of the information given, varies from one dictionary to another, but there are general tendencies. If we limit the type of dictionary

under consideration to those prepared specially for learners of the language, these tendencies become more uniform.

The meaning of the headword is obviously dealt with, together with guidance on its pronunciation, any peculiarities of the other forms of the headword, guidance on its syntactic behaviour and, where appropriate, its lexical relations with other words. As an example, the entry for the headword 'drink' in the *Collins Cobuild Student's Dictionary* (*CCSD*, Sinclair, 1990, p. 164) is:

drink /drɪŋk/, drinks, drinking, drank /dræŋk/, drunk /drʌŋk/. **1** VB WITH OR WITHOUT OBJ When you **drink** a liquid, you take it into your mouth and swallow it. *We sat drinking coffee…* He drank eagerly. **2** COUNT N A **drink** is an amount of a liquid which you drink. *I asked for a drink of water.* **3** VB To **drink** also means to drink alcohol. *You shouldn't drink and drive.* drinking UNCOUNT N *There had been some heavy drinking at the party.* **4** UNCOUNT N **Drink** is alcohol, for example beer, wine, or whisky. *He eventually died of drink.* **5** COUNT N A **drink** is also an alcoholic drink. *He poured himself a drink.* **6** See also **drunk**.

drink to. PHR VB If you **drink to** someone or something, you raise your glass before drinking, and say that you hope they will be happy or successful. *They agreed on their plan and drank to it.*

Some of the information relates to the entire headword, such as:

> the base form of the word, the principal basis of alphabetic reference within the dictionary
> pronunciation guides
> other forms of the headword, showing both regular and irregular morphology

Both the *Oxford Advanced Learner's Dictionary of Current English* (*OALDCE*, Cowie, 1989a, p. 370) and the *Longman Dictionary of Contemporary English* (*LDOCE*, Summers, 1987, p. 313) provide similar details, although only irregular word forms are printed.

Other pieces of information are specific to the individual senses dealt with in the entry, and these usually include:

a sense number
a grammar code
a definition
one or more examples of usage

In the Cobuild dictionaries senses appear in order of their perceived importance, using the frequency of occurrence together with the centrality, independence and concreteness of meaning of the individual senses, as described in the introduction to the original *Collins Cobuild English Language Dictionary* (*CCELD*, Sinclair, 1987, p. xix). This means that the order of treatment of senses preserves the semantic flow between them, and that sense numbers give a rough guide to the relative likelihood of specific senses being encountered by the user. The grammar codes usually specify the word class that a particular sense falls into, and sometimes, as with sense 1 of drink above, contain additional information on possible syntactic combinations. The definition sentences explain the meaning of the word by incorporating it within them, distinguished from the other words of the sentences by being in bold type. The examples of usage, taken from the corpus on which the dictionary is based, are selected to show the user how senses have been used in real English text.

Again, the organisation of these sense specific details is similar in *OALDCE* (p. 370) and *LDOCE* (p. 313), although the order of the senses is different. In *OALDCE* the senses of 'drink' are split between two headwords, the first for the noun, the second for the verb. In *LDOCE* the same split is used, but the verb is given first. The Cobuild arrangement, in which all possible senses of the same sequence of characters 'drink' are shown under the same heading, is unusual enough for Nuccorini (1993, p. 101) to discuss it as the main feature of the Cobuild macrostructure:

> Infatti questo dizionario non distingue gli omografi: ciò significa che vi è sempre soltanto un'entrata per tutti gli omonimi, senza distinzioni semantiche né di parte del discorso.[1]

The senses given in *CCSD*, with their corresponding senses in *OALDCE* and *LDOCE*, are shown below:

CCSD	OALDCE	LDOCE
1	2.1	1.1
2	1.1(b)	2.1
3	2.3	1.2

4	1.2(a)	2.2
5	1.2(b)	2.2

The definition of the individual senses is the main focus of the present study. Each definition uses lexical relations to convey the meaning of each of the senses. In the first sense of drink, given above from *CCSD*, the physical details of the main components of the process are given in the words which form the second half of the explanatory sentence:

> you take it into your mouth and swallow it

The meaning conveyed by these words is very similar to that given under sense 1 of the entries for drink as a verb in *OALDCE* (p. 370):

> take (liquid) into the mouth and swallow

There is also a reasonable similarity to the meaning given for sense 1 of the verb entry in *LDOCE* (p. 313):

> to move (liquid) from the mouth down the throat

All of the examples shown so far are from dictionaries designed for learners of English. Other general purpose dictionaries may contain different elements of information for their headwords. As an extreme example, the *Oxford English Dictionary* entry for 'drink' occupies nearly two pages, and, in addition to the information provided by the learner's dictionaries, has full notes on historical spelling variations and etymology, and deals with 18 main verb and 9 main noun senses. These are organised first by part of speech, and then on broadly historical principles.

The information contained in published dictionaries is no doubt interpreted and isolated by the dictionary's human users with varying degrees of ease and success, and hopefully assists their processing of the language being described. The encoding systems described above are designed for each different dictionary to facilitate this as much as possible within the normal commercial constraints of publishing. They are not specifically designed for use in computerised natural language processing systems, and need extensive analysis before they can be used in them. It is also assumed that the human user draws on significant amounts of world knowledge in decoding dictionary information, and that because of this the information available from dictionary entries would be inadequate for use by a natural language processing system without the explicit addition of this knowledge.

Boguraev & Briscoe, in the introduction to their description of the construction of lexica for natural language processing systems (1989, pp. 4–5), divide the knowledge relevant to such systems into five categories:

1. Phonological
2. Morphological
3. Syntactic
4. Semantic
5. Pragmatic or 'encyclopaedic'

They differentiate pragmatic knowledge from the other types as being 'least related to the lexicon', but admit that 'there is no clear division between lexical semantic knowledge and more general pragmatic knowledge'. This suggests that at least some of the knowledge of the world needed by a natural language processing system will be available directly or recoverable from dictionaries.

The same work also discusses the general advantages and disadvantages of using machine-readable versions of existing dictionaries as the starting-point for the construction of natural language processing lexica, and states that the main disadvantage, as suggested above, is that:

> published dictionaries are produced with the human reader in mind and therefore make many inconvenient assumptions from the point of view of processing by machine; for example, the assumption that the user can understand definitions of word senses written in English.
>
> (Boguraev & Briscoe, 1989, p. 2)

The nature of the understanding needed in different types of dictionary is discussed in Chapter 2, which investigates the history of monolingual English dictionaries, and in particular the vocabulary and structures used in them and the nature of the problems that they would pose for automatic analysis systems.

Despite these problems, it seems from the interest shown in the use of machine-readable published dictionaries that they have significant potential value for natural language processing systems. A system for analysing dictionary definitions automatically could convert machine-readable versions of dictionaries designed for the human reader into a form capable of being used directly by natural language processing applications. The ready availability of electronic texts of dictionaries, and the effort currently involved in the construction of lexica, combine to make this an attractive prospect. Boguraev & Briscoe (1989, p. 10) quote a survey of lexicons used in a group of projects

described by Whitelock et al.(1987) which gave an average lexicon size of 1,500 words. This group of projects contained one vocabulary for machine translation which contained between 5,000 and 6,000 words: the average size of the lexica used by the other projects was about 25 words. They also (p. 11) describe the problems of those large lexica which had been developed by 1989 when they needed to be extended to more generalised applications than the very specific tasks for which they had been designed. As explained in detail in Chapter 7, for these reasons and for many others, the potential applications of an analysis which made human readable dictionaries available for these purposes would extend far beyond the general requirements of natural language processing applications.

1.3 A representative sample — the Cobuild dictionary range

Automatic analysis systems have already been produced for specific dictionaries. Alshawi (1989) describes a set of routines developed to analyse part of the *Longman Dictionary of Contemporary English*. In general, these programs use the mark-up conventions of the dictionary's coding system to isolate a significant amount of the required elements of each definition. These elements have been determined by the lexicographer during the compilation process and are explicitly coded in the text. Such an analysis is a useful way of making this information available to a computer system, but it has a major drawback. The analysis is constrained by the original design of the dictionary, and cannot easily vary the level of detail of the information already available. The system is essentially closed, and the analysis program converts one form of coding into another.

This may make it difficult to access areas of information, which are present in the entry in an implicit rather than an explicit form. For example, consider the treatment of the headword 'prat' in *LDOCE*:

> a worthless stupid person (p. 808)

and in *CCELD*:

> If you call someone a prat, you mean that they are very stupid or foolish (p. 1125)

The *LDOCE* entry also includes the code:

> derog sl

which is explained in the list of short forms and labels as:

> derogatory slang.

Neither of these words is in the defining vocabulary listed on pp. B16-B22, and although both are defined in the dictionary and explained in the style and usage section on p. F46, there is not a direct explanation easily at hand when a user first encounters this label. It could, however, be accessed by a computer program, which could make an appropriate entry for usage.

The *CCELD* entry does not equate the meaning of 'prat' with 'very stupid and foolish' in the same way as *LDOCE* equates it with 'a stupid and worthless person'. Instead it describes what the user would mean if they called someone a prat. This puts the definition into a metalinguistic mode in which the normal method of usage of the headword is encoded implicitly within the definition text itself rather than explicitly as a separate, densely encoded abbreviation which the user may well ignore. *CCELD* adds the usage note:

> an offensive word, used in informal British English.

which makes explicit the information available from *LDOCE*, but gives it in the same typeface and style of text as the main definition, so that the user is much more likely to be aware of it.

So much for the human user: which approach is likely to be more useful for computer analysis? At first sight it might seem obvious that the explicitly coded information is easier to access using a computer and therefore more valuable. In fact, the Cobuild entry, although perhaps involving slightly more computational effort, can yield information which would not be available from the type of entry given in *LDOCE*. The *CCELD* entry begins with:

> If you call someone

None of these elements is present in the *LDOCE* entry. It could be argued that the note 'derog sl' means that this headword is one which falls into a general category of insults, and that this supplies the same information, but there is more subtlety and detail in the *CCELD* entry than might at first be apparent. Consider the definition of sense 2.1 of 'bastard' in *CCELD*:

> If someone calls someone else a bastard, they are referring to them or addressing them in an insulting way;

The usage note adds:

> a very offensive use.

Strictly speaking, this explicit usage note is unnecessary, because the definition strategy has already provided the information. The difference between the 'if you' at the beginning of the definition of 'prat' and the 'if someone' at the beginning of this definition is an implicit signal to the user that 'bastard' is likely to be regarded as a stronger and more offensive word: the direct possibility of the user of the dictionary calling someone a 'prat' is catered for, the possibility of them calling someone a 'bastard' is not. The only difference lies in the subject used for the verb 'calls', an element of the definition text referred to by Sinclair (1991) as the 'co-text', a concept which is dealt with in more detail in section 6.3.2 below.

This is not the only element of information provided by the structure of the definition itself. Both of these definitions begin with 'if'. Consider the definition of sense 1 of 'eat' from *CCELD*:

> When you eat something or when you eat, you put food into your mouth, chew it, and swallow it.

Calling someone a 'prat' or a 'bastard', whoever does it, is not an inescapable fact of life: eating is.

The accuracy and value of this further information in any specific Cobuild dictionary will of course depend on the effectiveness of controls over the lexicographers to ensure that these policies are carried out, but it is likely that even without rigid control, experienced lexicographers will tend to construct definitions with similar requirements in similar ways. A full analysis of the structural patterns, together with an awareness of the implications of particular elements of those patterns, could expand usefully on the explicit linguistic properties associated with the headwords in other types of dictionary.

As explained in more detail in Chapter 2, the format of the Cobuild dictionary entries, and the policies governing their compilation, combine to produce an explanation of the meaning of each headword which contains the essential semantic, syntactic and lexical information in a short piece of normal English text. The unique feature of the Cobuild range of dictionaries is the framing of these definitions within complete sentences formed following the normal grammar of English. The structure of each definition contains guidance for users similar to that given in the examples of usage, but in a more definitely structured form.

There is no explicit encoding of this information through abbreviation systems, and no apparent constraints on the language used to embody it, but the patterns observed by the lexicographers in their compilation of the entries are so consistent that, as explained in Chapter 3, they can be treated as if they form a separate sub-language. The grammar of this sub-language, the local grammar of definitions, is of crucial importance for the analysis of the implicit information, since in the Cobuild dictionaries both the form and the content of the definitions are used to describe the linguistic features of the headwords.

The primary objective of this study is to describe both the grammar of definition sentences and the parser which has been developed to analyse it. General language parsers, described in Chapter 3, need complex algorithms because they are required to deal with a huge variety of language forms. The restricted nature of the definition language, analysed in detail in Chapters 5 and 6, has allowed the development of a relatively simple specialised parser, also described in Chapter 6. This parser makes no attempt to analyse the text of the definitions into conventional general linguistic components. It is instead a functional parser which implements the functional grammar of the language and allows us to extract the information already described in section 1.2 above.

1.4 The sample dictionary

The Collins Cobuild English Language Dictionary (*CCELD*), the first of the Cobuild range, published in 1987, introduced the style of definition described in the previous section. The patterns established during the production of this work were refined, in some cases simplified, and perhaps applied more consistently in the dictionaries which followed. The Collins Cobuild *Student's Dictionary* (*CCSD*) is the smallest of the first edition set. Its list of headwords is restricted and its definition texts are relatively simple in comparison with the larger dictionaries which preceded it. This inevitably means that an investigation of the definition language based on CCSD may be incomplete in some senses, but it still provides a useful basis for the investigation of definition language for two main reasons.

Firstly, there are no grounds to suppose that the full range of definition structures are not present in CCSD despite the restricted number of headwords. The basis of headword selection for the smaller dictionary should not

result in the loss of particular word types, and the main forms of definition needed for those word types should be present and available for exploration in all editions. Consider the following examples of noun, verb, adjective and adverb definitions taken from pp. 960–961 of *CCELD*:

> Your **neck** is the part of your body which joins your head to the rest of your body. (p. 961, sense 1)
> If something **necessitates** an action, event, or situation, it makes it necessary; (p. 960)
> Something that is **neat** is made or kept very tidy, clean, and smart. (p. 960, sense 1)
> **Nearly** means almost, but not completely, totally or exactly. (p. 960, sense 1)

The corresponding definitions in *CCSD* are:

> Your **neck** is the part of your body which joins your head to the rest of your body. (p. 372, sense 1)
> If something **necessitates** a particular course of action, it makes it necessary; (p. 372)
> Something that is **neat** is tidy and smart. (p. 371, sense 1)
> **Nearly** means not completely or not exactly. (p. 371, sense 1)

These represent some of the most widely used definition structures in the Cobuild range, and they are as well represented in the *CCSD* as in the other dictionaries. Taking the *CCSD* as a corpus representing definition language, it seems likely that it would be fully representative of the main definition structures. Any deficiency in its representativeness compared to the other dictionaries in the range is more likely to become apparent in the less commonly used forms which are less significant parts of the language. Having said this, it is true to say that there are some differences between the structures used in *CCELD* and those in *CCSD*, but these lead to the second reason for the greater suitability of the smaller version.

The philosophy underlying the form of definitions chosen for the Cobuild range had been developed before the production of the first edition, but experience with the production of subsequent versions of the dictionary inevitably modified the detailed implementation of that philosophy in definition structures. To some extent this means that the forms of definition used in *CCSD* are likely to be more consistent than those in *CCELD*. The definition

forms for words with multiple senses are also often more complex in *CCELD*. For example, the definition of 'near miss' on p. 960:

> A **near miss** is 1 a bomb or shot which just misses the target, although it is very close. 2 a situation where you nearly had an accident or disaster but just avoided it. EG *Most aircraft accidents or near misses are caused by pilot error.* 3 an attempt to do something which nearly succeeds, but just fails to do so.

This branching structure with an embedded example would certainly demand a great deal more pre-processing before computer analysis could begin than is the case with *CCSD*, although the additional complexity in this case is of no great linguistic importance. These and similar considerations make *CCSD* more suitable as the starting-point for the development of a generally useful definition grammar and parser.

1.5 Specific objectives

This study, then, has two main objectives, which are ultimately dependent on each other. The first is a description of the structure of definition sentences in general based on a sample taken from the Cobuild definitions, in the form of a local grammar. The second objective is a practical application of the first: to find a means of parsing the definition sentences which implements this local grammar and allows us to extract information for use in natural language processing. The nature of monolingual dictionaries in general and the Cobuild range in particular is discussed in Chapter 2. The basic nature of grammars, parsers, sublanguages and local grammars is explored in Chapter 3. The overall methodology adopted in the investigation is described in Chapter 4. The taxonomy of definition structures, the first stage in the development of the grammar, is described in Chapter 5. The local grammar of definition sentences,. and its associated functional parser are both described in Chapter 6. Finally, Chapter 7 provides an evaluation of the results of the analysis and of the main possible future applications.

Note

1. In fact this dictionary does not distinguish homographs: this means that there is always only one entry for all the homonyms, without any distinction of meaning or of part of speech. (Author's translation)

Chapter 2

Monolingual English dictionaries

The source of the sample definition sentences used in this study is a monolingual English dictionary designed to be used by learners of the language. Monolingual English dictionaries have undergone considerable development since their origin in the late sixteenth and early seventeenth centuries, and the perceived needs of their users have obviously developed alongside them. Before considering definition structure in detail, it is important to clarify the general context in which the information contained in monolingual dictionaries, including definitions, is presented and used. This chapter contains an examination of the language used in monolingual English dictionaries in general, and in English learners' dictionaries in particular, together with a brief summary of their history and its relevance to their current state.

2.1 Linguistic information in monolingual English dictionaries

Dictionaries are being used in this study primarily as a source of examples of definitions in English. The analysis of definition structure, as already described in Chapter 1, has the twin objectives of providing a basis for the description of this area of language use and allowing the extraction of information for use in natural language processing systems and other applications. In order to explore the ways in which these twin objectives should be approached, it is first necessary to establish the basic nature of the monolingual dictionary and the complex role of language within it.

2.1.1 Object language and metalanguage in the monolingual dictionary

The fundamental difficulty facing anyone compiling a monolingual dictionary is described by Zgusta et al.(1971, pp. 248–9):

> 'In a monolingual dictionary, only one language is used in the entry. This circumstance should not, however, hide the fact that this single language has two

> different purposes: on the one hand, it is the object of the lexicographer's work (irrespective of whether the purpose of the dictionary is description, interpretation, or explanation etc.) but on the other hand, it is the instrument by which this work (description, explanation etc.) is done. This double purpose and double use must be constantly taken into consideration.'

As Harris (1988, pp. 2–3) similarly points out, other fields of study have an external metalanguage, 'a language of broader informational capacity than the given field', in which they can be investigated and defined, but this is not true of natural language. Zgusta, discussing the frequent overlap in monolingual dictionaries between glosses and examples (ibid., p. 270), points out the difficulty of separating these two applications of language:

> 'Indeed, it is within my experience impossible to make, in a monolingual dictionary, the neat distinction between the 'object language' and 'metalanguage' or 'language of description' which some theoreticians are inclined to postulate.'

The phrase 'object language' used here refers back to the first purpose of the language in which a monolingual dictionary is produced, described in the quotation at the start of this section: the object of the lexicographer's work.

The practical inability to separate the object of the description from the description itself has important implications for the form of analysis being developed in this work. Such separation as is possible between the language being described and the language used for its description is made to appear more definite in traditional dictionary formats. Here, much of the organisation of the description relies on the use of different type-faces, complex coding systems, heavily abbreviated technical terms, etc. In the method used in the Cobuild dictionary range, by contrast, the word being defined is literally embedded in the language used to define it.

As an example, consider the definitions of the senses of 'soap' in *CCELD*:

> **1 Soap** is a substance that you use with water for washing yourself or for washing clothes. It is made from oil or fats and alkali and is sold in small hard pieces, as a liquid, or as a powder.
> **2** If you **soap** yourself, you rub soap on your body in order to wash yourself.
> **3** A **soap** is the same as a soap opera; an informal use.
>
> (p. 1382)

Compare these with the corresponding entries in *OALDCE*:

> **1** substance used for washing and cleaning, made of fat or oil combined with an alkali

> **2** [C](infml) = SOAP OPERA
> soap *v*[Tn, Tn-p] apply soap to (sb|sth); rub with soap

The explicit syntactic information embedded in some parts of the definition in *OALDCE* (such as [C] for countable noun in sense 2) has its equivalent in *CCELD*, but is kept separate from the definition text. Schnelle (1995) draws a distinction between the use of a semantic metalanguage in formal linguistics representations and the classical definitional statement approach, which used 'sentences of the language itself' (Zgusta's 'object language'). In his analysis:

> 'Lexicographers were influenced by the classical view of sentential dictionary definitions. Being under the pressure of economizing printing space, however, their formats merely provided hints at the classical form of definition. The citation forms given in the lexical entry were taken as sufficient for the identification of the definiens; the subject phrase of the classical definition could therefore be dispensed with. From the set of differentiae specificae sometimes only the genus proximum term was given, sometimes together with others which did not seem obvious to the users.'
>
> (Schnelle, 1995, section 1)

He contrasts this with the approach adopted by Cobuild which, at least in the case of the basic noun definition, 'rigorously followed the classical logic of definitions'. The consequence of the use of the language to describe its own meanings is that:

> 'The semantics of the language is thus provided by a *subset of the language* whose systematic interdependencies are determined by the rules of syntax and the logic of inference. Dictionary explanations in English, the syntax of English, and logic applied to English are sufficient for specifying the semantic interdependency of the meanings and of the non-metaphorical uses of words in English.'
>
> (Schnelle, 1995, section 1)

The technical terms normally used for the two main components of the dictionary definition, the source of the semantic information relating to the headword, are 'definiendum' and 'definiens'. Their definitions in the *Oxford English Dictionary* (*OED,* Murray, J.A.H. et al., 1989, p. 403) are, for definiendum:

> 'That which is, or is to be, defined; the phrase of which a definition states or purports to state the meaning; in *Mathematical Logic*, the word or symbol (or the formula devised to contain the symbol) that is being introduced by definition into a system.'

and for definiens

> 'The defining part of a definition; the phrase that states the meaning; in *Mathematical Logic*, the verbal or symbolic expression to which a word or symbol being introduced by definition into a system is declared to be equivalent.'

There are two important characteristics of these definitions from the viewpoint of an examination of definition techniques in monolingual dictionaries. In the first place, they assume that a clear distinction exists between the two elements, a distinction which most dictionaries, including *OALDCE*, incorporate into their page layout. In the second place, although the definitions make clear the difference between the meanings of the words in their specialised use within Mathematical Logic and their wider use in other areas, the potential for confusion between the two is evident. Perhaps the most important consequence of this in lexicography is the ultimately misguided concept of the definition as a form of equation, in which these two logical elements form the left-hand and right-hand sides, with some form of equality operator, usually implicit in the dictionary structure, set between them.

Although the earliest quotation given in *OED* for both words, T.M. Lindsay's translation of *Überweg's Systemic Logic*, dates from 1871, this interpretation of the nature of the definition seems to be well established in English dictionaries by the beginning of the eighteenth century. The implications of the need for equivalence implied by these definitions is examined in more detail in section 2.4.4.2 below. For the time being, the definiendum can be equated to Zgusta's object language, and the definiens to the metalanguage used to describe it. The nature of this metalanguage in full sentence definitions of the kind used in the Cobuild dictionary range now needs to be considered to establish its relationship with traditional dictionary definition structures and the special features that affect its usefulness as a source of linguistic information.

2.1.2 The nature of the metalanguage in full sentence definitions

The difference described in section 2.1.1 between the Cobuild dictionaries and their more traditional counterparts — the relationship between the metalanguage and its object of description — is fundamental to the purpose of this analysis. It is therefore necessary to examine both the general nature of the metalanguage used in full sentence definitions, to the extent that it can be

separated from its object language, and its effect on the information contained in the individual dictionary entries.

Lyons (1977, vol.1, pp. 5–6) describes the standard philosophical distinction between reflexive use of language and other possible uses, which assigns technical meanings to the terms 'use' and 'mention' to indicate respectively non-reflexive and reflexive use. Zabeeh, in his introduction to part I of Zabeeh, Klemke & Jacobson (1974, pp. 21–31), describes both the types of problems which this distinction, together with the related division between object language and metalanguage, may assist with, and the further difficulties that can arise from the use of these distinctions.

These difficulties arise from the fact that philosophers have found problems in distinguishing the terms, as shown in the extracts from papers putting forward conflicting views in Zabeeh et al. (1974, pp. 91–104). Quine's paper (extracted from Quine (1951)) sets out the necessary conditions for 'use' and 'mention' to be properly separated, while Garver's paper (extracted from Garver (1965)) undermines the concept of pure 'mention', claiming that 'in the paradigms of mentioning the word mentioned is also in some way used', but accepting that this 'in no way impugns the practical effectiveness of the use-mention distinction'.

Lyons describes the main problems that can arise for linguists in following this arguable distinction without a clear understanding of what is implied by it, and regards the distinction between object language and metalanguage as potentially different from that between use and mention. Despite these reservations, the concept of use and mention provides a useful basis for examining the differences between the Cobuild use of metalanguage and the conventions of the other dictionaries.

Piotrowski (1989, pp. 73–74) suggests two ways of considering the meaning of lexical items which seem in some ways to parallel the use-mention distinction:

> 'Thus, on the one hand meaning can be seen as a sort of entity: concept, notion, prototype, stereotype, or fact of culture. On the other hand, meaning can be seen as a sort of activity: skill, knowledge of how to use a word.'

Using both these pairs of terms as a basis for a description of defining styles, the traditional approach within monolingual English dictionaries is to mention the word which is being defined, and so to give information about its meaning primarily as an entity. Any separate examples of usage that they give

do actually use the word (in the technical philosophical sense) and so give information about its meaning as an activity. When usage notes of some sort are also given, these generally preserve the separation between metalanguage and object language and mention the circumstances of normal usage rather than using the word directly. In contrast, Cobuild definition sentences clearly use their headwords within their normal linguistic context as an integral part of the process of mentioning them, and so deal with the meanings of the words being defined both as entities and activities. In the separate grammar notes, additional usage notes and examples, which are also given in the Cobuild range this basic information is supplemented by a mixture of use and mention, but these extra elements do not invariably provide information which cannot be deduced from the definition sentence itself.

Hanks (1987) introduces a further complication to this notion of the combination of 'use' and 'mention' in Cobuild definitions. He suggests that:

> Dictionaries are much concerned with accounting for what it is that an utterer may expect a hearer to believe. Whatever this is, it is in the form of a presumption rather than certain knowledge.
>
> (Hanks, 1987, p. 135)

As an example, he suggests that the definition given in *CCELD* for sense 1 of 'wash':

> If you **wash** something, you clean it because it is dirty, using water and soap or detergent. (*CCELD*, p. 1640)

is strictly a form of shorthand for a statement like:

> If you say that you are **washing** something, you probably intend to create the belief that you are cleaning it because it is dirty, and that you are using water and soap or detergent.

As he points out, this form of words is actually used for some meanings of words. Consider the definition of 'old school tie':

> In British English, when people talk about the **old school tie**, they are referring to the situation in which people who knew each other at public school or university use their positions of influence to help each other; usually used showing disapproval.

This direct metalinguistic comment on the usage of words is, nonetheless, still

a statement about meaning, as Hanks points out (Hanks, 1987, p. 135). The combination of the two modes of definition in one dictionary exposes Cobuild to the risk of confusion between them, and Fillmore (1989, p. 63) cites an example of a definition in *CCELD* in which this has been brought about by the unfortunate addition of an indefinite article:

> A **cunt** is a very rude and offensive word that refers to a woman's vagina.
> (*CCELD* p. 345, sense 1)

This confusion does not exist in the corresponding senses of *OALDCE* or *LDOCE*, both of which separate their usage comment from the definition:

> *n* (*offensive*[1]) **1** (*sl*) (**a**) vagina. (**b**) outer female sexual organs.
> (*OALDCE* p. 291)

> *n taboo* **1** VAGINA
> (*LDOCE* p. 252)

Taboo words of this level are not included in *CCSD*, but its treatment of the word 'bum' shows one alternative approach which avoids the confusion:

> Your **bum** is the part of your body which you sit on; an informal British use.

This, however, leaves unresolved the problem of taboo words, since the use of a similar form for a word like 'cunt' would give completely the wrong message in the first part of the definition, and the proper approach to problems of this kind is almost certainly the use of a fully metalinguistic definition.

Despite their potential dangers, these features, peculiar to full definition sentences, make them a uniquely rich source of information and show that their analysis could be an extremely valuable linguistic exercise. Before the potential of this analysis can be fully appreciated, it is necessary to examine the nature of dictionary definitions in general and the full implications of their realisation in the Cobuild dictionaries.

2.2 Defining the meanings of words

Definition is not a straightforward process. Senses 1 and 2 of the word in *CCELD* are defined in the following way:

> **1** If you **define** something, you show, describe, or state clearly what it is and what its limits are, or what it is like.

> **2** If you **define** a word or expression, you explain its meaning, for example in a dictionary.
>
> (*CCELD*, p. 370)

OALDCE, in senses 1 and 2, has:

> **1** ~ **sth** (**as sth**) state precisely the meaning of (eg words)
> **2** state (sth) clearly; explain (sth)
>
> (*OALDCE*, p. 314)

and *LDOCE*, in senses 1, 2 and 4, has:

> **1** to give the meaning of (a word or idea); describe exactly
> **2** to explain the exact qualities, limits, duties etc., of
> **4** [(**as**)] to show the nature of; CHARACTERIZE
>
> (*LDOCE*, p. 269)

There is substantial agreement between these dictionaries on the central process involved in their construction and usefulness, but there are also important differences between them. Both *OALDCE* and *LDOCE* include the notion of exactness or precision, treating the meaning of a word as something which can be 'stated precisely', 'given' or described 'exactly'. The Cobuild definition of sense 2, which is aimed specifically at words, paraphrases definition as the explanation of meaning: the relevant sense of 'explain' in the same dictionary is:

> **1** If you **explain** something, you give details about it or describe it so that it can be understood.'
>
> (*CCELD*, p. 495)

Giving details about a word or describing it so that it can be understood implies a process which is both looser and more open-ended than the rigidly precise terms demanded by the other dictionaries, and, crucially, this definition incorporates the purpose of the act of explanation. Explaining the meaning of a word is a process which is meant to lead to understanding. The conflict implied by these differences of approach between the Cobuild dictionary range and the others needs to be explored in detail to assess the effects it may have on the usefulness of the information contained in full definition sentences.

2.2.1 Lexicographical definition

Zgusta et al. (1971) distinguish between 'lexicographical definitions' and 'logical definitions', accepting that they overlap, but stressing the 'striking differ-

ences' that exist between them. In particular, the logical definition:

> must unequivocally identify the defined object (the *definiendum*) in such a way that it is both put in a definite contrast against everything else that is definable and positively and unequivocally characterized as a member of the closest class

whereas the lexicographic definition:

> enumerates only the most important semantic features of the defined lexical unit, which suffice to differentiate it from other units.
>
> (pp. 252–253)

In a footnote to this passage, (p. 252, note 86), the separate Russian terms for the two concepts are discussed: *definicija* for the logical definition, and, for the lexicographic definition, *tolkovanie*, which is translated as:

> something like "interpretation, explanation"

Bolinger's description (Bolinger, 1965, p. 572) of the implications of this separation for dictionaries reads like a prophetic summary of the difference between the Cobuild approach and traditional dictionary definitions:

> Dictionaries do not exist to define, but to help people grasp meanings, and for this purpose their main task is to supply a series of hints and associations that will relate the unknown to something known.

This presupposes that there is a difference between the two processes, a difference which the Cobuild definition of 'define', quoted in section 2.2, seems to deny. The precise nature of the definition process in dictionaries needs to be examined to identify this possible conflict and assess its implications. In order to understand the range of possibilities that exist for dealing with meaning in modern dictionaries, it is necessary to consider briefly the history of the development of the monolingual English dictionary.

2.3 Stages in the development of the monolingual English dictionary

The modern dictionary, and especially the learners' dictionary, as we have already seen, gives information which goes beyond the generally accepted concept of the meaning of each sense of its headwords. An examination of the stages in the development of the monolingual English dictionary should reveal how this information came to be selected as the most appropriate or

useful data to give about a word, and whether there have been any changes in the functions of such dictionaries.

Béjoint (1994, p. 92), considering the earliest origins of dictionaries, suggests that they 'are probably much older than is generally said.' He argues convincingly that all societies with writing systems, and at least some of those without, have produced dictionaries of some kind, though not necessarily all for the same reasons. These do not always convey meanings in the same way as a conventional modern dictionary.

As an example within our own culture it may be worth considering the contents of some of the 'listing' nursery rhymes such as 'The House that Jack Built', or 'The Twelve Days of Christmas'. It is at least possible that the relationships between the items on the list constitute devices for acquiring linguistic information. At the very least these songs give catalogues of lexically related groups of words. In the case of 'The House that Jack Built' the song also includes primitive defining strategies, best illustrated in the last verse:

> This is the farmer sowing his corn,
> That kept the cock that crowed in the morn,
> That waked the priest all shaven and shorn,
> That married the man all tattered and torn,
> That kissed the maiden all forlorn,
> That milked the cow with the crumpled horn,
> That tossed the dog,
> That worried the cat,
> That killed the rat,
> That ate the malt
> That lay in the house that Jack built.
>
> (Opie & Opie, 1951, pp. 229–231)

Every line of the cumulative verses of the rhyme, usually accompanied by appropriate illustrations on its first occurrence in printed editions, sets out some of the typical characteristics of the item introduced in the previous line as an integral part of the narrative. As an example, consider the *CCELD* definition for sense 1.1 of 'cat':

> A **cat** is a small furry animal with a tail, whiskers, and sharp claws that kills smaller animals such as mice and birds.
>
> (*CCELD* p. 214)

The line relating to 'cat' in the rhyme:

> That killed the rat

has significant echoes in this definition. Each line is almost a form of definition, and the cumulative nature of many of these catalogue rhymes in recitation could make them especially suitable for teaching the lexical, syntactic and even semantic properties of the words in their texts. Opie & Opie (1951) suggest that other similar accumulative rhymes, such as 'The Twelve Days of Christmas' (pp. 119–122) and 'The Wide-mouth waddling Frog' (pp. 181–183) would be played as forfeit games, with individuals responsible for each verse and paying forfeits for mistakes. The full title of a version of this latter rhyme, quoted by Opie & Opie from *The Top Book of All*, published around 1760, is 'The Play of the Wide-mouth waddling Frog, to amuse the mind, and exercise the Memory', an explicit statement of a pedagogic role concealed in the fun.

Early spelling books use similar techniques to distinguish between words which can easily be confused with each other: they place their subject words in a suitable context to provide the necessary information. The following consecutive groups of words are taken from R. Browne's *English School Reform'd* (1700, pp. 68–69), which is arranged in approximate alphabetical order:

> Pair of Shooes.
> Pare your Nails.
> Pear, a sort of Fruit.
> Peer of a Realm.
>
> Plot not against the King.
> Plod, or Walk.
>
> Pray to God.
> Prey, or Covet.
>
> Queen of England.
> Quean, a Harlot.
>
> Roof of a House.
> Rough, or Course.
> Ruff for the Neck.

A similar technique is used in Cocker (1696) to differentiate between 'Words which bear the like Sound, and Pronunciation, yet are of different Signification and Spelling, and are apt to cause mistakes in Writing' (p. 100). The entries under 'L' show the general range of techniques used:

> *Lick* honey if you *like* it.
> *Lock* the door; *Look* for good *Luck*.

Lanch the ship; *Lance* the Wound.
Leash of hounds; *Lease* of a House.
Less than another; *Lest* you suffer for it.
Learn this *Lesson*, not to *Lessen* or despise any.
Listen, and you may hear ye *Listed* Souldier.
Look to the *Lamb*, for he is *Lame*.
Loud the Oxe *Lowed*.
Lowr and frown; *Lower* than before; *Lour*, a *French* Palace.
Lot in *Sodom*; *Loth* and unwilling; *Loath* and abhor.
Louse bites, *Loose* and unty; *Lose* nothing.
Lice and Fleas; *Lies* are often reported.
Liturgy, or Common-prayer: *Lethargy* sleeping.
Line for a Jack: A *Loyn* of Veal.
League of Peace: *Leg* of the body.
Lattice of a window: The Maid *Lettice* fetcht some *Lettuce*

(Cocker, 1696, p. 103)

In most of the examples from both Browne and Cocker the setting of the words in some form of typical context establishes the method of treatment of them as 'use' rather than 'mention', so that the knowledge being presented relates to the word as activity, not only as entity. In some cases given above (e.g. 'Pear', 'Plod', 'Prey', 'Quean' and 'Rough' from Browne, 'Lour' and 'Liturgy' from Cocker) brief definitions or equivalents are given, so that use and mention, entity and activity, are mixed. One other important element is exhibited by the set of examples from Browne, two of which, 'Plot' and 'Pray', act partly as moral exhortations rather than neutral linguistic statements. The inclusion of this moral element is an explicit feature of many of the later dictionaries, most notably and self-consciously Johnson's.

2.3.1 English Dictionaries before Johnson

Histories of monolingual English dictionaries normally begin towards the end of the 16th century, and Cawdrey's *A Table Alphabeticall*, produced in 1604, is usually cited as the first fully recognisable specimen. This work is dealt with in detail in the next section. Glosses and bilingual dictionaries certainly existed before that date, together with spelling books and language manuals which contain some of the information normally associated with monolingual dictionaries.

As an example, Edmund Coote's *The English Schoole-maister* contains a

twenty page vocabulary list in alphabetical order, in which most of the words are given a brief gloss. He describes this as:

> a true Table conteining and teaching the true writing and understanding of any hard english word, borrowed from the Greeke, Latine, or French, and how to know the one from the other, with the interpretation thereof by a plaine English word
>
> (Coote, 1596, introductory note 12)

This extract shows its main features:

> *Garboile* **hurly burly**
> **garner. corne chamber**
> gem **precious stone**
> gentilitie)
> generositie) **gentrie**
> gentile **a heathen**
> generation **offspring**
> **gender**
> genealogie g.**generation**
> genitor **father**
> **gesture**
> **gives fetters**
> **ginger**
> **gourd *k* plant**
>
> (Coote, 1596, p. 84)

A detailed key to the conventions adopted is given in his introduction to the table: Roman letters are used for 'words taken from the Latine or other learned languages', italics for those from French, and 'those with the English letter, are meerly English, or from some other vulgar tongue.' The 'English letter' or black letter is shown above as bold type. Further annotations are 'g.' for Greek and 'k' for 'a kind of' (Coote, 1596, pp. 73–75)

The alphabetic arrangement of Cawdrey's work is lacking in most of the other earlier works, but the concept of a list of words arranged with their equivalents is established very early. The most important feature of Cawdrey's book is that it is purely a list of words and definitions and specifically monolingual. However, like its ancestors the glosses, it deals exclusively with the words which are likely to be difficult to understand.

2.3.1.1 *Hard word dictionaries*

The title page of the first edition of Cawdrey's book echoes Coote's introductory note:

> A table alphabeticall, conteyning and teaching the true writing and understanding of hard usuall English wordes, borrowed from the Hebrew, Greeke, Latine, or French, &c.
> With the interpretation thereof by *plaine English words, gathered for the benefit & helpe of Ladies, Gentlewomen, or any other unskilfull persons.*
> Whereby they may the more easilie and better understand many hard English wordes, which they shall heare or read in Scriptures, Sermons, or elswhere, and also be made able to use the same aptly themselves.
> *Legere, et non intelligere, neglegere est.*
> As good not to read, as not to understand

This is a very explicit description of the purposes and the method of the work. It is interesting to note that it is aimed at a very specific market, the word 'unskilfull' presumably describing their lack of knowledge of classical languages, although in practice it seems likely that its full readership would extend beyond the exclusively female examples given. It is also intended both for interpretation and production. In the traditions of the time, much of its contents were, of course, taken from existing works. Starnes and Noyes (1991, p. 13) draw attention to his extensive use of Coote (1596) both for general inspiration and for substantial portions of the word-list, definitions and surrounding text. They also stress the information that he incorporated from elsewhere, especially Thomas' Latin-English Dictionary of 1588. The tradition of near-plagiarism as a means of creating new dictionaries is established at the outset.

The defining method adopted by Cawdrey is stated on the title page as using 'plaine English words'. In the examples given below similar conventions are used to those in the extracts from Coote (1596) given in the previous section: black letter printing is shown in bold type, (g) after a word means that it is derived from Greek, § before it means that it is from French, and (k) means 'a kind of'. Cawdrey's spelling has been preserved, but no attempt has been made to show the use of the long form of s or the special character for a doubled o.

> abdicate, **put away, refuse, or forsake**.
> aggrauate, **make more grieuous, and more heauie:**
> agilitie, **nimblenes, or quicknes.**
> alacritie, **cheerefulnes, liuelines**

apologie, **defence, or excuse by speech.**
auburne (k) **colour**
§barke, **small ship**
capitall, **deadly, or great, or woorthy of shame, and punishment:**
celebrate, **holy, make famous, to publish, to commend, to keepe solemnlie**
circumspect, **heedie, quicke of sight, wise, and dooing matters advisedly.**
delectation, **delight, or pleasure**
diminution, **lessening**
effect, **a thing done, or to bring to passe**
§enhaunce, **to lift up, or make greater:**
expert, **skilfull**
fabricate, **make, fashion**
foraine, **strange, of another country**
gargarise, **to wash the mouth, and throate within, by stirring some liquor up and down in the mouth**
genius, **the angell that waits on man, be it a good or euill angell**
glee, **mirth, gladnes**
hononimie, **when diuers things are signified by one word**
idiot, (g) **unlearned, a foole**
implacable, **that cannot be pleased or pacified.**
iudaisme, **worshipping one God without Christ.**
laborious, **painfull, full of labour**
magistrate, **governour**
§malecontent, **discontented**
nauigable, **where ships may safely passe, or that may be sailed upon.**
notifie, **to make knowne, or to giue warning of.**
odious, **hatefull, disdainfull**
omit, **let passe, ouerslip.**
palinodie, **a recanting or unsaying of anything**
passeouer, **one of the Jewes feasts, in remembrance of Gods passing ouer them, when he slewe so many of the Egiptians**
persecute, **trouble, afflict, or pursue after.**
pomegarnet, or pomegranet, (k) **fruite**
preposterous, **disorder, froward, topsiteruie, setting the cart before the horse, as we use to say**
racha, **fie, a note of extreame anger signified by the gesture of the person that speaketh it, to him that he speaketh to**
represent, **expresse, beare shew of a thing**
scurrilitie, **saucie, scoffing**
sympathie,(g) **fellowelike feeling.**
transferre, **conceiue ouer**
transparent, **that which may bee seene through**
truculent, **cruell, or terrible in countenance**
veneriall,) **fleshly, or lecherous,**

venerous,) **giuen to lecherie**
§vpbraid, **rise in ones stomach, cast in ones teeth:**

Even in this relatively small sample (50 words) it is possible to see certain characteristics of Cawdrey's defining style. Some words, such as 'barke', 'diminution', 'expert', 'magistrate' and 'malecontent', are given one-word synonyms. Others, such as 'aggrauate' and 'gargarise', are defined by simple phrases which are almost capable of replacing the single word in its normal contexts. Some, notably 'hononimie', 'nauigable' and 'palinodie', have more complex definitions, which would be much more difficult to use as straight substitutes. Some words, such as 'passeouer' and 'iudaisme' are plainly encyclopaedic entries. Many words, such as 'abdicate', 'capitall', 'celebrate' and 'effect' have several senses, which are given as an unannotated list. In the case of two words in the sample, 'veneriall' and 'venerous', their similarity of meaning is such that they effectively share a dictionary entry.

In considering these examples it must be remembered that this form of definition is still effectively a type of gloss, a list purely of words thought unfamiliar enough to the projected user of the dictionary to warrant inclusion, replaced by the most appropriate 'plaine English' word. No examples of usage are given, no guidance is given on selection of meaning where more than one sense is possible. There is a sense, therefore, in which the description of this dictionary and its immediate successors as 'monolingual English dictionaries' is inappropriate. Their purpose is to gloss words from a particular subset of English lexis, the new words derived from other languages, using words chosen from the mainstream of commonly used English lexis. Cawdrey in his prefatory address 'To the Reader' warns against the possible division of English:

> Therefore, either wee must make a difference of English, & say, some is learned English, & othersome is rude English, or the one is Court talke, the other is Country-speech, or els we must of necessitie banish all affected Rhetorique, and vse altogether one manner of language.
>
> (Cawdrey, 1604, p. 2 of 'To the Reader')

The *Table Alphabeticall* is, of course, a tool designed to help promote the unity of the language under these difficult circumstances. The general approach used by Cawdrey remains the norm until dictionaries begin to deal with a more general vocabulary in the early eighteenth century, as described in section 2.3.1.2 below.

The style of definition used by Cawdrey is, however, by no means con-

fined to the 17th century. Many of its features have been preserved in at least the smaller monolingual dictionaries being published now. Using *The Oxford Popular Dictionary*, a typical pocket-sized general purpose dictionary published in 1993, as an example, it is interesting to compare some modern definitions with Cawdrey's. Obviously, this is only possible where the word is dealt with in both dictionaries, and where both the word and the sense have survived relatively unchanged. From the first few entries in the sample of headwords from Cawdrey we find:

> **abdicate** *v.i.* renounce a throne or right etc. **abdication** *n.*
> **aggravate** *v.t.* make worse; (*colloq.*) annoy. **aggravation** *n.*
> **agile** *a.* nimble, quick-moving. **agilely** *adv.*, **agility** *n.*
> **alacrity** *n* eager readiness.
> **apology** *n.* statement of regret for having done wrong or hurt; explanation of one's beliefs; poor specimen.
> **celebrate** *v.t./i.* mark or honour with festivities; engage in festivities; officiate at (a religious ceremony). **celebration** *n*
> **circumspect** *a.* cautious and watchful, wary. **circumspection** *n.*
> **delectation** *n.* enjoyment
> **diminution** *n.* decrease

There is certainly a little more syntactic information, but the overall amount of detail given and the concept of what constitutes the definition of meaning is almost identical.

The general dictionary model set up by Cawdrey and his predecessors, and indeed their complete entries, continued to be used well into the 17th century: Bullokar's *The English Expositor* (1616), Cockeram's *The English Dictionarie* (1623), Blount's *Glossographia* (1656), Phillips' *The New World of English Words* (1658) and Coles' *An English Dictionary* (1676) all deal with 'hard' or 'difficult' words. There does seem to be a trend towards greater verbosity in the definitions, perhaps in the pursuit of greater precision or a greater usefulness. Starnes & Noyes (1991, p. 23) give a comparison of Cawdrey and Bullokar which shows a general tendency to add words to the definitions, often making them less terse and cryptic in the process. As an example, consider Bullokar's definition of 'aggravate' in comparison to Cawdrey's given above:

> To make any thing in words more grievous, heavier or worse than it is.

The extra elements in this definition restrict the operation of the word to 'anything in words' and add the concept 'to make worse'. This may not in

practice be any more accurate, precise or helpful than Cawdrey's original: what is important is that this tendency to give more information, especially on restrictions of operation of meanings, continues as the hard word dictionary develops. Alongside the increase in size of entries there is also a steady increase in the total numbers of words included, from around 3,000 in Cawdrey to 25,000 in Coles, who also includes dialect words, but no pretence is made to cover the more usual words of the language. Apart from the other limitations of these early dictionaries, this restricted scope would make them unsuitable for use in natural language processing systems. Most modern monolingual dictionaries are more comprehensive, and J.K.'s *A New English Dictionary* (1702), which covers about 28,000 words, is one of the first to attempt this development.

2.3.1.2 *Comprehensive dictionaries*

The title page of *A New English Dictionary* (K[ersey], 1702) explicitly draws attention to the extent of its departure from the hard words tradition:

> A New English Dictionary: Or, a Compleat Collection Of the Most Proper and Significant Words, Commonly used in the Language; With a Short and Clear Exposition of Difficult Words and Terms of Art.
> The whole digested into Alphabetical Order; and chiefly designed for the benefit of Young Scholars, Tradesmen, Artificers, and the Female Sex, who would learn to spell truely; being so fitted to every Capacity, that it may be a continual help to all that want an Instructor'

Starnes & Noyes (1991, p. 71) refer to the fusion attempted in J.K.'s work between the spelling and grammar books, with their lists of ordinary words, usually without definition, and the dictionary, with its treatment only of hard words. The improvement of spelling is the main declared aim of this dictionary, and even the brief summary on the title page makes clear the difference between the treatment of hard words, which are given a 'Short and Clear Exposition', and the 'Compleat Collection Of the Most Proper and Significant Words, Commonly used in the Language'. The common words in the dictionary are often simply listed, as in a spelling book, although attempts are made to put them in a useful and informative context, as with these examples taken from the first two pages:

> A-board, as *a-board a Ship*
> Above, as *above an Hour*

About, as *about Noon*
A-broach, as *a vessel a-broach*
To sit abrood *upon eggs, as a bird does*
To accustom, *himself to a thing*
A-cross, as *arms folded a-cross*
An Adamant-*stone*
Addle, as, *an addle egg*

These look remarkably like ancestors of the Cobuild explanatory style, especially in their use of a different typeface to highlight the headword within surrounding text and their insertion of the headword into something like normal English phrases. Starnes and Noyes (1991, p. 73) point out the similarity of their structures to examples taken from contemporary spelling-books (already quoted in section 2.3).

Most of the examples of definitions given in Starnes & Noyes (1991, p. 74) from the revised 1713 edition of J.K.'s *New English Dictionary* are more genuinely definitions, rather than slightly random examples of usage, and the comparison shown there between the earlier and the later edition entries indicates that this is a conscious change of policy. These changes bring them even closer to the Cobuild style:

A *Gad*, a measure of 9 or 10 feet, a small bar of steel.
The *Gaffle* or Steel of a cross-bow.
A *Gag*, a stopple to hinder one from crying out.
A *Gage*, a rod to measure casks with.
To *Gage* or *Gauge*, to measure with a gage.
To *Gaggle*, to cry like a goose.
A *Gallop*, the swiftest pace of a horse.

Only the lack of a connective 'is' or 'means' prevents most of these definitions from reading almost exactly like the simplest forms of Cobuild definitions, for example:

A **gag** is a stopple to hinder one from crying out.
To **gaggle** means to cry like a goose.

Slightly more rearrangement of the definition of 'gaffle' would produce:

The **gaffle** of a cross-bow is its steel.

While this exercise may seem a little contrived, it seems important to point out that the principles used in this very early inclusive dictionary may have more in common with those applied in the Cobuild range than either ap-

proach has with the dictionaries produced during the 18th, 19th and earlier 20th centuries.

Some hard word dictionaries were still produced in the early 18th century, such as *Cocker's English Dictionary*, largely based on Coles' 1676 work and other earlier dictionaries, but the trend was now generally towards inclusiveness. Bailey's *Dictionarium Britannicum*, 1730, covers about 48,000 words and gives guidance on stress and details of etymology as well as definitions and examples of usage. This is not the first dictionary to include etymology: Blount provides details of either the original word adapted into English, or, where the word has been adopted without modification, of the source language; even Coote's brief table shows language of origin, as already described in section 2.3.1. It forms the sole subject of some earlier dictionaries: the *Etymologicon Linguae Anglicanae* (1671) deals exclusively with the etymology of English words, and purely etymological dictionaries continue to be produced up to the present day (e.g. Onions, 1966). The degree of importance attached to etymology as a source of information about headwords is, however, greatly increased from Bailey's time onwards, and it needs to be considered in some detail.

2.3.1.3 *The role of etymology in monolingual English dictionaries*

Etymology has a complex and sometimes doubtful relationship with the description of meaning in monolingual dictionaries. It has in the past been given great prominence in general purpose monolingual dictionaries, but seems to be given less importance in modern dictionaries that do not concern themselves specifically with historical descriptions. None of the learner's dictionaries so far referred to comments on the etymology of its headwords, presumably because it is not regarded as useful information for learners of the language, and it is not included in Boguraev & Briscoe's list outlined in section 1.2 above. Landau (1989, pp. 102–3) considers whether etymological information should be given in any synchronic dictionaries, and only decides that it should on the basis that, despite the inherent danger of misunderstandings arising from it, it provides a necessary cultural context for the present meaning of a word. Its main danger, of course, is that it can be seen as providing the 'correct' meaning, in a way which does not even need to rely on the lexicographer's intuition.

The origin of the word 'etymology' itself reflects this problem: the Greek word 'ετυμος' simply means 'true' (Liddell & Scott, 1869, p. 616), and in many

cases the original meaning of the source of a word has been considered to be the only possible true meaning of that word. Presumably this is because it can be considered as its first meaning, departures from which are regarded as a form of linguistic decay. The concept of a fixed, 'real' meaning of a word, central to any prescriptive form of lexicography, means that semantic changes are seen as regrettable departures from an authoritative standard. Such an attitude ignores the whole process of language change, and especially the fact that almost all borrowings into English from other languages shift their meanings significantly as they enter the language, and continue to develop steadily thereafter. It also conveniently ignores the difficulty of establishing a definitive and fixed meaning for the actual or supposed roots of the word in the source language. In practice, even the details of semantic development within English are generally agreed to be clouded in obscurity in most cases. Nuccorini (1993, pp. 103–4), discussing the impossibility of distinguishing between homonymy and polysemy, describes the problems that native speakers have with this area:

> Gli stessi parlanti nativi sono spesso in disaccordo se richiesti di individuare relazioni di significato tra supposti omonimi e in genere incapaci o impossibilitati a trovare radici etimologiche, comunque non "percepite", che li spieghino.[2]

Despite these significant problems, during the 18th and 19th centuries etymology was seriously treated as a major source of absolute meaning, and the idea is not entirely dead even now. Perhaps its apparent certainty and relative ease of determination, both in practice likely to be spurious, are somehow seen as compensating for its lack of any necessary practical connection with the likely range of current usages. It is also certainly the case that the attraction of the history of a word as an explanation for its current use and the reverence still felt for classical texts were strong factors in its continued prominence. The main problem posed by the influence of etymology on views of semantics in the use of dictionary information by natural language processing systems is the probable discrepancy between the information provided by the dictionary and real language use. To see how far this influence affected the nature of dictionary definitions, we need to consider the next major stage in the development of the monolingual English dictionary: Johnson's *Dictionary of the English Language*, first published in 1755.

2.3.2 Johnson

Lexicographers before Johnson usually make definite claims for the contents of their works once they are published: Johnson is probably the first to state in advance and in detail, in *The Plan of a Dictionary of the English Language* (Johnson, 1747), what he thought his dictionary should set out to do, and how he intended to achieve it. The *Plan* is addressed to the Earl of Chesterfield, and is plainly intended to obtain patronage from him. Despite this, Johnson's statement of his aims and projected methodology provides an extremely valuable insight into the attitudes to lexicography of one of its most influential practitioners. Although, as we shall see, he did not succeed in carrying out all of his objectives, his stated intentions, generally without the detailed descriptions of the problems that he foresaw in achieving them, have probably had more influence on the aims and approach of later monolingual English dictionaries than the actual dictionary that he eventually published.

2.3.2.1 *The Plan*

The Plan of A Dictionary of the English Language (Johnson, 1747) states quite explicitly what Johnson wants his dictionary to do, and the reasons for the choices that he intends to make. Summarising his intentions at the end of a detailed scheme of work, he describes the scope of his proposed dictionary:

> This, my Lord, is my idea of an English dictionary, a dictionary by which the pronunciation of our language may be fixed, and its attainment facilitated; by which its purity may be preserved, its use ascertained, and its duration lengthened.
>
> (*Plan*, p. 32)

It covers, in some detail, the principles which he intends to apply to:

> the selection of the word-list
> the choice of an appropriate standard spelling
> the contents of each dictionary entry; and
> the use of illustrative quotations and the basis of their selection.

The value of this to an investigation of dictionaries in general and the Cobuild definition style in particular lies in its contribution to our understanding of what lexicographers have thought they were doing when they produced dictionaries.

For a hard word list, which, as explained earlier in section 2.3.1.1, is

effectively the same exercise as the provision of a gloss for foreign words, there is little need to consider in detail either the objectives or the method adopted to achieve it. Hard words need to be explained in as much detail as the user needs in simple words, words which the user should already know and understand. For a comprehensive monolingual dictionary the whole purpose of the exercise is much more elusive. Among other questions the lexicographer needs to consider the reasons for including common words, and to devise a method for dealing with them so that their meanings and usage become clearer. The nature of the dictionary's users and the demands that they will make on it are obviously crucial elements in its design, but these factors are by no means straightforward or easy to determine.

Johnson, of course, has a definite aim, as already quoted from the *Plan*. His dictionary is to be the means of fixing the characteristics of a language whose instability caused serious writers embarrassment and reduced its effectiveness as a means of communication. He equates linguistic instability with moral and cultural weakness, and intends to deal with them both by the same process. His dictionary is to be unequivocally prescriptive: even those elements which are not direct comments on the language, the illustrative quotations, are to be selected for their moral uplift as well as for their appropriateness to the perceived correct usage of a word. The whole purpose of the dictionary is a moral one, capable of being determined in advance.

2.3.2.2 *The Dictionary*

The Preface to *A Dictionary of the English Language* (Johnson, 1773) shows that, in practice, he did not find the exercise quite so straightforward:

> When we see men grow old and die at a certain time one after another, from century to century, we laugh at the elixir that promises to prolong life to a thousand years; and with equal justice may the lexicographer be derided who being able to produce no example of a nation that has preserved their words and phrases from mutability shall imagine that his dictionary can embalm his language, and secure it from corruption and decay, that it is in his power to change sublunary nature, or clear the world at once from folly, vanity and affectation.'
>
> (Johnson, 1773, p. xi)

Despite this retraction, the fundamental notion of the dictionary as a prescriptive and authoritative source of the standard spelling, the correct meaning and even the inherent validity of a word as a piece of English vocabulary seems firmly entrenched in this dictionary and many of its successors, including

those being published today. Johnson himself goes on to make a case for an attempt at prescription:

> It remains that we retard what we cannot repel, that we palliate what we cannot cure. Life may be lengthened by care, though death cannot be ultimately defeated: tongues, like governments, have a natural tendency to degeneration; we have long preserved our constitution, let us make some struggles for our language.'
>
> (Johnson, 1773, p. xii)

If his dictionary cannot be wholly prescriptive, it will at least exercise as much linguistic conservatism as it can to slow the changes that it cannot wholly prevent.

This attitude means that, as already suggested in section 2.3.1.3, current usages may not coincide with those that lexicographers wish to preserve in their dictionaries. In Johnson's *Dictionary*, the quotations are chosen to illustrate meanings that he has already selected for the words: they are attestations of authority for that meaning, but do not necessarily form the basis for it. The primary source of meaning is Johnson himself, relying on his own superior grasp of the language and embodying it in the dictionary as part of his 'struggles for our language'.

It is important, then, to realise that although he specifies the body of text that he has used for his quotations, these should not be regarded as in any way equivalent to the corpus used for modern dictionary production. Béjoint (1994, p. 97) stresses the main difference:

> In modern corpus-based lexicography, all the words in the word-list, all their meanings, and all the quotations that illustrate them come exclusively from the corpus. The 'corpus' of eighteenth-century lexicographers was not closed or in any way meant to be representative of all the varieties of the language.

He also points out on the same page that 18th century lexicographers adapted their corpora 'to suit their needs', a point particularly relevant to Johnson. In his definition of sense 3 of 'universal' Johnson uses the quotation:

> An universal was the object of imagination, and there was no such thing in reality.
>
> (Johnson, 1773, p. 2151)

As McDermott (1995, pp. 145–146) points out, the original text reads:

> An universal was not the object of imagination, and there was no such thing in reality.

Johnson seems to have misunderstood the meaning of the text, and has altered it to remove what he saw as an inconsistency.

This equation of the meaning of a word with the lexicographer's own actual or idealised usage exposes a major problem of lexicography. Even the lexicographer who relies on etymology for meaning is using an outside source whose authority, doubtful though its validity might be, has at times been generally agreed. The lexicographer who acts not as discoverer of meaning, but as the source of it, risks more than mere inaccuracy. Inaccurate dictionaries may not directly affect the ways in which native speakers use their mainstream vocabulary, but they are capable of misleading language learners, including even the native speaker in search of the meanings of more obscure words, and would significantly impair the usefulness of information extracted for natural language processing systems.

It is probably true to say that modern monolingual dictionaries are widely regarded as the main source of authority for the meaning of a word, and that this respect for the dictionary depends on a widely held belief in the notion of 'correct' meanings for words. In many people's minds, conflicts between the meanings of specific words enshrined in dictionaries and their own usage of the same words are often assumed to imply that they are using the words wrongly. This probably does not affect their use of those words, but there are important negative implications for their use in natural language processing if they cannot be relied upon to reflect normal usage rather than the lexicographers' own prejudices. Hopefully, modern dictionaries, especially those produced on the basis of large representative language corpora, should be relatively free from this defect.

2.3.2.3 *Johnson's definition strategies*

The sample of definition texts below is taken from the fourth edition of Johnson's *Dictionary*, and shows a range of his definition strategies. It has been stripped of the other elements of the dictionary text — the etymology, illustrative quotations, authorial comment etc.

> FICKLE. 1. Changeable; unconstant; irresolute; wavering; unsteady; mutable; changeful; without steady adherence.
> 2. Not fixed; subject to vicissitude.
> FICKLENESS. Inconstancy; uncertainty; unsteadiness.
> FICKLY. Without certainty or stability.
> FICO. An act of contempt done with the fingers, expressing a fig for you.
> FICTILE. Moulded into form; manufactured by the potter.

> FICTION. 1. The act of feigning or inventing.
> 2. The thing feigned or invented.
> 3. A falsehood; a lye.
> FICTIOUS. Fictitious; imaginary; invented.
> FICTITIOUS. 1. Counterfeit; false; not genuine.
> 2. Feigned; imaginary.
> 3. Not real; not true; allegorical; made by prosopopoeia
> FICTITIOUSLY. Falsely; counterfeitly.

The list of meanings given for 'fickle' sense 1 is of interest. Although they are all close in meaning to each other, they are not precisely synonyms. The user of the dictionary is being given a range of associated meanings, all recognisably within the same semantic area, with no indication of a method for differentiating between them. This method is widely used in the other definitions in the sample. Its effect is to give a series of roughly substitutable equivalents of the headword, leaving users to disambiguate from their own knowledge of normal contexts. A comparison with some modern dictionaries might be useful. *CCSD* (p. 203) has only one sense, specifically restricted to a person:

> A **fickle** person keeps changing their mind about what they like or want;

CCELD (p. 529) gives two senses:

> 1. Someone who is **fickle** keeps changing their mind about what they like or want;
> 2. If a wind or the weather is **fickle**, it changes often and suddenly.

OALDCE (p. 450) has only one entry:

> often changing; not constant

which echoes Johnson's list of undifferentiated meanings, although in the usage examples given for the word it includes:

> *a fickle person, lover etc.*, i.e. not faithful or loyal

LDOCE (p. 377) manages to cover both the *CCELD* senses together in one definition:

> likely to change suddenly and without reason, esp. in love or friendship

Hanks (1987, p. 120) describes this tendency of Johnson and later lexicographers to construct lists of approximately substitutable terms as the 'multiple-bite' strategy. In terms of Johnson's avowed aims it may be a reasonable thing

to do. Johnson is, after all, simply trying to describe the range of meanings over which a word's use is valid. For a modern learner's dictionary such a method seems unhelpful and uninformative, but the legacy of Johnson and his predecessors is obviously very powerful.

The implications of this approach for the use of dictionary definitions in NLP systems are obvious: the more differentiation that a definition provides between alternative senses within a specific semantic field the higher the quality of the information that can be extracted. Johnson's approach demands an informed human user to select the most appropriate meaning. The NLP system cannot rely on this intervention and needs as much precise information as it can get.

2.3.3 The Oxford English dictionary

The last major work to be considered in this brief survey of the development of monolingual English dictionaries is *The Oxford English Dictionary*, although in many ways it is a mistake to think of it as being in the mainstream of the process. Originally conceived by the Philological Society as a supplement to update the major existing dictionaries, such as Johnson's *Dictionary* and Richardson's *A New Dictionary of the English Language*, it became apparent very early in its development that a substantial work would be needed which would actually replace these other works. Trench (1857) laid down the basis for construction of such a dictionary, and a massive reading project was set in motion by the Society to collect data for it.

Under the chief editorship of James Murray until his death in 1915, *A New English Dictionary on Historical Principles*, later *The Oxford English Dictionary*, was published between 1879 and 1928. A supplement was needed almost immediately, and was published in 1933. A further four volume supplement was produced by a completely new editorial team between 1957 and 1986, and a reset, reordered and enlarged Second Edition was published in 1989. A completely revised Third Edition is currently under construction and partially available through the World Wide Web (at www.oed.com).

The scale of the *OED* is prodigious and overwhelming, but it is still very much a 19th century dictionary. Although it represents a magnificent achievement for its time, it suffers from the inherent impossibility of the task that its compilers set themselves, at least at the time at which the original work was carried out. Given the full involvement of computer technology the problems

involved in its production are likely to be far less intractable, though still by no means easy to overcome. The *OED* sets out to document the development of the entire vocabulary of English from the 12th century onwards, including as many obsolete and non-standard dialect terms as possible. For each word sense dealt with in the dictionary its entire life cycle needs to be shown, from its entry into English, including its ultimate discernible etymological origins in older forms of English and other languages, to the 'present' day (often the mid-nineteenth century) or the point at which it became obsolete. In addition to the definitions, past and present variants in spelling are shown and, where possible, dated quotations are given for every sense identified. Senses of the same word form are grouped together to give an indication of the likely route taken by the word during its semantic development.

This is, then, the ultimate descriptive English dictionary. Whether it is strictly monolingual is another matter: English can hardly be regarded as one language from the 12th century to the present day, and the differences are greater than merely dialectal or varietal. Certainly, its special requirements impose on the *OED* a structure more complex than any other dictionary with more modest aims could ever need. The sample of definition texts from Johnson's *Dictionary* given in section 2.3.2.3 shows the over-formalisation of entries, often with unnecessary repetition of elements that apply to several forms of the same headword, which can beset dictionaries that try to do too much. The *OED* has no choice: the complexity of its entries is forced on it by the function it is trying to perform. Sweet (1899, p. 141), in a discussion of the ideal dictionary for language teaching purposes, says that it 'is not, even from a purely scientific and theoretical point of view, a dictionary, but a series of dictionaries digested under one alphabet.'

The complexity of its structure is not entirely a bad thing. Although there are some inconsistencies inevitable in the construction of such a vast work entirely by manual means, this monument to nineteenth century perseverance performed amazingly well during its computerization. The section of the preliminary material to the Second Edition that deals with the History of the Oxford English Dictionary (*OED*, p. liii) describes the approach adopted to convert the dictionary text to a database:

> The structure devised by Sir James Murray and used by him and all his successors for writing Dictionary entries was so regular that it was possible to analyse them as if they were sentences of a language with a definite syntax and grammar.

This regularity allowed the use of an automatic entry parser as part of the conversion process, and the results of that process now allow computer readable versions of the *OED* to be accessed in a wide variety of different ways, providing scope for fairly sophisticated computer analysis. The accessibility of the data in the *OED* is already being exploited by researchers exploring the history of the English language. While this exploitation is unlikely to provide suitable information for use in NLP systems dealing with modern forms of English, its potential applications in research emphasise the value of making dictionaries of modern English equally accessible.

2.3.4 Learners' dictionaries

Dictionaries designed to help learners of a language obviously have very different objectives from those designed to act as reference books for native speakers, and their strategies would be expected to reflect these objectives. Despite their more limited scope and simplistic approach to definition, the original hard word dictionaries have significant elements in common with learner's dictionaries. It is also true to say that all of the dictionaries quoted so far, with the exception of the *OED*, regard themselves as having a pedagogic role. O'Kill (1990) points out that even Johnson's *Dictionary*, although 'implicitly addressed to a more sophisticated audience' was published in an abridged form and became 'a popular pedagogic tool for many years' (O'Kill, 1990, p. 10). Nuccorini (1993) extends the teaching role to all dictionaries:

> Ogni opera di lessicografia ha un aspetto didactico. Nel consultare un dizionario si cerca prevalentemente qualcosa che non si sa o di cui non si è sicuri, ed è in questo senso, nel rispondere alle domande o alle incertezze di chi li consulta, che i dizionari insegnano sempre qualcosa, anche se questo qualcosa varia da lingua a lingua, da situazione a situazione, da epoca a epoca, e, sopratutto, da dizionario a dizionario.[3]
>
> (Nuccorini, 1993, p. 39)

This places every user of a dictionary in the role of a learner. The crucial question for the use of any given dictionary as the source of a lexicon for an NLP system must then depend on the nature of 'questo qualcosa', 'this something' which the dictionary can provide as an answer to the user's questions. In the case of learners' dictionaries, changes in the nature of 'this something' can be traced to the end of the nineteenth century.

McArthur (1989, pp. 54–55) identifies a change in the approach to lan-

guage teaching in Europe and the USA around 1880, mainly as a reaction to three perceived negative aspects of existing methods:

a) a dependence on the classical languages
b) a bias towards literary and textual study
c) the use of formal drills and artificial translation exercises

The leaders of this change, including Henry Sweet, Paul Passy, Otto Jespersen, Wilhelm Vietor and Maximilian Berlitz, developed a system of teaching by immersion in the target language which helped create the appropriate conditions for the development of the learners' dictionary as a separate specialised form.

Sweet (1899, pp. 140–163) lays down the principles on which dictionaries ought to be constructed if they are to be useful for language learning. He deals with the scope of the dictionary, which 'should be distinctly defined and strictly limited' (p. 141), the usefulness of separate pronouncing dictionaries (p. 144), the need to avoid the superfluity of the contents of some dictionaries, which 'heap up useless material', usually in the form of obsolete words, rare and spurious coinages and encyclopaedic entries (pp. 145–146), the need for conciseness to be taken 'as far as is consistent with clearness and convenience'. In the section dealing with meanings he states: 'The first business of a dictionary is to give the meanings of the words in plain, simple, unambiguous language.' (p. 148). He also stresses the need for quotations (p. 149) and grammatical information relating to the constructions in which words are used.

The modern learners' dictionaries being considered in this chapter seem to incorporate at least some of these principles. They developed, according to Béjoint (1994, p. 66), from West and Endicott's *New Method English Dictionary* (*NMED*), published in 1935, and Hornby, Gatenby and Wakefield's *An Idiomatic and Syntactic English Dictionary*, published in Japan in 1942, which became the *Oxford Advanced Learner's Dictionary of Current English* (*OALDCE*), one of the dictionaries under consideration. Sweet's requirements for the treatment of meaning in learners' dictionaries are the most relevant for the present study, and it is now necessary to consider the options open to dictionary compilers for a basic concept of word meaning, and the methods used in learners' dictionaries to describe meaning.

2.4 The concept of meaning in dictionaries

The notion of the meaning of a particular word dealt with in a dictionary can, as has been shown, include the purely functional glosses of the hard word dictionary, which perhaps is strictly speaking a form of bilingual dictionary, the prescriptive formulation of correctness based on the lexicographer's intuition, etymological meaning, etc., found in most dictionaries of the eighteenth and nineteenth century, and many from the twentieth, and the neutral description of observed usage of the *OED*, often with notes on the main variations that can be encountered and their normal environments. Explicit choices between these options and their intermediate possibilities have been a major consideration of dictionary construction since the production of the very first monolingual English dictionaries.

It is interesting to consider whether this is such a significant issue in the construction of bilingual dictionaries, where a notion of prescriptiveness which does not reflect actual usage should certainly be considered a real defect. All too often, in fact, problems do arise in the use of bilingual dictionaries because of an inadequate consideration of the most useful notion of meaning. Consider the definitions of the Italian word 'punto' in the *Cambridge Italian Dictionary* (Reynolds, 1975, p. 204):

> **punto**[1] *part.* of **pungere**; *adj.* pricked, stabbed, punctured; (fig.) goaded.
> **punt-o**[2] *m.* dot, spot, point, mark; – *fermo*, full stop; (needlew.) stitch; *pl.* blackheads; (fig.) blemishes; *di – in bianco*, point blank; *in –*, *a –*, in order; state, condition; *far –*, to leave off, to stop payment; detail, item, particular; particle.
> **punt-o**[3] *neg.* not at all; no, not any

These meanings may all, in some sense, be accurate, but an examination of the occurrence of the word 'punto' in a corpus[4] of written Italian shows that they are not the most useful. The participial use quoted as the first sense did not occur at all in the 2,463 concordance lines for 'punto'. The concrete meaning of 'dot' or 'point' is also badly represented in the corpus, although its figurative meaning occurs in 452 instances of the phrase 'punto di vista', viewpoint or perspective. The most common single meaning occurs in various forms of the phrase 'mettere a punto', put in order, which is some way down the list.

The selection of the most appropriate meaning for use in a dictionary is obviously problematic, and is also of the utmost importance for the usefulness of dictionary information in language processing. It is now necessary to consider the sources of the semantic information used in the dictionaries.

2.4.1 Sources of semantic information for monolingual English dictionaries

If a dictionary is to provide useful semantic information for language processing it must derive it from appropriate and reliable sources. Because NLP systems are usually required to deal with real examples of language this automatically excludes prescriptive notions of meaning which do not reflect normal usage. Bindi et al. (1994, p. 29), discussing the essential role played by corpora in NLP, make this clear:

> If an NLP system is to process successfully a given language for a given purpose, it must be based on evidence of how language is really used. The analysis of corpora … is the main source of obtaining this evidence. As such it is irreplaceable.

On this basis, if a dictionary is to provide useful information for NLP systems, that information must be based directly on representative corpus evidence. The Cobuild range of dictionaries certainly meets the corpus requirement, and was the first major dictionary series to do so.

Even the *OED*, despite its comprehensively descriptive aims, suffers from the lack of a properly representative corpus. The army of volunteer readers recruited by the Philological Society in the first days of the project, whose work provided much of the raw material for the compilation of the dictionary, simply selected usage examples which appealed to them. Detailed instructions were given to the readers in the later stages, but these make it clear that the basis of selection would not produce a fully representative sample. The directions given in 1879 were:

> Make a quotation for *every* word that strikes you as rare, obsolete, old-fashioned, new, peculiar, or used in a particular way.
> Take special note of passages which show or imply that a word is either new and tentative, or needing explanation as obsolete or archaic, and which thus helps fix the date of its introduction or disuse.
> Make as *many* quotations *as you can* for ordinary words, especially when they are used significantly, and tend by the context to explain or suggest their own meaning.
>
> (*OED*, p. xli)

The level of judgement required of the readers was obviously unavoidable in a task of this nature, carried out without any text processing technology, but it also makes the sample judgemental and therefore unlikely to be properly representative of the language under examination.

The dictionaries under consideration in this analysis are intended for use by learners. Learners' dictionaries are generally used both for interpretation and production of the target language. This imposes a difficult compromise on the compilers of such dictionaries, since interpretative needs are more likely to be met by a wide-ranging description of the usages that the learner could encounter, while the needs of language production almost demand some sort of normative, if not actually prescriptive account of preferred usage. Sinclair, in the introduction to *CCELD* (p. xx) describes the principle used in its compilation as a 'cautious reflection of modern usage', and expresses the hope that:

> the language presented in this book is above all reliable, not dated nor markedly avant-garde, nor unusual to the kind of person we think of as an average user.

If this compromise is achieved, and the description produced meets Sinclair's requirements, the resulting dictionaries should represent a wholly appropriate source of information both for learners and for automatic language processing.

2.4.2 Adequacy of detail of the definitions

The level of detail of the information available from a definition is also of the greatest importance: the simple gloss provided by the hard word dictionary is unlikely to be particularly useful since it will not provide enough detailed environmental information. Learners' dictionaries can obviously assume less detailed linguistic knowledge from their users than those intended for use by native speakers, and this in itself should make them more suitable as sources of information for natural language processing applications. The general range of information provided by these dictionaries — phonology, morphology, syntax and semantics — corresponds exactly to the perceived needs of the NLP system described in section 1.2 above. The structure of the full definition sentence used in the Cobuild range, which includes a normal linguistic environment for the word being defined, provides even more detail than is found in other learners' dictionaries and this makes them potentially the most valuable of all.

2.4.3 Definition strategies

Once the source of the meaning and the level of detail have been determined, methods of definition appropriate to each sense need to be established. In Cawdrey's *Table Alphabeticall* the basic defining strategy is the provision of a synonym, or list of synonyms, as shown in many of the examples in section 2.3.1.1. Johnson continues this approach for most of the words in his *Dictionary*, but now and then a slightly different pattern is found, as in the following definitions taken from the Fourth edition. Page references are to Johnson (1773).

> **Barrack** Little cabbins made by the Spanish fishermen on the sea shore; or little lodges for soldiers in a camp (p. 152)
> **Dogkennel** A little hut or house for dogs (p. 581)
> **Foolhardy** Daring without judgement; madly adventurous; foolishly bold (p. 776)
> **Maleadministration** Bad management of affairs (p. 1194)
> **Tassel** An ornamental bunch of silk, or glittering substances (p. 1986)

In these cases, which use definition strategies relatively rare in the *Dictionary*, the defining phrases do not list straightforward synonyms. Instead, they use superordinate terms with accompanying discriminating elements to limit their more general meaning and focus on the explanation of the word's usage. The definitions given above can be analysed into these components:

Discriminator	**Superordinate**	**Discriminator**
Little	cabbins	made by the Spanish fishermen on the sea shore; or
little	lodges	for soldiers in a camp
A little	hut or house	for dogs
	Daring	without judgement;
madly	adventurous;	
foolishly	bold	
Bad	management	of affairs
An ornamental	bunch	of silk, or glittering substances

This approach is much more widely used in *OED*. The definitions of the main senses of the word 'barrack', for example, are:

> **1.a.** A temporary hut or cabin, e.g. for the use of soldiers during a siege, etc.
> **b.** 'A straw-thatched roof supported by four posts, capable of being raised or lowered at pleasure, under which hay is kept.'

> **2.** A set of buildings erected or used as a place of lodgement or residence for troops. (*OED*, p. 108)

In the learners' dictionaries under investigation in this project the superordinate and discriminator model shown here has become the main method of definition of both nouns and adjectives, and even for some verbs. The *OALDCE* definition of 'nominate', for example, is:

> formally propose that sb. should be chosen for a position, office, task, etc. (p. 838)

This could be analysed to give 'formally' as a discriminator preceding the superordinate, 'propose' as the superordinate and 'that sb. should be chosen for a position, office, task, etc.' as the following discriminator. As is described in more detail in section 5.3.7, this is the basis of the analysis of the definiens within the definition sentences. The restatement of meaning in these terms should give access to simpler and more frequent words than the original headword, and the information that it provides about the place of the definiendum within the lexical hierarchy should be an extremely valuable contribution to the needs of NLP systems.

2.4.4 The language of definition

The construction of definitions of words for a monolingual dictionary demands that the words used in them will be in some way either more precise or easier to understand than the headword is by itself — a modern equivalent of Cawdrey's 'plaine English words' or J.K.'s 'Short and Clear Exposition', (see sections 2.3.1.1 and 2.3.1.2) — if they are, in Sweet's words, 'to give the meanings of words in plain, simple, unambiguous language' (Sweet, 1899, p. 148: see section 2.3.4). Some dictionaries, such as *OED*, inevitably place more stress on technical accuracy of description than on ease of understanding. In others ease of understanding is so crucial to the usefulness of the dictionary that it has been formalised into a policy for the compilers. There may be a requirement that the language used to explain a meaning should only contain words which are more frequent than the headword being dealt with, or that it should be constructed using only words which belong to a specially selected defining vocabulary.

2.4.4.1 *Definition vocabularies in learners' dictionaries*

According to Cowie (1989b) West and Endicott's *New Method English Dictionary* (*NMED*) used a simplified defining vocabulary of 1,490 words to deal with its 24,000 entries, and a similar principle is applied in *LDOCE* (which claims a defining vocabulary of 2,000 words, listed in the back of the dictionary). The *Student's Dictionary* lists all words which are used ten times or more in definitions (1,860 words with 2,591 forms), and states that:

> The dictionary editors were asked to keep their explanations simple, but they were left free to choose any words they needed. When the dictionary was being revised, the computer was used to check carefully which words had actually been used, and then to produce a final list of these words. The result is a natural and economical word list.
>
> (*CCSD* p. 660)

No explicit description of a defining vocabulary is given in *OALDCE*, but a small sample of definitions examined (from 'ninety' to 'nobble', *OALDCE* p. 836) revealed the following words which were not in either the *LDOCE* or the *CCSD* lists:

bliss
chilly
claw
concrete
corrodes
corrugated
crab
debate
doorpost
fertilizers
frost
glycerine
grease
gripping
individual
intelligent
lobster
louse
majority
nimble
nitre
nitric
nitrogen

parasitic
petty
pincers
pinch
projection
saltpetre
seam
squirting
sulphuric
supreme
tunnel
umpire
unlawfully

Some of the names of chemical elements or compounds, such as 'glycerine', 'nitre', 'nitric', 'nitrogen', 'saltpetre' and 'sulphuric', are probably unavoidable, and the same may be true of 'crab', 'lobster' and 'louse', but it is interesting to see words like 'bliss', 'chilly', 'nimble', 'petty', 'projection' and 'supreme' being used within a defining vocabulary. *OALDCE*'s estimation of its users' abilities is obviously very different from that of the compilers of the other two dictionaries.

An examination of the words used in the definition elements of entries in the fourth edition of Johnson's *Dictionary*, using the letter 'f' as a sample[5], shows a total vocabulary of about 2,300 words, nearly 1500 of which are used once only. This suggests that Johnson's entire defining vocabulary is much larger than that of the learners' dictionaries, which is consistent with the rather different approach adopted for that dictionary. The main implication of the reasonably restricted vocabulary used for Cobuild definitions is that the lexis of the language used in them is reasonably small. By no means all of the words used most frequently within the definitions will have a structural significance, but the list does give a reasonable idea of the scope of the work needed to analyse them.

The vocabulary used to define words is an important aspect of the definition process, but the structural aspects of the definition also need to be considered. In particular, the nature of the equation which is set up between the two components of the dictionary definition, the definiendum and the definiens (described in section 2.1.1), dictates the extent to which the learner or the NLP system can use the appropriate part of the definition text as a direct substitute for the definiendum, and the extent of any linguistic processing needed to make that use possible.

2.4.4.2 *Substitutability of the definiens for the definiendum*

Consider the following definition from *CCSD*:

> **Acrimonious** words or quarrels are bitter and angry;
>
> (p. 6)

The definition of 'acrimonious' provides a set of words which could in theory be used to replace the headword, so that:

> acrimonious words or quarrels

could become:

> bitter and angry words or quarrels

LDOCE does not have a separate definition for the adjective (though it covers the noun, acrimony), but *OALDCE* has:

> (esp of quarrels) bitter
>
> (p. 11)

This is obviously also substitutable in the same way.

Hanks (1987, p. 119) ascribes the importance that lexicographers have attached to the preservation of this substitutability in their definitions to Leibniz's notion that:

> two expressions are synonymous if the one can be substituted for the other 'salva veritate' — provided that the truth remains unaltered.

This, according to Hanks, led lexicographers to believe that their defining text, the definiens, must be capable of substitution in any context for the definiendum, the lexical unit being defined.

Consider the following definitions from *CCSD* of the four senses of 'artificial':

> An **artificial** state or situation is not natural and exists because people have created it. (p. 27, sense 1)
> **Artificial** objects or materials do not occur naturally and are created by people. (sense 2)
> An **artificial** arm or leg is made of metal or plastic and is fitted to someone's body when their own arm or leg has been removed. (sense 3)
> If someone's behaviour is **artificial,** they are pretending to have attitudes and feelings which they do not really have. (sense 4)

None of these, strictly speaking, observes the substitutability requirement. It is certainly possible to construct the phrase:

> an arm or leg made of metal or plastic, fitted to someone's body when their own arm or leg has been removed

This suggests that the problem here is purely syntactic: the definition has used a different construction which cannot be substituted in exactly the same sequence, but in fact the huge difference in length between the definition and the original word means that the one could never be a substitute for the other in any real sense. With the other definitions there are much deeper problems of rearrangement. As an example, sense 4 can only become substitutable on the basis that:

> someone's behaviour is artificial = someone is pretending to have attitudes and feelings which they do not really have

The change of subject between the two sides of the equation makes any idea of substitution rather absurd.

The corresponding definitions in *OALDCE* are:

> made or produced by man in imitation of sth natural; not real (p. 56, sense 1)
> affected; insincere; not genuine (sense 2)

and in *LDOCE*:

> made by humans, esp. as a copy of something natural (p. 47, sense 1)
> lacking true feelings; insincere (sense 2)
> happening as a result of human action, not through a natural process (sense 3)

Despite the standard 'lexicographese' of these definitions, they are only marginally more substitutable than the elements of the Cobuild definitions.

Lack of substitutability may at first sight be a problem within NLP applications. In practice, however, the concept of substitutability in all circumstances is unattainable regardless of the efforts of the lexicographer. The infinite number of potential co-texts means that any definiens, however carefully constructed, could be an inappropriate combination for some realisations. It is also likely that the information which can be extracted from the dictionary will be adequate to provide the necessary syntactic information for any process of rearrangement that might be needed, as well as the semantic information normally expected.

2.4.4.3 *Explaining function words*

The concepts of the explanation of meaning discussed so far do not have an obvious application to the function words the, of, etc. These seem to represent

a special case within all dictionaries, where the information given is not strictly speaking an explanation of meaning, so much as a set of guidance notes outlining the circumstances under which the words are used. In this context, it is worth noting the view expressed by Hanks that 'all statements about word meaning are statements about word use' (Hanks, 1987, p. 135). As an example, he suggests that a definition such as:

> A **boy** is a male child

is strictly a form of shorthand for the full explanation:

> If you use the word **boy**, you can expect to be presumed to be talking about a male child.

There is still, however, a difference between this 'statement about word use', considered in a slightly different context in section 2.1.2, and the sort of thing that is needed for a function word. Consider for example sense 1 of 'the' from *CCSD*:

> You use the word **the** in front of a noun in order to indicate that you are referring to a person or thing that is known about or has just been mentioned, or when you are going to give more details about them.
>
> (*CCSD* p. 587)

and its corresponding entries in *OALDCE* and *LDOCE*:

> (used to make the following *n* refer to a specific person, thing, event or group) 1 (when it has already been mentioned or implied)
>
> (*OALDCE* p. 1329, sense 1)
>
> (used for mentioning a particular thing, either because you already know which one is being talked about or because only one exists)
>
> (*LDOCE* p. 1099, sense 1 of first entry for 'the')

Note the bracketing of both these definitions to show that the entire text constitutes a usage note rather than a normal definition, a fact which is specified explicitly and naturally in the text of the Cobuild definition, but which needs to be shown by a special code in the other two dictionaries to prevent the entries from being taken as definitions of meaning.

The exact difference between this usage information and the definitions given for other headwords is hard to define precisely. Perhaps the most useful way of describing it is to use the normal distinction between content or lexical words and function words. In the definition of a lexical word like 'meat':

> **Meat** is the flesh of a dead animal that people cook and eat.
>
> (*CCSD*, p. 347)

information is being given about what the word itself means. In Hanks' terms it is still a 'statement about word use', but when the word 'meat' is used it has a genuine semantic content in itself. In the case of a function word like 'the', the information relates to its effect on the meanings of the words following it, in other words to the function of the word 'the'. In the terms already considered in section 2.1.2, the definition of 'meat' explicitly uses the word while also implicitly mentioning it. The definition of 'the' employs a construction which explicitly mentions the word as a way of providing information about use. In both of them, despite their different approaches, information is given about meaning both as entity and activity. The dual method of definition, incorporating both use and mention, and the dual nature of the information provided about meaning, incorporating both entity and activity, should make the full sentence definition style especially productive for use as a source of linguistic information in NLP systems.

2.4.5 Overall assessment of the Cobuild dictionaries

Throughout this consideration of the characteristics of monolingual English dictionaries, the demands of NLP systems have formed the basis for determining the suitability of Cobuild dictionaries for the objectives of this research. As discussed throughout section 2.4.4.3, it is at least arguable that the full sentence definition is capable of providing information about meaning with a greater degree of subtlety than other more rigidly constructed dictionary definition formats because of the flexibility and rich information content of its range of defining strategies. The adequacy of the contents of any individual dictionary is a separate consideration. It is unlikely that a dictionary produced specifically for the human user would be completely suitable for any significant NLP application without modification.

In the case of learners' dictionaries, the provision of too much information is likely to be at least as unhelpful to the user as the provision of too little, and it is important, as Sweet (1899, pp. 141–146) points out, to establish an appropriate level of detail. Inevitably, this will involve some simplification so that the amount of information provided about any given headword matches the needs of the dictionary's user. Ultimately this is likely to reduce the usefulness of the learner's dictionary for language processing applications,

although during the development of a system working with a particular subset of real language is likely to be an advantage. In the end the type of dictionary needed as a source of information for a given NLP system will need to be specifically designed for that system.

2.5 Summary

The most important feature of the Cobuild dictionary range is that the object language and the metalanguage are not separated, so that within the definition sentences dictionary headwords are generally *used* as working units of language as well as being *mentioned* in the process of definition. This not only makes the definitions likely to be fairly close to the general subset of the language which is under consideration, it also makes it possible to extract a potentially more useful set of information from full sentence definitions than from those in other dictionaries with more rigid structures. Much of the information provided in the Cobuild definitions is implied by the structure of the sentence rather than being explicitly selected and encoded in a separate metalanguage. The process of lexicographic definition, especially as it operates in a learner's dictionary, should provide a useful basis for the study of definition as a general function of the language, and for the extraction of information needed by NLP systems.

Before the detailed analysis of definition language can be described, we need to consider the nature of grammars and parsers and their relationship with definitions and with the English language in general. This is dealt with in the next chapter.

Notes

1. In the dictionary this is preceded by a special 'warning triangle', not reproducible here.

2. Even native speakers often disagree if asked to detail semantic relations between supposed homonyms and are generally incapable or made incapable of considering etymological roots which, even if not 'perceived', might explain them. (Author's translation)

3. Every lexicographic exercise has a didactic aspect. In consulting a dictionary you most often seek something which you do not know or of which you are not sure, and it is in this sense, in answering the questions or the uncertainties of those who consult them, that dictionaries teach something, even if this something varies from language to language, from

situation to situation, from age to age, and, above all, from dictionary to dictionary. (Author's translation)

4. A 3.5 million word sample from the Mondadori corpus held at the Istituto di Linguistica Computazionale in Pisa. For a description of the contents see Ball (1995, pp. 2–3).

5. This analysis was carried out on a computer readable version of the fourth edition of Johnson's *Dictionary*, prepared at the University of Birmingham for the Johnson Project under the direction of Anne McDermott.

Chapter 3

Grammars, parsers, sublanguages and local grammars

This chapter deals with the nature of the grammar that will be used to describe the language of definition sentences, and of the parser that will be used to analyse them. Grammars and parsers are each considered first in general terms, then in relation both to the English language as a whole and to the definition language itself. Finally, the relationship between the definition language and the English language in general is considered using the approaches of the sublanguage and of the local grammar.

The concept of a parser is inseparable from the concept of a grammar. Grune & Jacobs (1990, p. 13), who deliberately avoid restricting the process to any specific concrete realisation, including that of language, define parsing as 'the process of structuring a linear representation in accordance with a given grammar'. The definition sentences under consideration form a subset of the linear representations known as sentences in English. They are constructed according to the normal grammar of English, the nature of which is both substantially undocumented and beyond the scope of this work. However, because of their restricted nature, which can be explored through the concept of the sublanguage, they can also be described by a local grammar which makes no attempt to describe the general set of English sentences. These specific approaches to the development of a grammar and parser for the definition sentences are described in sections 3.4 to 3.7 below.

3.1 What is a grammar?

Chomsky (1965, p. 4) describes the grammar of a language by saying that it 'purports to be a description of the ideal speaker-hearer's intrinsic competence'. In the case of any sentence of English, including the definition sentences, this competence would presumably be understood in terms of the English language as a whole and of its general uses rather than the special

competence of the lexicographers and the special functions of their language use. Such a grammar would not be especially useful for the purposes for which the definition parser is required. The information to be extracted from the definitions relates specifically to the meaning and usage of the words which they define. The language which is to be analysed is effectively a subset of the metalanguage described by Harris (1968, pp. 125–128). As a result of the special characteristics of the metalanguage, described by Harris later in the same work (p. 152) its grammar is not the grammar of the language as a whole. Obviously, both the grammar used to describe the definition language and its associated parser must reflect both this difference and the special analysis requirements already described in section 1.2 above.

At the simplest level of difference, the special status of the word or words shown in bold type in the dictionary definition sentences, the headword, would be ignored by a general-purpose grammar. In such a grammar the headword's function in each definition sentence would be seen to vary according to the sentence's structure, while within the definition language it has a fixed role as part of the definiendum, highlighted by the bold type. As an example, consider the definitions of the three senses of 'drunk' on p. 165 of *CCSD*:

> **Drunk** is the past participle of **drink**.
> If someone is **drunk,** they have drunk so much alcohol that they cannot speak clearly or behave sensibly.
> A **drunk** is someone who is drunk or who often gets drunk.

If the purpose of the analysis of these definitions is the extraction of the information they contain about word meaning and usage, the only meaningful structures within them are those which relate all other words in each sentence to the definiendum. The fact that in sense 1 and 3 the word 'drunk' could be described, in a general purpose grammar, as the subject, or part of it, in a free clause, while in sense 2 it is the complement in a bound clause, is almost completely irrelevant.

Instead, if the information extracted from the definition sentences is to be relevant and useful, the grammar must reflect the required analysis. In each case, the definiendum must be identified and treated appropriately regardless of the function it may be thought to fulfil as a linguistic unit in terms of general purpose grammars. Once this has been done, the functions of the other words in the definition are determined by their relationship with the definiendum. The main components of each definition sentence are the definiendum and its

associated definiens, already described in some detail in section 2.1.1. Both elements are common to more conventional dictionary definition forms, but in the Cobuild definition form there is also a 'hinge' element which links them, and which is usually implied in other dictionaries rather than being stated explicitly. The hinge element is of crucial importance in the Cobuild dictionaries, since it specifies the nature of the relationship between the definiendum and the definiens, which is not always one of simple equality.

As an example consider the definition of sense 1 of 'legacy':

> A **legacy** is money or property which someone leaves to you when they die. (p. 320)

In this definition, the definiendum 'A legacy' is linked to its definiens 'money or property which someone leaves to you when they die' by the hinge 'is'. Other similar definitions use hinges which are obviously directly related to 'is', such as:

> The **buffers** on a train or at the end of a railway line are two metal discs on springs that reduce the shock when they are hit. (p. 66, sense 2)
> **Mammoths** were animals like elephants with very long tusks and long hair. (p. 340, sense 2)
> A **purse** is also the same as a handbag; (p. 450, sense 2)

These hinges, 'are', 'were' and 'is also', could easily be categorised in a conventional general purpose grammar as forms of the verb 'to be', although the inclusion of 'also' in the last example may be problematic. However, there are other possible hinges in similar definitions which are less obviously related:

> **Brushwood** consists of small branches and twigs that have broken off trees and bushes. (p. 65)
> **Freestyle** refers to sports competitions, especially swimming and wrestling, in which competitors can use any style or method they like. (p. 221)

The identification of these hinges — 'consists of' and 'refers to' — as components which are parallel to the previous forms of the verb 'to be' would be both unlikely and over-complicated using general purpose grammatical descriptions. The grammar developed for the definition sublanguage, described in detail in Chapter 6 below, only identifies those distinctions between definition components which are necessary for the extraction of the required information from the definition texts. The general purpose grammar must describe the full range of possibilities of the language as a whole, and its utterances cover an enormously wide range of communicative purposes. The definition

sentences, the utterances of the definition language, have only one communicative purpose: the provision of information describing the meaning and usage of the dictionary's headwords. In Chomsky's terms, the linguistic competence which is to be described by the definition grammar is limited to this communicative purpose, and to the community of 'ideal speaker-hearers' represented by the lexicographers and the dictionary's users.

3.2 What is a parser?

The relationship between a grammar and the parser that works from it is described in De Roeck (1983, p. 8). While the grammar contains all of the rules needed to generate the sentences of the language, the parser is a procedure which carries out a dual function: it will 'not just recognise the sentence but also discover how it is built'. Similarly, Grune & Jacobs (1990, p. 62) say that to 'parse a string according to a grammar means to reconstruct the production tree (or trees) that indicate how the given string can be produced from the given grammar'. While the fundamental role of a grammar as a complete set of generative rules for a language is of the utmost importance within formal linguistics, it is less important in the context of this project than the need to describe and extract the information contained in the definitions. Because of this, the act of parsing the definition sentences may seem inadequate and incomplete in formal linguistic terms, but this represents a fundamental misunderstanding of the parser's purpose. Any apparent incompleteness is not the result of shortcomings in the specification of the definition grammar or the development of the parsing software. It is the result of the relatively restricted analysis needed to extract the required information and the restricted range of possible sentence structures found within the definitions.

Among other things, this choice allows the formulation of a rather more open definition structure than would otherwise be the case, one in which, for example, the boundaries of the functional components are more definitely specified than the exact contents of the components themselves. While most parsing systems depend on a full knowledge of the functions of all of the components of the text before a structural interpretation can be given, the definition parser operates on a minimal knowledge of individual words. The relatively few words used by the system are typically:

(a) those which form restricted closed classes within the definitions, or
(b) those which mark the division between one definition component and the next.

As an example of category (a), consider the definition structure typically used for verb headwords, exemplified by:

> When you **answer** a question in a test, you give the answer to it. (p. 20, sense 7)
> When the police **breathalyze** a driver, they ask the driver to breathe into a special bag to see if he or she has drunk too much alcohol. (p. 61)
> If you **muck about** or **muck around,** you behave in a stupid way and waste time; (p. 365)
> If someone **tutors** a person or subject, they teach that person or subject. (p. 608, sense 3)

In each of these definitions, the first word, 'when' or 'if', constitutes the 'hinge' element which links the definiendum to the definiens. Within the definitions which use a form of this structure, not all of which are used to define verbs, this is an invariable characteristic, and no other words can fulfil this function. Similarly, this function is restricted to the use of one of these two words in the initial position: in the definition of 'breathalyze' the use of the word 'if' within the definiens does not have the same structural significance.

As an example of the second category, consider the definition structure most often used for nouns:

> **Biology** is the science which is concerned with the study of living things. (p. 48)
> A **cabin** is a small room in a boat or plane. (p. 70, sense 1)
> A **cushion** is a fabric case filled with soft material, which you put on a seat to make it more comfortable. (p. 129, sense 1)
> A **fence** is a barrier made of wood or wire supported by posts. (p. 202, sense 1)
> A **match** is also a small wooden stick with a substance on one end that produces a flame when you pull or push it along the side of a matchbox. (p. 344, sense 2)

These definitions have a structure in which the definiens can be seen as consisting of a superordinate with optional discriminators preceding and following it. All of these examples have at least one following discriminator, and the boundary marker between this and the superordinate is realised by the words 'which' (biology), 'in' (cabin), 'filled' (cushion), 'made' (fence) and 'with' (match). This may seem a rather ill-assorted group at first sight, but the parser can identify these boundaries using a combination of three elements: a general rule, which identifies regular past and present participles; a list of less than 100 words containing closed class members such as 'which', 'that', 'in',

etc., together with irregular past participle forms; and an exclusion list of words likely to be wrongly treated by the general matching rule.

3.3 Formal linguistics and practical analysis

The brief outline of the nature of grammars and parsers given in sections 3.1 and 3.2 should be sufficient to show that there is a significant gap between the generally accepted nature of grammars and parsers in formal linguistics and their practical application in this research. It is not enough simply to dismiss this gap as an inevitable discrepancy between theory and practice. If the approach adopted in the development of the grammar and its associated parser is to be understood properly, the exact nature of the discrepancy should be identified and, if possible, the practical approach adopted should be reconciled with the underlying theories.

3.3.1 The scope of the definition grammar and parser

The main discrepancy between the theoretical approach and the practical analysis being carried out has already been referred to in section 3.1 above: the scope of both the grammar and the parser which implements it is restricted to the information needs of the definition analysis. The grammar does not describe the full linguistic characteristics of the definition sentences. This is very different from the approach of general purpose grammars and parsers within formal linguistics. The reason for this discrepancy should, however, be clear. The definition grammar and parser are only intended to provide an accurate description of the sentences as definitions and an effective and efficient way of recovering the required information from them at an appropriate and meaningful level.

This leaves an important question unanswered. While it is obviously appropriate for the parser to recover only the required elements of the linguistic structure of the definition sentences, if the definition grammar does not describe the basis on which the sentences are constructed, which grammar does so? Where the specific definition grammar breaks off, constraints on the formation of definition sentences obviously remain . The definition grammar provides no information about them and the parser ignores them. The solution to this apparent problem is provided by the basic nature of the definition

sentences. They are all constructed in the same way as any other normal sentences of English, using a grammar which, although it is not yet fully documented, is generally acknowledged. The definition grammar describes the special features of these sentences when they are regarded as definitions. It represents the constraints which led the lexicographers to choose those forms of sentence from all the possible forms allowed by the general language grammar. In terms of the production of the definition sentences, it ensures that they conform to the sequences of functional components recognised and allowed by the definition language. It does not determine the sequence of linguistic units within those components, since this is a normal feature of the general grammar of English.

This is best explained by means of an example. Consider the definition of 'caterpillar':

> A **caterpillar** is a small, worm-like animal that eventually develops into a butterfly or moth. (p. 78)

The functional components of this sentence in terms of the definition grammar can be fairly easily identified (the nature of the components and a more formal representation is described in detail in Chapter 6):

Article:	A
Headword:	caterpillar
Hinge:	is
Matching article:	a
Discriminator 1:	small, worm-like
Superordinate:	animal
Discriminator 2:	that eventually develops into a butterfly or moth.

Some of these functional definition components contain more than one word. Discriminator 2, for example, consists of a unit which could be referred to in the whole language grammar as a relative clause, the phrase 'that eventually develops into a butterfly or moth'. While the nature and interrelationships of the functional components of the definition sentences are fully specified within the definition grammar and its associated parser, the permitted sequences of words which make them up are dictated by the whole language grammar.

This dual grammatical constraint is also true of the sequences of the functional components themselves when they are being considered as words within the whole language rather than as linguistic units with special functions within the definitions. In Harris's terms, the definition grammar and the

whole language grammar intersect (Harris, 1968, p. 155), while the definition sentences form a subset of the whole language. Because of this duality, it would be possible to attempt to analyse the definition sentences using any general purpose parser of English which is available and sufficiently reliable. However, as described in more detail in the following section, the resulting analysis would not necessarily provide the most suitable information for use in natural language processing systems, and it would inevitably abandon the enormous advantage of the restrictions inherent in the definition language.

3.3.2 Levels of analysis

The design of the parser for the definition sentences demands a choice of level of detail of analysis. Perhaps the minimum level that would constitute a form of analysis would be the division of each of the dictionary definitions into their two traditional components, the definiendum and the definiens, and any linking text. This would at least reflect an important aspect of the nature of definition texts, but it would be unlikely to yield adequate information for the types of application for which the parser is being developed. It is also by no means certain, in the case of the Cobuild definitions being used as a sample, that such a simple division would always be possible. The conventional lexicographic equation, described in detail in section 2.1.1 above, has already been shown in section 2.4.4.2 above to be of doubtful validity even in the more traditional dictionaries. The problems of its application to the Cobuild dictionaries, described in the same section, are much greater.

At the other extreme, as already suggested in section 3.1, the definition sentences could be parsed according to a selected general grammar. This approach may seem attractive because it would provide a full account of the use of natural language in the definitions which would not be restricted by the fact that they are constructed as definitions. It would also, however, ignore the fact that the definition sentences form a restricted subset of the language as a whole. An analysis which takes account of the nature of the basic components of the definition sentences and the rules governing their combination seems almost certain to provide a more useful source of information than a generally based grammatical analysis, simply because it can reflect and exploit those restrictions.

The detailed implications of the restrictions inherent in the construction of definition sentences are considered in section 3.4 below, but their general

characteristics can be dealt with here. In his treatment of the grammars of science sublanguages, Harris provides a useful theoretical basis for the semi-intuitive view expressed above. He points out that:

> …the sublanguage grammar contains rules which the language violates and the language grammar contains rules which the sublanguage never meets. It follows that while the sentences of such science object-languages are included in the language as a whole, the grammar of these sublanguages intersects (rather than is included in) the grammar of the language as a whole.
>
> (Harris, 1968, p. 155)

This statement, already referred to at the end of the preceding section, raises important problems for a parsing approach which begins with a grammar of the whole language. The analysis which could be produced by a general grammar of the whole language would not simply be inefficient because it would go into more detail than was necessary and would not take account of restrictions within the definition sentences. It would be likely to analyse the sentences incorrectly in terms of their linguistic purpose, and thus fail to meet the information needs of the analysis process.

The parsing strategies developed in this work were therefore aimed at a level of detail which would accurately reflect the distinctive grammar developed for the definition language. As is described in more detail in Chapters 5 and 6, the definition structure taxonomy and the grammar and parser derived from it have been developed to identify recurrent features of the definition texts and to determine their status as linguistic units purely on the basis of their use within the sentences, with little or no reference to their possible descriptions in general language grammars.

3.3.3 The grammar, the parser and formal linguistics

Now that the distinctive character of the definition sentence grammar and parser has been established, it is important to consider how they fit within the framework of the formal linguistics which underlies most general language grammars and parsers. The wider scope of general language description and analysis inevitably leads to much greater complexity, but it must be remembered that the restrictions imposed on the scope and the level of detail of the description and analysis performed by the definition grammar and parser are intentional, and do not represent limitations on their effectiveness for the purposes for which they have been developed. Both arise from the restricted

nature of the definition sentences and the highly specific analysis requirements of the applications which would exploit the linguistic information contained in them. It may, however, still be a useful exercise to compare the basic characteristics of the definition grammar and parser with those associated with formal linguistics.

3.3.3.1 *The grammar and formal grammars*

Grune & Jacobs (1990, p. 28) describe Chomsky's hierarchy of grammars, and the cumulative restrictions that distinguish the Type 0, pure phrase structure grammar from Types 1, 2 and 3. These theoretical categories of grammars are derived from a consideration of the properties of theoretical languages, and the restrictions imposed on them are no doubt extremely significant within the context of the formal languages covered by this study. The definition sentence grammar described in Chapter 6 has not been derived from this rigid theoretical background and none of these grammar types were considered during its specification. The only criterion adopted for assessing the adequacy of the grammar was its suitability for the purposes of the investigation. Having said that, it might be useful to consider the grammar in the same terms as those specified by formal linguistics so that any major differences of approach can be identified. The best way to do this would seem to be to express part of the grammar in one of the formalisms commonly adopted for the theoretical grammars.

The grammar of type A1 definitions is specified in the formal summary in section 6.7.2 by the standard sentence form:

$$(A)_1\ (M^r)\ H^d\ (Q)\ H^i\ (A_m)_1\ (D^{r1})\ S\ (D^{r2})$$

Symbol	Meaning
A	Article
M^r	Modifier, preceding a noun
H^d	Headword
Q	Qualifier, following a noun
H^i	Hinge
D^{r1}	Preceding discriminator
S	Superordinate
D^{r2}	Following discriminator

In this notation, which is more fully explained in section 6.7.1, the items enclosed within brackets are optional, so that the minimal form is:

H^d H^i S

This does not provide the full generative description normally given for formal grammars. Using the conventions of phrase structure grammars, it could be restated as follows:

$D^n{}_S \rightarrow Part_1$, $Part_2$, $Part_3$
$Part_1 \rightarrow A$, M^r
$Part_2 \rightarrow H^d$
$Part_3 \rightarrow Q$, H^i , D^{r1} , S , D^{r2}
$A \rightarrow$ a | an | the | ε
$M^r \rightarrow M^r$ | ε
$Q \rightarrow Q$ | ε
$H^i \rightarrow$ SimpleHinge , Also | ComplexHinge
SimpleHinge → is | are | was | were
ComplexHinge → Can , Also , (be | Consist | Refer)
Can → can | ε
Also → also | ε
Consist → consist of | consists of
Refer → refer to | refers to
$D^{r1} \rightarrow D^{r1}$ | ε
$D^{r2} \rightarrow D^{r2}$ | ε

The conventions used in this notation are as follows:

$D^n{}_S$ shows that the start symbol is the definition sentence
→ stands for 'can be replaced by'
ε stands for the empty component
| stands for 'or'
Non-terminal symbols begin with an upper case letter (e.g. '$Part_1$'), terminal symbols begin with a lower case letter (e.g. 'a')

The most important characteristics of this grammar from a formal linguistics perspective are:

a) the relative absence of terminal symbols

The grammar given above contains relatively few terminal symbols: only those making up 'A' and 'H^i' and the empty item, 'ε'. No indication is given of the set of terminals that can replace, for example, the symbol D^{r2}. In the description of the parser in Chapter 6 the means by which the boundary between the superordinate and its following discriminator, the equivalent of

D^{r2}, is made clear. However, this boundary generally consists of a single word at the beginning of the phrase which constitutes the following discriminator, and the remaining contents are not capable of being predicted.

Sager (1981, pp. 17–18) describes the benefits and likely disadvantages of a computer grammar which uses this approach:

> This would be a tremendous advantage in applications of the program, since the dictionary burden — the necessity of classifying text words in advance of processing — is one of the heavy costs in using linguistic processing. Unfortunately, it turns out that the program that does not have a considerable number of the text words preclassified (particularly the verbs) yields many incorrect analyses for each sentence.

This rather gloomy note must be understood, however, in the context of the general language grammar which Sager is describing. Because of the restrictions found in the construction of definition sentences this lack of specification does not cause any weakness in the definition grammar or inaccuracy in the parser's output. Instead it enhances the analytical power of the grammar and parser, allowing them to deal with the full range of sentences likely to be produced as definitions. The arrangement of the words after the boundary marker within the following discriminator is not produced by the definition grammar: as explained earlier in section 3.3.1 it is produced by the constraints of the grammar of the whole language. As a consequence, individual words within these units do not form basic components of the grammar except in the case of the restricted elements mentioned above and where they identify boundaries between other units.

b) the presence of 'ε'

In the rules for the production of A, M^r, D^{r1} and D^{r2} items, the empty item ε appears as an alternative to the items themselves. This is a feature of certain types of formal grammars, usually referred to as non-monotonic. From a formal viewpoint they are often thought to cause problems for parsers since the shrinkage that they allow in the right hand side of the rule makes the recognition of the items in the sentence theoretically difficult. The parser developed for the definition sentences, working as it does mainly on item boundaries, has no difficulties with this feature, and is able to recognise the omission of optional elements and deal properly with those elements which are present.

c) context sensitivity

The rule for the production of the definition is given above as:

$$D^n_S \rightarrow Part_1 \text{ , } Part_2 \text{ , } Part_3$$

The identification of the various elements in the lower levels of the grammar by the definition parser depends on a knowledge of the part of the definition that is being dealt with, and generally uses pattern-matching based on relatively small closed classes of words or morphemes, both often specified in terms of their position within the string or word under consideration. This general context sensitivity marks the grammar out as most similar to the group of Type 1 context-sensitive grammars with non-monotonic rules described in Grune & Jacobs (1990, p. 53).

3.3.3.2 *The parser and formal parsing methods*

In a similar way to the classification attempted above for the definition grammar, the parser can be considered in the context of the categories used for formal types of parser, but it produces rather less illuminating results. The basic distinction made between parser types in Grune & Jacobs (1990, pp. 64–68) distinguishes the top-down and the bottom-up approaches. There are aspects of both, however, in the parser developed for the definition sentences.

The initial stages of analysis for all the definition types involve the identification of the components of the definition sentence referred to in the previous section as $Part_1$, $Part_2$ and $Part_3$. Separate sub-routines then perform the required analysis within these items. This appears to be a straightforward top-down parsing system, but in fact the later stages of analysis often depend on the identification of terminal items, such as the realisation of A in $Part_1$, or of H^i in $Part_3$. Once the position of these elements (or, in the case of the realisation of A, possibly their absence) can be established the other elements of that section of the definition can be isolated for further analysis. This approach owes rather more to the bottom-up method.

The other main peculiarity of the parser is the sequence adopted for processing. In some ways it is non-directional, since the order in which the three sections of the definition are analysed is of no particular importance: later stages of processing do not, generally, depend on earlier stages. However, in the analysis of $Part_3$ of the Type A1 definition, the blocks of text which potentially contain Q, D^{r1}, S and D^{r2} can only be determined by the identification of the text which realises H^i. This splits the Q text block, if any, from

the rest. The subsequent analysis of the rest into the superordinate and its discriminators is achieved by first identifying the boundary marker for D^{r2}, and then splitting the remainder of the text which potentially contains both S and D^{r1}. This rather unconventional approach is possible, and indeed necessary, because the definition grammar only deals with a deliberately restricted level of analysis and the parser is designed to perform this analysis.

3.4 Restrictions on the definition language and the sublanguage approach

To summarise, then, the grammar and parser for the definition sentences have not been developed using the same principles that would be needed for the English language as a whole. Instead, they are based on the hypothesis that the definition language forms a relatively restricted subset of English and that the nature of the restrictions allows the formulation of a specific grammar to describe its operation. This grammar seeks to describe both the limits within which the compilers of the definitions use the language available to them, and the specific functions performed by the definitions.

The main dictionary used in this study, the *Collins Cobuild Students Dictionary* (*CCSD*), explicitly refers to its own observable lexical restrictions. The word list given at the end of the dictionary 'of all the words that are used ten times or more in the dictionary explanations' (*CCSD*, p. 660) contains only 1860 words (2591 separate forms). The notes on the method of explanation in the Guide to the Use of the Dictionary part 5 (*CCSD* pp. viii-ix) do not deal explicitly with the linguistic structure of definition texts, but they do claim that:

> The explanations of words show you what other words are typically used in association with them, and what kind of structures they are used in.

It would probably have been counter-productive if attention had been drawn too explicitly to the nature of the structures used to do this, since part of the virtue of the Cobuild definition style lies in the fact that the information needed by users is contained implicitly in the definition styles adopted for individual words. If users became too aware of the available range of definition strategies and the reasons for the use of a particular approach in defining a particular word, it might reduce the effectiveness of this technique. Nevertheless, as Hanks makes clear in his description of the definition style adopted for the first Cobuild dictionary (Hanks, 1987, p. 117) the process of compilation

depended on the development of 'an inventory of strategies that look remarkably similar to ordinary English prose', and conscious use is clearly made by the lexicographers of this range of possibilities.

It is also possible to see these strategies as the realisation in language of the restrictions adopted, consciously or unconsciously, by the lexicographers. The following sections consider the appropriateness and usefulness of the sublanguage approach in describing and analysing the definition sentences.

3.4.1 What is a sublanguage?

Harris (1968, p. 152) defines a sublanguage in terms of set closure:

> Certain proper subsets of the sentences of a language may be closed under some or all of the operations defined in the language, and thus constitute a sublanguage of it.

Kittredge (1982, p. 110) points out that this property is not sufficient in itself to resolve all of the questions arising from the need for an empirical definition of the term 'sublanguage', partly because the strict application of the condition by itself would identify too many subsets, including many trivial examples, as sublanguages, but mainly because the concept of closure depends on an intuitive recognition of the boundaries of the sublanguage.

Harris's definition is part of a mathematical description of language structure, and forms an important part of a theoretical model of language. For practical applications, a more empirically-based concept is needed. Having said this, it is worth considering the relevance of Harris's definition to the range of realisations within the dictionary of the definition types described in Chapter 5. The creation of a new definition which meets the membership requirements of a specific definition type comes about because the lexicographer has selected an existing defining strategy and is adapting it to the needs of the headword. This is a practical example of the transformation of a prototypical definition form. The maintenance of set closure is demonstrated by the fact that the new definition can be allocated to an existing definition type without rewriting the membership conditions. By extension, what applies to the individual type groups within the definition sublanguage should apply to the sublanguage as a whole.

Harris (1968, p. 152) suggests as a particularly important and interesting example of a sublanguage the metalanguage which he derives in an earlier part

of the same chapter (pp. 125–128). In this derivation he establishes an important aspect of the nature of the metalanguage: the set of all metalinguistic sentences is a subset of the sentences of the object language itself. In his introduction to the concept of sublanguages (1968, p. 152) he shows that this metalanguage can only form a coherent grammar of the language that it describes if it is a subclass of the language as a whole which consists of all the sentences containing the terms of the metalanguage. Because the language as a whole has no basis for recognising this subclass as a separate entity:

> we can say that the grammar of the metalanguage is characterised by a certain grammatical property which the language as a whole does not satisfy.

In Harris, 1988, (pp. 35–36) he takes the notion of the metalanguage to its logical conclusion, pointing out that there exists 'an interesting regress of metalanguages'. The metalanguage has its own grammar, which is also a metalanguage, and which therefore also has its own grammar, and so on. None of these further abstractions of metalanguages are contained in the metalanguage that each of them sets out to describe. In view of the reservations about the practical usefulness of Harris's definition of sublanguage, it is interesting to note that in another paper (Harris, 1982, pp. 234–5), in which he sets out to clarify the distinction between discourse and sublanguage, he bases the notion of differences between the grammars of a sublanguage and the language as a whole more firmly on a practical and intuitively recognised example:

> if we take as our raw data the speech and writing in a disciplined subject-matter, we obtain a distinct grammar for this material.
>
> (Harris, 1982, p. 235)

In general terms Harris seems to see the properties of sublanguages as characteristic of the languages of science: 'sets of sentences devoted to describing... particular areas of structured phenomena' (Harris, 1968, p. 152). In a later discussion of sublanguages (Harris, 1988, pp. 272ff.) he distinguishes grammar-based sublanguages, 'composed of sentences which satisfy certain grammatical conditions which are not satisfied by all other sentences of the language' (p. 273), the metalanguage already described (p. 274), subject-matter sublanguages 'composed of sentences which deal with a more or less closed subject matter — one in which a limited vocabulary is used and in which the occurrence of other words is rare' (p. 278) and science sublanguages as a special case of this last category (p. 283). Sager (1982, p. 9) describes a practical application within this sub-category:

> We have found that the research papers in a given science subfield display such regularities of occurrence over and above those of the language as a whole that it is possible to write a grammar of the language used in the subfield, and that this specialised grammar closely reflects the informational structure of discourse in the subfield. We use the term sublanguage for that part of the whole language which can be described by such a specialized grammar.

The conditions described by all of these attempts to specify the general nature of sublanguages seem reassuringly close to those encountered in the definition sentences. It is now necessary to consider the detailed practical features normally associated with sublanguages and the typical applications that have been developed using the concept to determine how well the definitions are likely to correspond with them, and how successful the use of the concept is likely to be for the objectives of this research.

3.4.2 Distinguishing features of sublanguages

There seems to be general agreement among those who have worked with or commented on sublanguages that their primary distinguishing feature arises from their subject matter. Kittredge and Lehrberger (1982, p. 2), after discussing Harris's theoretically based sublanguage definition, point out that:

> Actual instances of sublanguages that have been recognized and studied are the result of discourse in particular subject matter fields. The term *sublanguage* has come to be used not just for any marked subset of sentences which satisfies the closure property, but for those sets of sentences whose lexical and grammatical restrictions reflect the restricted set of objects and relations found in a given domain of discourse.

These restrictions have been described in a number of ways. Kittredge (1983, p. 49) suggests as the most widely accepted the following:

> restricted domain of reference;
> restricted purpose and orientation;
> restricted mode of communication and
> community of participants sharing specialized knowledge

These conditions certainly match the science sublanguages referred to by Harris (1968, 1986 and 1988, references in section 3.4.1 above). Lehrberger (1982, p. 102), in a more linguistically specific description, lists the factors 'which help to characterize a sublanguage' as:

(i) limited subject matter
(ii) lexical, syntactic and semantic restrictions
(iii) 'deviant' rules of grammar
(iv) high frequency of certain constructions
(v) text structure and
(vi) use of special symbols

Sager (1986, p. 3) elaborates on point (ii) in the above list by noting that:

> The distinguishing feature of sublanguage is that over certain subsets of the sentences of the language the phenomenon of selection, for which rules cannot be stated for the language as a whole, is brought under the rubric of grammar.

These factors are used, in the next section, to explore the validity of treating the Cobuild definitions as a sublanguage. Examples of applications which have made use of some or all of these restrictions are given in section 3.6.

3.5 Definition sentences as a sublanguage

The approach adopted in this investigation relies on the treatment of the definition sentences as a sublanguage of English. It is now necessary to consider the validity of this approach in detail. As mentioned above in section 3.4.1, Harris's definition of a sublanguage in terms of set closure is not empirically useful. To some extent, the question of the validity of a particular sublanguage definition is one which can only be resolved at a practical level: if the sublanguage concept can be applied successfully within a specific area, it is valid, at least for that area. It would, however, be useful to consider the ways in which the language of the definition sentences conforms to or departs from the generally accepted characteristics of sublanguage described in section 3.4.2. In sections 3.5.1 to 3.5.6, the six factors described by Lehrberger (1982, p. 102) as characterising sublanguages, already quoted in section 3.4.2, are considered individually and assessed for their validity as characteristics of the definition language.

3.5.1 Limited subject matter

It is difficult to decide whether or not this restriction applies to the definitions. On the one hand, the subject matter is the meaning and usage of a subset of

English vocabulary, ranging in size from 'over 70,000 references' in *CCELD* to 'almost 40,000 references' in *CCSD* (both quoted from the back covers of the publications). This certainly seems at first sight to be a significant restriction. However, the nature of the information which needs to be covered for the range of words included in even the smallest of these dictionaries is not restricted. The explanation of the meanings of a very small vocabulary can involve reference to information related to a wide range of areas of knowledge. As an example, the entries for one page (p. 284) of *CCSD* contain the following definition sentences:

1 If people are **inconsistent,** they behave differently in similar situations;
2 Something that is **inconsistent** with a particular set of ideas or values is not in accordance with them.
3 Something that is **inconspicuous** is not at all noticeable.
4 Someone who is **incontinent** is unable to control their bladder or bowels.
5 If something causes **inconvenience,** it causes problems or difficulties.
6 If you **inconvenience** someone, you cause problems or difficulties for them.
7 Something that is **inconvenient** causes problems or difficulties for you.
8 If one thing **is incorporated** into another, it becomes a part of the second thing.
9 If one thing **incorporates** another, it includes the second thing as one of its parts.
10 Something that is **incorrect** is wrong or untrue.
11 Someone who is **incorrigible** has faults that will never change;
12 Someone who is **incorruptible** cannot be bribed or persuaded to do things that they should not do.
13 If something **increases,** it becomes larger in amount.
14 An **increase** is a rise in the number, level, or amount of something.
15 If something is **on the increase,** it is becoming more frequent.
16 You use **increasingly** to indicate that a situation or quality is becoming greater in intensity or more common.
17 Something that is **incredible** is amazing or very difficult to believe.
18 **Incredible** also means very great in amount or degree.
19 If someone is **incredulous,** they cannot believe what they have just heard.
20 An **increment** is an addition to something, especially a regular addition to someone's salary.
21 If something **incriminates** you, it indicates that you are the person responsible for a crime.
22 When a bird **incubates** its eggs or when they **incubate,** it keeps them warm until they hatch.
23 The time that an infection or virus takes to **incubate** is the time that it takes to develop and affect someone.

24 An **incubator** is a piece of hospital equipment in which a sick or weak newborn baby is kept.
25 If you **inculcate** an idea in someone, you teach it to them so that it becomes fixed in their mind;
26 If it is **incumbent** on you to do something, it is your duty to do it.
27 An **incumbent** is the person who is holding a particular post.
28 If you **incur** something, especially something unpleasant, it happens to you because of what you do;
29 An **incurable** disease cannot be cured.
30 You can use **incurable** to describe people with a fixed attitude or habit.
31 An **incursion** is a small military invasion;
32 If you are **indebted** to someone, you owe them gratitude for something.
33 If you are **indebted,** you owe someone money;
34 Something that is **indecent** is shocking, usually because it relates to sex or nakedness.
35 If writing is **indecipherable,** you cannot read it;
36 **Indecision** is uncertainty about what you should do.
37 If you are **indecisive,** you find it difficult to make decisions.
38 You use **indeed** to emphasize what you are saying.
39 You also use **indeed** when adding information which strengthens the point you have already made.
40 You can also use **indeed** to express anger or scorn;

On the basis of an extremely loose and *ad hoc* taxonomy of topics, these forty definitions can be regarded as dealing with at least eleven different subject areas. These are summarised below, followed by a list of the definitions included under each heading:

Topic	**Frequency**	**Definitions**
General human behaviour	13	(1, 6, 11, 12, 19, 25, 26, 27, 28, 30, 32, 36, 37)
Inanimate objects	6	(3, 5, 7, 8, 9, 35)
Measurement	6	(13, 14, 15, 16, 18, 20)
Medicine	4	(4, 23, 24, 29)
Language usage	3	(38, 39, 40)
Logic	3	(2, 10, 17)
Finance	1	(33)
Legal matters	1	(21)
Military matters	1	(31)
Morality	1	(34)
Zoology	1	(22)

This is not, by any means, an exact analysis, but it is more likely to err in being over-inclusive rather than in being over-analytical. Such a wide range of subjects encountered in such a small sample of definitions suggests that the defining language does not deal with a restricted subject matter. However, although the range of subjects may appear to be rather wide, the level at which each is covered is, of necessity, extremely superficial. The absolute minimum of information is provided to enable the meanings of the words to be conveyed, and the initial selection of frequently occurring words restricts the vocabulary associated with each subject area to the commonest and simplest terms. The penetration of each subject is relatively shallow, and this restriction on the depth of knowledge involved may be sufficient to compensate for the perceived horizontal diffusion.

3.5.2 Lexical, syntactic and semantic restrictions

Sublanguages are characterised as restricted in their selection and combination of words. The next three sections explore the ways in which the use of the definition sentence vocabulary conforms to these restrictions.

3.5.2.1 *Lexical restrictions*

The introduction to the Cobuild Word List (*CCSD*, p. 660) makes it clear, as has already been stated in section 2.4.4.1, that there was no specifically restricted vocabulary set up for the lexicographers in advance: any restrictions found in the word list arise from the general requirement that explanations should be simple. The list of words used ten or more times, already described in section 3.4, contains only 1,860 'words' (including different morphological realisations of the same word together), or 2,591 different forms if morphological variation is included. As a comparison, a sample of exactly the same number of words (402,792) taken from a corpus containing the text of several editions of 'The Times' had 4,456 word forms occurring ten or more times. The overall word frequency list for the definition texts, excluding the headwords, shows 8,579 different types, a token to type ratio of 46.95. The 'Times' sample had 27,814 types, or only 14.48 tokens per type. There are only 2,501 hapax (single frequency) words in the definition texts, against 12,107 in the 'Times' sample.

These differences between the definition text and the sample of more general language can also be brought out by the calculation of a measure taken

from Information Theory, the uni-gram perplexity measure. Sekine (1994) provides a formula for the calculation of this characteristic from a text's word frequency list. The first element of the calculation, called uni-gram entropy (designated here by H) is given by:

$$H = -\sum p(w) \log_2 (p(w))$$

where $p(w)$ is the proportional frequency of each word, and the final measure of perplexity (PP) is calculated from the formula:

$$PP = 2^H$$

This produces a measure of the dispersal of the text's lexis. The extreme possible values for the perplexity measure are 1, if the entire text consisted of only one word form, and the number of tokens in the text if every type occurred only once, producing a completely even dispersal. The value for the dictionary definition text is 308.55, which compares with 1509.7 for the 'Times' sample. The lower the perplexity figure, the greater the uniformity of the tokens.

The comparisons carried out in this investigation of lexical restriction have all been made against text taken from articles published in one newspaper, and while it is probably a fairly representative sample of written English of a particular kind, it would be worth considering the fact that journalism, as a specialised form of language in its own right, may also be affected by some sublanguage restrictions. Bearing this in mind, the significant differences shown by all the measures of lexical dispersion considered above seem remarkably convincing. There seems no doubt that the requirement for lexical restriction within the sublanguage has been fully met.

3.5.2.2 *Syntactic restrictions*

The syntactic restrictions operating within the definition language sentences are not capable of such straightforward analysis, but they do exist. Perhaps the most obvious restriction is that all definitions consist of statements. Interrogative and imperative forms exist in the example texts, but not in the definitions themselves. The majority of noun definitions (and many adjective definitions) are simple statements of equivalence between the definiendum and the definiens, most often using the appropriate part of the verb 'to be' or the word 'means' as the link between them, as in:

> A **bolt** on a door or window is a metal bar that you slide across in order to fasten the door or window. (p. 54, sense 3)
> A **compartment** is also one of the separate parts of an object used for keeping things in. (p. 102, sense 2)
> **Decided** means clear and definite. (p. 136)
> To **fix** something means to repair it. (p. 209, sense 4)
> **Traffic lights** are the coloured lights at road junctions which control the flow of traffic. (p. 600)

Further restrictions apply to the verb 'to be' itself. The form 'was' appears 147 times, against 21,256 occurrences of 'is', and 'were' appears 100 times against 6,142 occurrences of 'are'. There are obvious reasons for this. Definitions normally describe current meanings of currently used words. The use of the past tense is largely restricted to the definitions of words which describe historical events and situations, or have meaning only in reference to past circumstances, as in the following examples:

> A **mummy** is a dead body which was preserved long ago by being rubbed with oils and wrapped in cloth. (p. 366, sense 2)
> A **native** of a country or region is someone who was born there. (p. 370, sense 2)
> **Warriors** were soldiers or experienced fighting men in former times; (p. 636)

There is one other significant syntactic restriction, which operates between definition sentences rather than within them. Each unit of definition text in the dictionary is a single sentence, parts of which may form register notes. These sentences are almost completely independent of each other within the dictionary text, so that virtually none of the normal cohesive devices found in all connected forms of text occur in the dictionary. Because of this, almost all references to entities already introduced into the text operate entirely within the individual sentence. Ellipsis, the use of pronouns to replace repeated elements and the other normal features of cohesion are only found on a very limited scale and are almost always contained within the single definition sentence.

As an illustration, consider the following definitions:

> If someone **defuses** a bomb, they remove the fuse from it so that it cannot explode. (p. 139, sense 2)
> If you **demonstrate** something to someone, you show them how to do it or how it works. (p. 141, sense 2)
> The **key** to a map, diagram, or technical book is a list of the symbols and abbreviations used in it, and their meanings. (p. 308, sense 5)

> If a piece of writing or speech **ranges** over a group of topics, it includes all those topics. (p. 458, sense 5)
> **Theoretical** means based on or concerning the ideas and abstract principles of a subject, rather than the practical aspects of it. (p. 587)

All of these sentences contain examples of pronouns which refer anaphorically to other elements of the definition, but the whole nature of the dictionary's organisation, the fact that it is normally accessed by individual sense entries which contain only one definition sentence, ensures that this reference is limited to the sentence itself.

This seems to be true throughout *CCSD* despite the fact that where more than one sense exists for a single headword the definition sentences are not set out completely separately but are organised into one or more paragraphs under the same headword. Almost the only exception to this rule is the occasional occurrence of sentences after the first in such a paragraph where minimal reference is made to the fact that an alternative sense has just been described using the same definition structure. Consider the various senses of 'fork' on p. 218:

> A **fork** is a tool that you eat food with. (sense 1)
> A **fork** is also a tool that you dig your garden with. (sense 2)
> A **fork** in a road, path, or river is the point at which it divides into two parts in the shape of a 'Y'. (sense 3)
> If something such as a path or river **forks,** it divides into two parts in the shape of a 'Y'. (sense 4)

The word 'also' in sense 2 seems to be the only point at which reference is made to any other definition sentence. Because this is an entirely trivial and predictable manifestation of reference beyond the individual sentence it can easily be dealt with during parsing by treating the phrase 'is also' as an alternative to 'is' within the sublanguage grammar.

This extremely limited use of cohesion is a major syntactic restriction, caused, obviously, by the nature of dictionaries and the way in which they are accessed. The definition of the meaning of a specific sense of a headword is treated as if it is independent of the definitions of other senses, although it may be useful for the dictionary user to consider them in order to reach a clearer understanding of the precise meaning of the sense which is being considered. The difference between this and a normal piece of text can be demonstrated by the short extract from 'The Times' of 13th March 1989

given below. To make reference easier, each sentence has been numbered and placed on a separate line.

> **1.** The Queen today takes the opportunity of her annual message to the Commonwealth to add her voice to the Royal Family's increasing concern for the environment.
> **2.** She calls for a common partnership to conserve the world "not only across the oceans but also between generations".
> **3.** Her Commonwealth Day message echoes the themes spelt out by the Prince of Wales and the Duke of Edinburgh in two speeches last week.
> **4.** The Prince called for the total and immediate elimination of chlorofluorocarbon gases (CFCs) which are destroying the ozone layer that protects the Earth from harmful radiation from the sun.
> **5.** The Duke, who was giving the Dimbleby Lecture, said the Earth's resources were under strain because of the pressures facing farmers and agriculturalists to produce increasing amounts of food for growing populations.
> **6.** The Queen's message, underlining her own personal commitment, comes a month to the day after Buckingham Palace delighted environmentalists by announcing that the royal fleet of cars is to be converted to lead-free petrol.
> **7.** In her speech, to be broadcast across the Commonwealth by the BBC World Service, the Queen says that perhaps nothing during the past year has underlined world interdependence more forcefully than the 'dramatic growth' in awareness of the serious dangers man's own activities pose to the environment.

In this text, sentence 1, which begins the news item, only has internal reference, using 'her' twice to refer back to 'the Queen'. Sentence 2 replaces 'the Queen' in sentence 1 with 'she'. Sentence 3 replaces 'annual message to the Commonwealth' in sentence 1 with 'Commonwealth Day message', which is evidently an alternative description. Sentence 4 uses 'the Prince' in place of 'the Prince of Wales' in sentence 3, and sentence 5 similarly uses 'the Duke' to replace sentence 3's 'the Duke of Edinburgh'. Sentence 6 replaces 'her annual message to the Commonwealth' in sentence 1 with 'her message' and sentence 7 replaces the same item with 'her speech'. As would be expected from the normal cohesive use of language, every sentence is connected to others within the text.

The other less obvious syntactic restrictions, which came to light during the development of the definition type taxonomy, are shown in detail in the descriptions of the research methodology, the taxonomy, the grammar and the parser in Chapters 4, 5, and 6.

Apart from the specific restrictions described above, it is evident from the fact that a relatively simple sentence structure taxonomy can be constructed

for the definition texts that the range of possible sentence structures, and hence the syntactic range of the language used within the sentences, is significantly restricted as compared with the language at large.

3.5.2.3 *Semantic restrictions*

The lexical restrictions, already described in section 3.5.2.1, seem to be matched to some extent by semantic restrictions on the already very limited vocabulary. A detailed examination was made of the use of the word 'system' in definition texts. The dictionary lists six senses of this word, on p. 576:

> **1.** A **system** is a way of organizing or doing something in which you follow a fixed plan or set of rules.
> **2.** A **system** is also a particular set of rules, especially one in mathematics or science which is used to count or measure things.
> **3.** You use **system** to refer to a whole institution or aspect of society that is organized in a particular way.
> **4.** People sometimes refer to the government or administration of a country as the **system.**
> **5.** You also use **system** to refer to a set of equipment, parts, or devices, for example a hi-fi or computer, or the set of pipes or wiring which supplies water, heat, or electricity.
> **6.** A **system** in your body is a set of organs or other parts that together perform a particular function.

An analysis of the 212 occurrences of this word showed the following distribution of senses between definition texts:

1.	167
2.	1
3.	4
4.	14
5.	20
6.	6

This shows that while all the dictionary senses of the word are present in the definition language, there is a very strong tendency to use sense 1, which is the most general. This tendency towards the most general use of defining words seems to be borne out by a random sample of twenty-five definitions selected from those containing the word 'people', the single most common lexical word in the definition texts:

Calculation is behaviour in which someone thinks only of themselves and not of other people. (p. 71, sense 2)
A **castle** is a large building with thick, high walls, built by important people in former times, for protection during wars and battles. (p. 77, sense 1)
A **charitable** organization or activity helps and supports people who are ill, handicapped, or poor. (p. 83, sense 2)
When people **creep** somewhere, they move quietly and slowly. (p. 123, sense 1)
If you **stand** or **hold** your **ground,** you do not retreat or give in when people are opposing you. (p. 246, phrases)
If two people **go halves,** they divide the cost of something equally between them. (p. 251, phrases)
Heroin is a powerful drug which some people take for pleasure, but which they can become addicted to. (p. 262)
If you try to **ingratiate** yourself with other people, you try to make them like you; (p. 289)
Something that is an **instrument** for achieving a particular aim is used by people to achieve that aim; (p. 293, sense 3)
Interpersonal means relating to relationships between people. (p. 296)
The **key** things or people in a group are the most important ones. (p. 308, sense 6)
A group of people who are close **knit** or tightly **knit** feel closely linked to each other. (p. 310, sense 3)
Middle-aged people are between the ages of about 40 and 60. (p352.)
A **minority** of people or things in a group is less than half of the whole group. (p. 355, sense 1)
If you **take part** in an activity, you are one of the people involved in it. (p. 405, sense 7)
When people **reach** an agreement, decision, or result, they succeed in achieving it. (p. 460, sense 6)
If something or someone has a particular kind of **reception,** that is the way people react to them. (p. 463, sense 3)
You say that people are **rough** when they use too much force. (p. 488, sense 5)
If you are **sensitive** to other people's problems and feelings, you understand and are aware of them. (p. 509, sense 1)
Sharks are very large fish with sharp teeth that sometimes attack people. (p. 514)
Subject people are controlled by a government or ruler. (p. 564, sense 5)
people use **this** to introduce a person or thing into a story. (p. 590, sense 2)
If you do something **undetected,** people do not notice you doing it. (p. , sense 2)
If a group of people do something **in unison,** they all do it together at the same time. (p. 617)
Upright people are careful to behave in a way that is moral and socially acceptable. (p. 622, sense 3)

The senses of the word 'people' given in the dictionary on p. 411 are:

1. People are men, women, and children.
2. The **people** are ordinary men and women, as opposed to the upper classes or the government.
3. A **people** consists of all the men, women, and children of a particular country or race.
4. If a place **is peopled** by a particular group of people, those people live there.

All of the definitions in the sample shown above use sense 1, again the most general sense of the word. This suggests a very significant degree of semantic restriction in the use of this important word. The list of the ten most frequent lexical words in the definition text, given below, shows a set of similarly general words, most likely to be used in a similar way to 'people':

people	2743
person	1604
things	1533
particular	1319
say	1227
used	1128
place	1119
other	1081
way	1078
thing	1006

Many of these words perform structural functions within the definitions, such as generalised co-texts (e.g. people, person, place), higher level superordinates (e.g. thing), boundary markers for discriminators (e.g. used) and so on. In order for them to do this, their semantic range needs to be severely restricted: another major sublanguage requirement is met.

3.5.3 'Deviant' rules of grammar

The definitions are written in natural English sentences, constructed to give learners guidance on usage at the same time as explaining meaning. This means that the unusual linguistic features found in some sublanguages, for example the 'telegraphic style' identified by Lehrberger (1982, p. 84) in aviation maintenance manuals and by Kittredge (1983, p. 46) in weather bulletins, and analysed in some detail in Fitzpatrick, Bachenko & Hindle (1986), do not appear in the definitions. All the definition sentences, even those which are explicitly metalinguistic, conform to the grammatical norms of the English language as a whole.

Despite this, however, the grammar which is being proposed within this project, and which forms the basis of the parser which has been developed, deviates significantly from the grammar of normal English usage. As is shown in more detail in Chapter 6, the functional components of the definition sentences are no longer those of normal English grammar, and some of the most basic elements of normal English grammar, such as the membership of wordclasses, are largely irrelevant in the functional analysis of the definitions. The definition sentences used in the dictionary could of course be described using a general grammar of English and parsed using a general parser, but for the special linguistic purposes for which they have been constructed the functional grammar and parser developed in this project provide a more useful description and analysis. If the functional definition grammar were to be applied to non-definition sentences, on the other hand, the results would be absurd. The definition sentences are a subset of all English sentences, but their grammar is not a subset of general English grammar.

This asymmetry demonstrates that the language used in the dictionary is indeed deviant, and at the same time exposes the inadequacy of the notion of deviance generally used in the identification of sublanguages. The deviance of the definition sentences does not lie directly in their grammatical structure, but in the functional analysis which can be carried out on them.

3.5.4 High frequency of certain constructions

The development of the definition structure taxonomy, described in detail in Chapter 4, depended on the high frequency of a restricted number of sentence constructions. Each of the seventeen types in the taxonomy represents a group of definitions which conform to a limited structural pattern. The groups were all identified initially on the basis of the linguistic patterns which they displayed, and which were evident to a significant extent, as described in section 4.2.3, on the basis of the initial words of definitions. The seventeen types finally identified, outlined in section 5.1, range in numbers of members from 10,494 (type A1) to 14 (type B4), but the eight types which account for more than 1,000 definitions each contain between them 28,928, or over 92%, of the 31,407 definitions.

3.5.5 Text structure

Even within the text of the definitions themselves there is a highly specialised text structure which affects the meanings and functions of individual words and constructions. Definiendum elements of the definitions are delineated in the dictionary by mark-up codes which are realised in the printed edition as bold type. The positions of these codes in the definition text have been used in the development of the parser to help with decisions on the boundaries of functional units. As an example, consider the following definitions:

> You can use **bottle** to refer to a bottle and its contents, or to the contents only. (p. 56, sense 2)
> Nuclear weapons are sometimes referred to as the **bomb.** (p. 54, sense 2)
> **Duck** refers to the meat of a duck when it is cooked and eaten. (p. 166, sense 2)
> You can refer to any pleasant place or situation as an **oasis** when it is surrounded by unpleasant ones. (p. 382, sense 2)

There are obviously some differences in the forms of the verb 'refer' encountered in these definitions, but a more immediately accessible means of differentiating between them is provided by the knowledge that in the definitions of 'bomb' and 'oasis' the verb precedes the definiendum, while in the definitions of 'bottle' and 'duck' it follows it. This establishes the direction of the equivalence being created by the definition, and allows the different areas within which the functional components of the definition are to be identified to be correctly treated. The operation of the first version of the parser relied very heavily on this and similar forms of restricted text structure.

3.5.6 Use of special symbols

The parser has been developed specifically to analyse the text of the definitions themselves, and with the exception of the definiendum markers described in the previous section there are no special symbols within this text. However, the software currently used to identify the parsing algorithm to be used on the definition does make use of other information in some circumstances. The most common definition structures for nouns and adjectives are very similar:

> A **door** is a swinging or sliding piece of wood, glass, or metal, which is used to open and close the entrance to a building, room, cupboard, or vehicle. (p. 160, sense 1)
> The **outer** parts of something are the parts which contain or enclose the other parts, and which are farthest from the centre. (p. 395)

In order to differentiate properly between these, the special grammar codes in the dictionary are checked. The code for sense 1 of 'door' is COUNT N and that for 'outer' is ATTRIB ADJ. These special symbols, which lie outside the definition text itself, but may still be considered as an aspect of the definition sublanguage, allow proper differentiation between noun and adjective definition types.

3.6 Examples of sublanguage applications

Kittredge & Lehrberger (1982) and Grishman & Kittredge (1986) both contain several papers which describe the exploitation of the restricted linguistic properties of sublanguages, and it is useful to consider these in some detail to establish any marked similarities or differences between their objectives and approaches and those of the current work.

3.6.1 The Linguistic String Project

Part 3 of Sager (1981), chapters 1 and 2 of Kittredge & Lehrberger (Sager, 1982; Hirschman & Sager, 1982), chapters 1, 6 & 12 of Grishman & Kittredge (1986) (Sager, 1986; Friedman, 1986; Hirschman, 1986) and Sager, Friedman & Lyman (1987) all describe work carried out within the Linguistic String Project at New York University, aimed at parsing and reformatting scientific texts and medical records for information retrieval. They describe a variety of analysis methods which rely on one or more of the linguistic restrictions described in section 3.4.2.

As an example of the approach, Sager (1982, pp. 10–14) describes a taxonomy of 'elementary sentences', produced by collecting related science specific nouns into sets appropriate to the sublanguage subject matter, such as 'pharmacological agents' (e.g. glycosides, digitalis), 'tissue' (e.g. muscle, epithelium) and so on. These sets were then used to classify the verbs used in the sublanguage sentences on the basis of the noun environments in which they normally occur, and this yielded a reasonably compact and reliable description of the possible uses of the verbs within the sublanguage, corresponding to the main subtypes of elementary sentences. The taxonomy was then summarised by creating more inclusive noun classes, reducing the overall number of different sentence subtypes. The resulting sublanguage grammar was then tested against actual sentences to check its validity.

These co-occurrence patterns within sublanguages are described by Hirschman and Sager (1982, p. 27) as 'central to processing sublanguage texts'. Sager (1986, pp. 5–11) describes a more sophisticated computer assisted version of the same method of analysis, also based on co-occurrence patterns, and Hirschman (1986, p. 215) describes a portable method of sublanguage analysis which adopts the same approach. The analysis method used for the definition sentences began, as described in Chapter 4, with a frequency analysis of initial words of those sentences to reveal the most basic patterns, and although further analysis differs in important respects from the techniques used by Sager and Hirschman, the overall approach is similar. The differences arise mainly because the wider range of subject matter in the dictionary makes it more difficult to use subject-specific nouns as a starting-point for structural exploration: instead the functional words which form the framework of the definition structures define the sentence's co-occurrence patterns.
Friedman (1986) describes an application within the Linguistic String Project which maps the narrative portions of patient documents into a structured database format. The output stage of the definition sentence parser can carry out a similar mapping, as described in section 7.6.2.

3.6.2 TAUM-METEO and TAUM-AVIATION

The Traduction Automatique Université de Montréal (TAUM) project and its offspring METEO (for the automatic translation of weather reports) and AVIATION (for translating aircraft maintenance manuals) are described in Lehrberger (1982, pp. 81–106) and Kittredge (1982, pp. 107–137; 1983, pp. 46–47). Both are designed to perform automatic translation from English into French, and both need to parse their original sublanguage sentences in order to do this, but these original sublanguages are very different from each other. The METEO parser relies on the telegraphic style of weather bulletins. Kittredge (1982) quotes as an example:

> RAIN OCCASIONALLY MIXED WITH SLEET TODAY CHANGING TO SNOW THIS EVENING
>
> (p. 46)

The METEO parser is designed to reject sentences in which the restrictions of this telegraphic style are breached, and to refer them for manual translation. There is a fundamental difference between this sublanguage and the definitions, since the latter are all perfectly well formed sentences of natural English.

The restrictions exploited by the definition parser, already described in section 3.5, are rather different in detail, but they share many general sublanguage characteristics with the texts dealt with by METEO. The sublanguage addressed by the AVIATION project seems to share more of the definition characteristics. Lehrberger comments on the relative success of AVIATION:

> In view of the complexity of the domain, it is perhaps surprising that these texts should be relatively amenable to automatic translation. That this is so appears attributable to the fact that the domain is quite well-defined.
>
> (1982a, p. 47)

This seems to be true to an even greater extent of the definition sentences: their very closely defined linguistic aims significantly reduce the number of possible sentence structures and makes it possible to adopt the taxonomic approach to parser development described in Chapters 4 and 5.

3.6.3 The Speech Understanding Project

The analysis of task-oriented dialogues under the Speech Understanding Project at Stanford Research Institute, described by Grosz (1982), aims 'to characterize the language used when people communicate for the purpose of solving a problem' as part of an investigation of the language needs of people who use computers as problem-solving aids. This is a discourse analysis exercise, and most of the detailed analysis described in the paper relates to the varying discourse structures produced by different physical relationships between participants rather than the detailed structure of individual sublanguage sentences.

There is, however, a brief description of the lexical restrictions encountered in the analysis (pp. 167–169), which confirms the general characteristic suggested for sublanguages in section 3.4.2. Only 520 word forms were found in the four core dialogues described in the paper, although the total number of words is also relatively small. The overall size of the dialogues is given as about 8000 words 'not including occurrences of the articles "a" and "the"' (p. 167). Only 100 words are used more than 10 times in the dialogues.

While these characteristics of the dialogues are not directly comparable with the lexical restrictions found in the definitions because of the huge difference in size of the two bodies of texts, an interesting parallel between the two projects emerges from Grosz's comment on the dialogue vocabulary:

> 'Our results suggest that, in a given discourse context, even if people are allowed unrestricted use of language, they will use only a small number of words.'
>
> (Grosz, 1982, p. 167)

This echoes the discussion of the vocabulary used in the definition sentences in sections 2.4.4.1and 3.5.2.1.

3.6.4 The study of legal language

Charrow, Crandall and Charrow (1982) set out an account of the claims of legal language to be regarded as a sublanguage. They do not describe an analysis project for legal language: instead their paper is roughly the equivalent of the justification set out earlier in section 3.5 for treating the definitions as a sublanguage. They take the characteristics of legal language which differentiate it from ordinary usage, but rather than exploring the potential provided by these differences for some form of automatic analysis they investigate the historical and other reasons for their development and preservation, and the problems posed by the special nature of the legal sublanguage for non-lawyers. Perhaps the most interesting point made by the authors is the comparison between the concepts of jargon and sublanguage, and the exploration of the idea that many variants of language, assumed to be characterised by purely lexical variation and so referred to as jargons, in fact possess distinctive syntactic and discourse features which make them worth investigating as sublanguages (p. 175).

The main conclusion of the paper deals with the prospects for changing the legal sublanguage into a more accessible form and the implications of any such change for the various communicative purposes of the legal profession. In doing so it considers the need for the legal profession, the 'gate-keepers' of legal language, to respond to lay demands for comprehensibility (p. 188). This raises interesting questions of the self-consciousness of the users of a sublanguage, and the extent to which conscious choices can be made to adjust its characteristics, which again echo the relationship between the lexicographers, the requirements of dictionary users, and the language used in the definitions (see sections 2.4.4.1 and 3.5.2.1 above).

3.6.5 Summary of application examples

The range of subject matter and applications found in this very small sample demonstrates both the general usefulness of the concept of the sublanguage and its importance as an approach to some of the major problems of natural language processing. The automatic reformatting of science and medical information described in Sager (1982 and 1986), Hirschman & Sager (1982) and Hirschman (1986) uses the relatively limited range of possibilities encountered in the sublanguages to produce a fixed database format for information originally expressed in natural language. This concept is explored in detail for the Cobuild dictionaries in section 7.6.2. The TAUM-METEO and TAUM-AVIATION projects described in Lehrberger (1982) use the restrictions of their sublanguages to enable the parsing necessary for translation to be carried out with reasonable success. A possible application of the Cobuild dictionaries in computer assisted translation is outlined in section 7.7.2. The analysis of task-oriented dialogues described by Grosz (1982) has been carried out to establish the scope and nature of the language that might be needed in similar interactions with a computer-based expert system, and the investigation of the legal sublanguage described by Charrow, Crandall and Charrow (1982) seeks to establish the main problems involved for non-specialists in trying to understand an important professional jargon. Similar considerations underlie the possible use of the parser to improve dictionary production, described in section 7.7.1.

It is fairly obvious from these brief descriptions that the present study has most in common with the Linguistic String Project's work on the reformatting of science and medical information and the TAUM translation work, although the implications of the analysis of the definition language for an assessment of its suitability for the learners of English who are the main intended users of the dictionaries overlap with the objectives of the Speech Understanding Project described in section 3.6.3 and the legal language analysis described in section 3.6.4. It thus unites all of the main aspects of these representative exercises in the analysis of restricted languages.

3.7 Local grammars

Given that the definition language can be regarded, in some ways, as fulfilling the requirements of the sublanguage model, another concept becomes useful:

that of local grammar. This was proposed by Gross (in, for example, Gross (1993)), to deal with different forms of text organisation which occur within otherwise normal text. In the dictionary, for example, all the different elements of each entry could be seen as having their own local grammar. In the case of the definitions their local grammar describes the behaviour of the subset of normal language, the sublanguage, represented by the definition sentences. As noted in Barnbrook and Sinclair (2001), other areas have been explored using this concept since the definition grammar was produced. Hunston and Sinclair (2000) have applied it to evaluation sentences, and Allen (1998) to sentences which describe causality.

Hunston and Sinclair (*op. cit.*) explicitly link the concepts of the sublanguage and the local grammar:

> It is possible, then, to see the items described by local grammars as small (but not insignificant) sub-languages, and sub-language descriptions as extended local grammars. Since the search for genuine sub-languages in text of ordinary occurrence has proved singularly unsuccessful to date, there could be point in building up a view of specialist uses of a language from the humble levels of local grammars.
>
> (Hunston and Sinclair, *op. cit.*, p. 77)

On this basis the grammar developed for the definition sentences is a local grammar, reflecting only the behaviour of those sentences seen as definitions, and the sentences themselves, again when seen as definitions, can be said to form an authentic sublanguage.

3.8 Summary

The concept of a sublanguage is an extremely powerful approach to the practical analysis of texts which show a restricted use of linguistic features or have special organisational properties. From an examination of the main characteristics of the definition sentences the distinguishing features described by Lehrberger (1982, p. 102) seem to be largely present, with the exception of the first, limited subject matter. However, as already described in section 3.5.1, the range of subjects found in the dictionary definitions is compensated for to some extent by the low level of detail of its coverage. Significant lexical, syntactic and semantic restrictions have been demonstrated. The frequently recurring structural patterns of the definitions, de-

scribed in detail in Chapters 4 and 5, the specialised nature of the functional grammar, described in Chapter 6, and the special structure of the text itself, using special symbols to delineate the definiendum, all qualify the definition language for sublanguage status on an empirical basis.

The formal definition proposed by Harris (1968, p. 152) is less easy to apply, since the concepts of set membership and transformation depend on a prior definition of membership criteria and acceptable transformation rules. It can, however, be shown that the range of actual definition sentences found within each definition type (described in detail in Chapter 5) show a form of closure under transformation which may also satisfy this definition.

The projects which have already used the concept of a sublanguage, described in section 3.6, suggest that it is a sound practical basis for the development of a functional grammar and parser which will allow the extraction of linguistic information from the definitions, and which can be used in the whole range of applications found in these projects. The next chapter describes the approach used in their development.

Chapter 4

Methodology

The theoretical background to this study, the concept of the definition sentence and the restricted nature of the language used for it, has now been established, and this chapter moves on to describe the practical development of the grammar and parser. It details the methodology adopted for the construction of a taxonomy of definition sentences, based on the structural patterns of their texts, and for the exploitation of the taxonomy in the formulation of the definition language grammar and in the development and application of its associated parser. First it may be useful to consider the general requirements for a structural taxonomy capable of supporting the development of the definition grammar and parser, and the main problems encountered in using a computer to carry out the basic exploration needed for its construction.

4.1 Requirements for a taxonomy

Definition sentences could be automatically categorised in many different ways, any of which could be useful or significant for specific research or data retrieval purposes. As examples, it would be possible to group them according to the parts of speech of the words they define, on the basis of the potential ambiguity of their headwords, using the number of senses defined for each in a selected dictionary, into some system of semantic fields using explicit cross-referencing within definitions, and so on. The main objective of this study is the production of a grammar to describe the definition sentences and their automatic parsing into the functional components of the grammar. Because of this, the taxonomy was constructed on the basis of sentence structures. In order to identify these structures and classify them into the most appropriate groups for grammar and parser development extensive use was made of the computer's pattern-matching and sorting abilities. The problems encountered in the development of an appropriate method of analysis are described in the next section.

4.1.1 Identifying recurrent patterns

The whole basis of the approach adopted in this research, explained in detail in Chapter 3, is that the definitions in the dictionary, although freely composed by lexicographers to meet the needs of the senses of individual words, form a discrete sublanguage which has its own local grammar. The extraction of useful linguistic information from the definitions depends on the establishment of the grammar of this sublanguage, and its use as the basis for the development of the parsing algorithms. The sublanguage grammar can be derived in turn through a process of abstraction of general structural principles from the text patterns found in the definitions, and the starting point for an exploration of the grammar was therefore an investigation of the nature and distribution of recurrent text patterns.

The first stage of this process was the grouping together of definitions with similar text patterns as the basis for the formulation of a taxonomy of definition structure types. The main shortcomings of the computer as a tool for this stage of the investigation arise from the need to differentiate between variations in the definition texts which are significant aspects of definition structure, and those which are unlikely to affect grammatical features or parsing strategies and which can therefore be disregarded in the construction of the taxonomy. The difference between these two types of variation would obviously not be apparent to the computer without specific programming, which demands a knowledge of the distinguishing features of the two types of variation within specific definition patterns.

As an example, one of the first patterns to be identified was a common verb definition structure which is shown in the following definitions:

> If you **acquire** something, you obtain it. (p. 6, sense 1)
> If you **alienate** someone, you make them become unfriendly or unsympathetic towards you. (p. 14, sense 1)
> If you **carry on** an activity, you take part in it. (p. 75, sense 2)
> If you **copy** something that has been written, you write it down. (p. 116, sense 2)
> If you **explode** a theory, you prove that it is wrong or impossible. (p. 191, sense 3)
> If you **honour** someone, you give them public praise or a medal for something they have done. (p. 268, sense 5)
> If you **skin** a dead animal, you remove its skin. (p. 526, sense 5)

In all of these definitions, the fixed elements are 'if you' at the beginning of the sentence and ', you' after the headword and before the explanatory text. Apart from the obvious variation in the headword and its associated explanatory

text, there is a further variable element which comes after the headword and before the ‘, you’. In the context of the investigation it seemed most useful to deal with these definitions as examples of a single pattern, in which ‘something’, ’someone’, ‘an activity’ etc. represented different realisations of the same structural component. The generalisation involved in the establishment of this pattern was based on the nature of the output which would ultimately be needed from the definition parser.

To show how this approach was developed a stage further, consider the similar patterns found in the following definitions:

> If one room, place, or object **adjoins** another, they are next to each other; (p. 8)
> If a disease **affects** you, it causes you to become ill. (p. 10, sense 2)
> If someone **assumes** power or responsibility, they begin to have power or responsibility. (p. 29, sense 2)
> If people in a position of authority **enforce** a law or rule, they make sure that it is obeyed. (p. 178, sense 1)
> If a substance **marks** a surface, it damages it and leaves a stain. (p. 342, sense 5)
> If someone in authority **sanctions** an action or practice, they officially approve of it and allow it to be done. (p. 495, sense 1)
> If a house **sleeps** a particular number of people, it has beds for that number. (p. 528, sense 4)
> If someone **tutors** a person or subject, they teach that person or subject. (p. 608, sense 3)

Two more elements are now varying: the piece of text after the initial ‘if’ and immediately before the headword, such as ‘one room, place, or object’, ‘a disease’, ‘someone’, people in a position of authority’ and so on, and the corresponding pronoun replacing this element after the comma, realised in these examples by ‘they’ or ‘it’ rather than ‘you’. Again, this does not alter the parsing strategy. Another element of the definition is capable of being realised by more than one piece of text, and that realisation in any given definition needs to be recognised and analysed accordingly. A further, apparently trivial development is illustrated by the definition:

> When the police **breathalyze** a driver, they ask the driver to breathe into a special bag to see if he or she has drunk too much alcohol. (p. 61)

One of the last remaining fixed elements, the initial ‘if’, has now been replaced by ‘when’, leaving the rest of the structural pattern unchanged. The pattern could now be described as:

'if' or 'when'
first variable text element
verb headword
second variable text element
comma
pronoun matching first variable text element
explanatory text.

Once this pattern was established, it became possible to consider the functional relationships between these structural elements and to carry out a more detailed and rigorous search for other structural variations which could be included within the same group for grammatical and parsing purposes. Similar processes were used to establish the other definition groups.

This was not the only problem encountered in the identification of recurrent text patterns. The variations described so far affect the contents of specific items which are found within the definition text. It became apparent early in the investigation that some of the structural components of particular definition patterns were optional. Consider the definition of sense 5 of 'divide':

> If you **divide** a larger number by a smaller number, you calculate how many times the smaller number can go exactly into the larger number. (p. 157)

The text between the headword and the pronoun, the second variable text element in the description above, can be split into two elements, 'a larger number' and 'by a smaller number', each of which contributes separately to the headword's normal context. By contrast, consider the definition of 'baby-sit':

> If you **baby-sit,** you look after someone's children while they are out. (p. 34)

Here there is no second variable text element between the headword and the pronoun because the verb typically has no further context.

Similar types of variation are reflected in the main definition pattern used for noun headwords, as shown in the following examples:

> An **array** of different things is a large number of them. (p. 26)
> Your **attitude** to something is the way you think and feel about it. (p. 31, sense 1)
> A person's **behaviour** is the way they behave. (p. 44, sense 1)
> Someone's **capacity** for food or drink is the amount that they can eat or drink. (p. 73, sense 4)
> **Denim** is a thick cotton cloth used to make clothes. (p. 141, sense 1)

> The **exclusion** of something from a speech, piece of writing, or activity is the act of deliberately not including it. (p. 188, sense 1)
> A **facsimile** of something is an exact model or copy of it. (p. 195)
> A sheep's **fleece** is its wool. (p. 211, sense 1)
> A **hatchet** is a small axe. (p. 256)

The variations in the first element are now much more pronounced than in the earlier verb headword examples, but they are of a similar nature. The main items capable of realising this element appear to be 'a', 'an' and 'the', which are obviously also very closely related under more general grammars, or some form of possessive, such as 'your', 'someone's' 'a sheep's' and so on. In the definition of 'denim', however, another feature becomes apparent: this first element can, under some circumstances, be omitted. The reason for the lack of a first element in this definition is fairly clear from the general grammar information provided in the dictionary: 'denim' is marked 'UNCOUNT N OR MOD', while 'array', 'facsimile' and 'hatchet' are all marked 'COUNT N'. The definition structure itself, in these cases, provides this general grammatical information.

These definition examples contain a further optional element. In the definitions of 'behaviour', 'denim', 'fleece' and 'hatchet' the headword is immediately followed by the word 'is', which links the definiendum to its definiens. In each of the other definitions there is an extra element between the headword and this link:

> **array** of different things
> **capacity** for food or drink
> **exclusion** of something from a speech, piece of writing, or activity
> **facsimile** of something

Both sets of optional elements need to be taken into account in analysing the definitions but do not affect the basic approach to be adopted, and so do not represent distinguishing characteristics of definition groups. It became obvious that to deal with these variations it would be necessary to devise parsing strategies which were capable of detecting the presence or absence of optional elements and treating them appropriately.

The precise point at which a variation in structural pattern would demand a change in parsing strategy could not be determined until the complete range of possible patterns was known, so that a preliminary investigation was needed to establish the limits of variation. Some form of manual examination was needed to identify the structurally important elements, but this by itself

would have lacked the rigour, exhaustiveness and objectivity of a computer-based analysis. Manual identification of structural patterns would be biased towards the more obvious recurrences and would be less likely to cover all possible structures equally and ensure that none are omitted. It would also be much more tedious: the computer's ability to produce crude analyses of the data very quickly was a major factor in the viability of this research, since it allowed hypotheses to be tested and refined quickly and easily. The combined manual and computerised search strategy described in more detail in section 4.2 below, developed as a compromise between these conflicting demands.

4.1.2 Identification of parsable structures

The identification and differentiation of recurrent patterns outlined in 4.1.1 was carried out in order to achieve an appropriate mapping between the types of definition structures described by the taxonomy and their associated parsing strategies. The basic aim of this mapping was to ensure that each group of definitions in the taxonomy could be dealt with by a single, coherent parsing strategy.

In the context of the research the development of the taxonomy can be seen as having two main purposes. In the first place, it was the means by which the developing grammar and parser were aligned with each other: it made it possible to identify the sublanguage components and their normal sequences of combination, together with the degree of importance of any variations from the main patterns within particular definition types from the perspective of the analysis to be carried out. Secondly, a piece of software based on the taxonomy, described in section 6.9, forms the first practical stage of the analysis process, the basis for allocating any individual definition sentence to its appropriate structural category, and hence its appropriate parsing strategy. The relationship between the taxonomy, the grammar and the parser is explored in more detail in section 5.5.

The development of the grammar and parser began from a close examination of the provisional definition types described by the taxonomy. Once a part of the taxonomy had been constructed, it became possible to abstract the nature of the functional components within the definition type covered by that group and the rules governing their combination. Together these formed the basis of the sublanguage grammar, and a starting-point for the development of the automatic parsing procedures. As an illustration of this process,

consider these examples of the two definition patterns already considered in section 4.1.1:

> If you **drive** someone somewhere, you take them there in a car. (p. 164, sense 2)
> A particular **slant** on a subject is a particular way of thinking about it, especially one that is biased or prejudiced. (p. 527, sense 4)

From an examination of the definitions matching these patterns these seemed to be maximal examples of their types: they contain realisations of all the elements normally encountered in definitions falling into these categories. There is no guarantee, of course, that any single definition will contain all such elements in the case of all definition types, but a combination of the characteristics of several different near maximal examples would achieve the same results. From these examples a preliminary description of the contents of the fullest possible versions of these definition types could be constructed. An analysis of the extent of variation within the definitions falling into each category allowed this description to include details of obligatory and optional items.

In the case of the definition type represented by 'drive' the pattern abstracted from the set of similar definitions is shown below, together with the realisation of each item in the definition of 'drive'. Items realised by members of a closed set are shown in single quotes, with alternatives separated by a vertical bar.

The abstracted pattern for the type represented by the definition of 'slant' is shown in the same way.

This preliminary form of description is, of course, by no means complete, nor is it specified in sufficient detail. Where the general description 'variable text' appears in the tables over it gives no indication of any limits on the range of possibilities, which is certainly not entirely unrestricted. Despite its simplicity, however, this rudimentary first statement of a definition text grammar did provide a basis for devising effective parsing methods. Generally, as described in more detail in Chapter 6, the parsing process either used membership of closed classes as a means of identification of complete items, as in the case of 'if' or 'when' in Item 1 of the table for 'drive', or else relied on the detection of item boundaries, as in the case of the comma beginning Item 6 in the same table. Using the preliminary descriptions of the definition contents, hypothetical grammars and associated parsing algorithms were constructed, tested and refined.

Item	Possible Realisation	Status	Actual Realisation
1	'if|when'	obligatory	If
2	variable text	obligatory	you
3	verb headword	obligatory	drive
4	variable text	optional	someone
5	variable text	optional	somewhere
6	',' + match for Item 2	obligatory	, you
7	variable text	obligatory	take
8	match for Item 4	optional	them
9	match for Item 5	optional	there
10	variable text	optional	in a car

Item	Possible Realisation	Status	Actual Realisation
1	'a|an|the'	optional	A
2	variable text	optional	particular
3	noun headword	obligatory	slant
4	variable text	optional	on a subject
5	'is|are|was|were'	obligatory	is
6	'a|an|the'	optional	a
7	variable text	optional	particular
8	variable text	obligatory	way of thinking
9	variable text	optional	about
10	partial match for item 4	optional	it,
11	variable text	optional	especially one that is biased or prejudiced.

4.2 A detailed description of the investigation methodology

The construction of a taxonomy intended to realise the two main objectives described above could have been carried out using the entire dictionary text file, but the superfluous information contained in it would have made the development process extremely inefficient and would probably have caused a great deal of unnecessary confusion. Since the objectives of the analysis involved the identification of the definition sublanguage grammar, only a small part of the total set of information was needed, especially in the earlier stages of the work. The next two sections describe the extraction of the necessary data from the machine readable version of the dictionary and the preprocessing needed to make it suitable for investigation. The main stages of the investigation itself are described in sections 4.2.3 and 4.2.4.

4.2.1 The extraction of definition data from the dictionary text

Before the process of pattern identification could begin, it was necessary to extract the data which formed the subject of the investigation from the full text of the dictionary. As has already been pointed out, the machine readable version of *CCSD* contains much more information than is needed for an analysis of the sublanguage. It is a database in which each dictionary headword forms one record, and in which the various mark-up codes act as field markers to provide information for typesetting and for other purposes. As an example, the full dictionary file entry for the headword 'drink', which is shown in its final printed form in section 1.2 above, is given below:

[EB]
[LB]
[HW]drink
[PR]/dr*!i!nk/,
[IF]drinks, drinking, drank
[PR]/dr!a!nk/,
[IF]drunk
[PR]/dr*%u!nk/.
[LE]
[MB]
[MM]1
[GR]VB [GS]with or without [GC]OBJ
[DT]When you [HH]drink [DC]a liquid, you take it into your mouth and swallow it.

[XB]
[XX]We sat drinking coffee.
[XX]He drank eagerly.
[XE]
[ME]
[MB]
[MM]2
[GR]COUNT N
[DT]A [HH]drink [DC]is an amount of a liquid which you drink.
[XB]
[XX]I asked for a drink of water.
[XE]
[ME]
[MB]
[MM]3
[GR]VB
[DT]To [HH]drink [DC]also means to drink alcohol.
[XB]
[XX]You shouldn't drink and drive.
[XE]
[ZB]
[ZH]drinking
[GR]UNCOUNT N
[XB]
[XX]There had been some heavy drinking at the party.
[XE]
[ZE]
[ME]
[MB]
[MM]4
[GR]UNCOUNT N
[DT][HH]Drink [DC]is alcohol, for example beer, wine, or whisky.
[XB]
[XX]He eventually died of drink.
[XE]
[ME]
[MB]
[MM]5
[GR]COUNT N
[DT]A [HH]drink [DC]is also an alcoholic drink.
[XB]
[XX]He poured himself a drink.
[XE]
[ME]

[MB]
[MM]6
[QQ][QS]See also [QH]drunk.
[ME]
[CB]
[VB]
[VW]drink to.
[GR]PHR VB
[MB]
[DT]If you [HH]drink to [DC]someone or something, you raise your glass before drinking, and say that you hope they will be happy or successful.
[XB]
[XX]They agreed on their plan and drank to it.
[XE]
[ME]
[VE]
[CE]
[EE]

The record delimiters in this extract are the 'entry begins' code ([EB]) and the 'entry ends' code ([EE]), and within the complete record there are several substructures, including the headword information delimited by [LB] and [LE], and sets of information for each meaning, delimited by [MB] and [ME]. These allow for variable amounts of data to be included within each of the main data structures.

The earliest investigations of the textual patterns of definition sentences, described in section 4.4.1 below, were carried out on a small file containing only the definitions themselves, extracted from the entire dictionary database. Lines were selected from the database records if they began with the [DT] marker, which signals the start of a definition text. For the headword 'drink' shown above, the file produced from this process would have included the lines:

[DT]When you [HH]drink [DC]a liquid, you take it into your mouth and swallow it.
[DT]A [HH]drink [DC]is an amount of a liquid which you drink.
[DT]To [HH]drink [DC]also means to drink alcohol.
[DT][HH]Drink [DC]is alcohol, for example beer, wine, or whisky.
[DT]A [HH]drink [DC]is also an alcoholic drink.
[DT]If you [HH]drink to [DC]someone or something, you raise your glass before drinking, and say that you hope they will be happy or successful.

Although this file was very valuable in the early stages of the investigation, it was soon found that it omitted some potentially interesting and useful information. A new file was extracted which contained the following five pieces of information:

the definition text, including any separate additional usage notes
a sense number
the grammar note
a sequential number representing the position in the dictionary of the individual definition
the forms of the headword.

As can be seen from the example of the full dictionary text given above, most of this is available in different places within the set of entries for the headword, and is easily identified by the mark-up codes at the beginning of each line. Some simple extraction programs were written, using the awk programming language, to collect this information and to convert the various dictionary database field delimiters contained within the definition texts (such as [HH], [DC] etc.) to a uniform "|" field separator, which was also used to delimit the other fields in each line of the resulting file. This greatly facilitated later processing, but did not in itself carry out any of the necessary analysis. The entries for 'drink' in the file which was used as the starting-point for the construction of the taxonomy, extracted from the full machine readable version of the dictionary, are:

When you |drink |a liquid, you take it into your mouth and swallow it.|1|VB with or without OBJ|8116|drink+drinks+drinking+drank+drunk||

A |drink |is an amount of a liquid which you drink.|2|COUNT N|8117| drink+drinks+drinking+drank+drunk||

To |drink |also means to drink alcohol.|3|VB|8118|drink+drinks+drinking+drank+drunk||

|Drink |is alcohol, for example beer, wine, or whisky.|4|UNCOUNT N| 8119|drink+ drinks+drinking+drank+drunk||

A |drink |is also an alcoholic drink.|5|COUNT N|8120|drink+drinks+drinking+drank+ drunk||

If you |drink to |someone or something, you raise your glass before drinking, and say that you hope they will be happy or successful.|6|PHR VB|8121|drink+drinks+drinking+ drank+drunk||

The only piece of information contained in these entries which is not present in the original dictionary is the sequential definition number, calculated by the extraction program to facilitate automatic reference to individual definition texts within the full file. The forms of the headword are taken from the text given in the dictionary at the start of the entry for the individual headword, and will not necessarily all apply to every sense of it. They make it possible to access individual definitions through all the forms which the word could take within a text, although this capability has not been fully exploited within the present research. The two fields at the end of each record, empty in these cases, are for additional usage notes, which are explained in more detail in section 4.2.2.1.

4.2.2 Preprocessing

During the preliminary stages of examination of the definition texts it became apparent that some features of their construction could obscure the underlying patterns of the sublanguage. The main elements which needed to be dealt with before a valid taxonomy could be constructed are described below. The programs which perform this preprocessing were developed on a trial and error basis during the exploratory investigations that led to the production of the taxonomy. As the taxonomy developed, new problems in the preprocessing were revealed and dealt with by revising the appropriate software.

4.2.2.1 *Additional notes*

The definition text printed in the dictionary for a particular sense of a headword is normally restricted to the words actually used to explain its meaning. In some cases, however, extra information is included within the definition sentence. This often details restrictions on the area of usage of the sense, or provides examples or further details of normal usage, and it is usually marked with a separate field label in the database as a register note. As an example, the entry for sense 2 of 'car' (*CCSD* p. 74) is split in the dictionary database file between a register note and the definition text proper:

> [RN]In American English,
> [DT]railway carriages are called [HH]cars.

In the entry for 'auto' (p. 32) however, information with an essentially similar function is embedded in the definition text section of the database entry:

> [DT]In North America, cars are sometimes called [HH]autos.

It was necessary to separate information of this sort from the rest of the definition text before the identification of the textual patterns distinguishing the definition types was attempted. The extra information can take several forms. It can be given as a note before the main definition text begins, usually separated from it by a comma, as in sense 3 of 'queen' (p. 454):

> In chess, the **queen** is the most powerful piece, which can be moved in any direction.

Alternatively, it may be appended to the definition text after a semi-colon, a colon or a full stop, as in 'abacus':

> An **abacus** is a frame used for counting. It has rods with sliding beads on them.

The pre-processing software automatically identifies these parts of the definition and puts them into separate sections of the record. After pre-processing, the file entries for the two definitions quoted above become:

> the |queen |is the most powerful piece, which can be moved in any direction. |3|COUNT N|21701|queen+queens||In chess
> An |abacus |is a frame used for counting.||COUNT N|5|abacus+ abacuses|It has rods with sliding beads on them.|

This reveals the underlying regularity of the definition text and enables proper exploration of structural features for later processing. The information contained in the notes is also preserved for later parsing. To ensure uniform processing, where register notes already existed as separate entries in the dictionary file, the initial extraction process was adapted to allocate them to these same two fields. Any embedded notes found in pre-processing were then concatenated as necessary with the separately marked text.

4.2.2.2 *Complex headwords*

The first characteristic of the definition sentences to become apparent from the initial investigations was the fact that many of them had the structure:

> text before headword |single or multiple word headword| text after headword.

In other words, the bars corresponding to the typesetting field labels in the database, which produce bold type in the printed dictionary text, often enclosed one continuous piece of text which was to be treated as the headword

and so divided the definition sentence into three sections. There are, however, some more complex definitions such as sense 1 of 'deal':

> **A good deal** or **a great deal** of something is a lot of it. (p. 134)

Here, there are two alternative pieces of headword text, in this case split by the word 'or', which is not in bold type in the dictionary. The definition sentence portion of the entry in the file extracted from the dictionary is:

> |A good deal |or |a great deal |of something is a lot of it.

To ensure that these definitions were treated properly during the construction of the taxonomy, and to make eventual parsing less problematic, the extra bars produced between the alternative headwords by the extraction programs were automatically replaced during pre-processing by asterisks. These asterisks could then be used during the parsing process to identify alternative headword elements within the definition text, but would not interfere with the identification of recurrent text patterns for the taxonomy. After pre-processing the above definition sentence became:

> |A good deal *or *a great deal |of something is a lot of it.

This restored the basic three section pattern, albeit with an empty first section, while preserving the original level of detail.

4.2.2.3 *Incomplete definition formats*

After the preprocessing described in the previous section, all definitions with complex headwords could be treated as if they matched the 'standard' three section form of definition. Some definitions, however, do not contain all three sections. In the first case, there is a special form of usage note, such as that found under sense 1 of 'long' on p. 331:

> used in questions and statements about duration

These notes are introduced in the dictionary database by the [DT] code for definition texts, and several similar items were originally extracted for processing by the extraction software. It later proved possible to treat them in the same way as the register notes already referred to in section 4.2.2.1, and to append them to the data extracted for their associated headword definitions. There are also definition sentences in which the headword is placed at the end of the text, such as 'listener', on p. 327:

> People who listen to the radio are often referred to as **listeners**

The problem with this type of definition arises partly as an artefact of the extraction software. Because there is no marker in the dictionary database to switch off bold type at the end of the definition sentence, the extraction program does not create a bar at the end of the headword, so that the record in the file of extracted definitions only has two definition sections. In these cases, the total number of sections in the definition text part of the record was made up to three during preprocessing by adding an extra bar at the end of the definition sentence.

The identification of this problem in the early stages of the development of the taxonomy led to the discovery of an important feature of some of the definition patterns. Consider the following examples of definitions which were originally extracted with only two definition text sections:

> You can refer to stormy weather as the **elements.** (p. 173, sense 6)
> Animals kept on a farm are referred to as **livestock.** (p. 329)
> Some government organizations are called **services.** (p. 511, sense 2)

These all use a reversed form of the normal definition sequence in which the definiens precedes the definiendum. They all offer a more explicit form of metalinguistic comment, in the sense described earlier in section 2.1.2 above, in that they directly describe the usage of their headwords rather than implying it within the definition. The variant structure seems to be a simple rearrangement of a form found in other definitions, for example:

> You use **mess** to refer to something that is very untidy and dirty or disorganized. (p. 350, sense 1)

The implications of this reversed form of definition for the development of the taxonomy, the grammar and the parsing algorithms, then, were initially highlighted by the simplest of structural features.

4.2.2.4 *The importance of the three section definition text structure*

The universal notional division of the definition text into three sections which resulted from the preprocessing described above proved extremely useful in the development of the taxonomy. Since the patterns found in the first and third sections differed significantly, it was possible to use this structure as a rough but effective method for localising pattern analysis techniques. This simple typographical distinction, devised within the dictionary database as a

means of highlighting the headword in the printed form of the dictionary, made the parsing process much easier than would otherwise have been the case. As a minimum, it provides direct evidence of the boundaries of one major component of the definition, the headword, and even this apparently trivial piece of information allows the analysis process to be oriented more accurately within the definition text.

It is, however, equally important to realise that this identification of the headword and its preceding text does not necessarily correspond directly with the split between the definiendum and the definiens already described in section 2.1.1. This distinction tends to be rather complex within Cobuild definition sentences, as has already been explained in section 2.4.4.2. In many cases, especially in those definitions beginning with 'a' or 'an', the definiendum simply corresponds to the first two sections of the definition. Consider the following definitions:

> **Defeat** is the state of being beaten in a battle, game, or contest, or of failing to achieve what you wanted to. (p. 137, sense 5)
> **Imports** are products or raw materials bought from another country for use in your own country. (p. 280, sense 2)
> **Pottery** is pots, dishes, and other objects made from clay. (p. 431, sense 1)

In each of these examples the definiendum corresponds exactly to the headword, which is treated by the parser as the second section of the definition text. The first section, before the initial bold type marker, is empty rather than nonexistent. The definiens in each case corresponds to the part of the third field that follows the word 'is' or 'are'. These forms of the verb 'to be' at the start of the third field simply act as a means of joining the definiendum and its definiens together in a simple version of the lexicographic equation. Some of the complexities of this equation as it applies in the definition sentences have already been described in section 2.4.4.2, and the same complexities interfere with a straightforward correspondence between the typographical sections of the definition text and the lexicographic components.

Many definitions of uncount nouns follow the pattern of the above definitions exactly, and many more definition structures behave similarly. Most count noun definitions, for example, follow a similar pattern to that shown in the following examples:

> A **bin** is a container that you use to put rubbish in, or to store things in. (p. 48)
> An **exit** is a door through which you can leave a public building. (p. 189, sense 1)
> A **trainee** is someone who is being taught how to do a job. (p. 600, sense 6)

For both 'bin' and 'exit' the definiendum could now be considered to include 'a' or 'an', the first section of the definition sentence, while the definiens for each begins with the matching element 'a' in the third section. In the case of 'trainee' the position is slightly different, since the initial 'a' is unmatched within the definiens, but this is a relatively trivial problem for the parser, which can simply test for the presence or absence of potentially matching elements found in appropriate sections of the definition and interpret the structure accordingly.

In many other definition structures, however, the correspondence between the three typographically determined fields of the definition and the definiendum and definiens is more problematic. In some, for example, there are elements of the definiendum in the third section of the definition text. Consider the following:

> If you **divulge** a piece of information, you tell someone about it; (p. 158)
> If you **manipulate** a piece of equipment, you control it in a skilful way. (p. 341, sense 2)
> If you **say** something in a letter or a book, for example, you express it in writing. (p. 497, sense 3)

In each of these examples, the definiendum is the whole construction beginning with the 'you' immediately before the headword and going on to the comma immediately before the second 'you', and the definiens is the whole construction beginning with the second 'you'. In a similar way to the word 'is' in the previous sets of definition examples, the 'if' at the beginning of each definition simply joins the definiendum and definiens together. In these cases, and many more with more complex patterns, the identification of the definiendum and the definiens could not be carried out entirely on the basis of the three typographically defined sections already available in the records extracted from the dictionary, but these sections have nevertheless proved an extremely useful starting-point for pattern analysis and parser development.

4.2.3 Initial word frequencies and sentence types

To make the exploration of definition structures as objective and rigorous as possible, the initial analysis was carried out with minimal operator interference. In the first place, a list of the first words of the definitions was produced in order of their frequency of occurrence. Only 122 different word forms are shown in this list, but this relatively small number is partly an artefact of the

analysis method. For the purposes of the production of the frequency list only the first section of the definition text, the text preceding the headword, was considered. Since the headword is in the second section, the 5,174 definitions which begin with the headword are treated in the list as starting with an empty string, which thus counts as only one of the 122 initial word forms. All of the following statistics are based on this approach. Of these 122 first word forms, only 45 occurred more than once, and only 17 occur more than 10 times. These words are shown, with their frequencies of occurrence, in the list below. As already explained, the 5,174 definitions which start with their headwords and so have no text in the first section are counted together under the heading '*no first word*' in the third line of the list. Between them the words listed introduce more than 99.5% of all the definitions in *CCSD*.

if	10206
a	6805
no first word[1]	5174
you	1908
when	1487
the	1472
an	1106
something	1026
to	670
someone	659
your	458
someone's	121
people	95
in[2]	22
some	20
things	15
food	12

Total: 31,256, or 99.52% of the total 31,407 definitions

This list provided a starting-point for the construction of a taxonomy of definitions based on simple linear patterns. The most obvious parallel in this list was between definitions introduced by 'if' and 'when'. There are 11,693 of these in the dictionary, so that they constitute over 37% of the total. A sample of these definition texts is shown below.

> If you **fend off** questions or requests, you avoid answering them.
> When wine, beer, or fruit **ferments** or **is fermented,** a chemical change takes place in it.

When something is done with **ferocity,** it is done in a fierce and violent way.
If you **ferret out** information, you discover it by searching thoroughly;
If someone has a **fertile** mind or imagination, they produce a lot of good or original ideas.
When an egg or plant **is fertilized,** the process of reproduction begins by sperm joining with the egg, or by pollen coming into contact with the reproductive part of a plant.
When a wound **festers,** it becomes infected and produces pus.
If an unpleasant situation, feeling, or thought **festers,** it grows worse.
If something **is festooned** with objects, the objects are hanging across it in large numbers.
If you **fetch** something or someone, you go and get them from where they are.

It should be clear from these examples that, although the basic sentence structure of each is very similar in conventional grammatical terms, they are defining different kinds of headword: 'fend off', for example, is a verb; 'ferocity' is an adverb; 'fertile' is an adjective. This changes the position of the headword within the definition sentence, both in the sense of its strict linear sequence and of its grammatical function, and changes the relationships between the functional components of the definition sublanguage at the same time. The problem for the construction of an adequate taxonomy is not simply the identification of basic sentence types, in itself almost a trivial matter, but the slightly more complex problem of identifying the type of definition for which a given sentence pattern is being used. This is determined mainly by the type of headword being defined within that sentence type, and this can be established by examining the structure of the definition sentence in more detail, or, where that leaves unresolved ambiguities, by using other information available from the dictionary such as the grammar code for the headword. Similar considerations apply to the other main groups of sentences headed by specific words. The next group to be considered were those beginning with 'a', 'an' and 'the', accounting for 9,383 definitions, or 30% of the total. A sample is shown below:

An **overt** action or attitude is done or shown in an open and obvious way.
An **overture** is a piece of music used as the introduction to an opera or play.
An **overview** of a situation is a general understanding or description of it.
An **owl** is a bird with large eyes which hunts small animals at night.
The **owner** of something is the person to whom it belongs.
An **ox** is a castrated bull.
An **oyster** is a large, flat shellfish.
The **pace** of something is the speed at which it happens or is done.

> A **pace** is the distance you move when you take one step.
> A **pack** is a rucksack.

In this case the range of definition types in the sample is slightly smaller. All of their headwords are nouns, except for the definition of 'overt', an adjective, but a similar shift can be seen in the relationships between the components of this definition when compared with the others.

A simple grouping based on initial words thus provided a very valuable basis for the construction of a structural taxonomy that would allow the development of the definition parser. Its refinement into such a taxonomy demanded, firstly, the identification of potential groups of structural patterns, followed by an assessment of their relative suitability for single strategy parsing to determine which generated the most effective functional taxonomy for the definitions and mapped most efficiently onto the potential grammatical structures and their associated parsing algorithms. The basis of selection was the need to achieve the optimum balance in the construction of the parsing software between the use of large numbers of highly specific parsing algorithms, dealing individually with very few definitions but capable of accurate analysis without the need for complex decision-making on variant structures, and the development of over-complex routines which could deal with large numbers of definitions only at the expense of accuracy or reliability. This required the taxonomy and the parser to be developed, to some extent, in parallel, so that the final version of the taxonomy represents a classification of definition types based on parsing strategies.

4.2.4 The identification of structural pattern groups

The groups of definitions based on initial words formed a general framework for the next stage of exploration, the identification and analysis of structural patterns to determine the most efficient parsing strategies. Some significant patterns were immediately apparent on an initial examination of the data. For example, the definition of the word 'abattoir':

> An **abattoir** is a place where animals are killed for meat. (p. 1)

typifies a very common definition structure which can be generalised as:

> A/An/The **noun headword** is/are a/an/the...

The relative simplicity of this structure and its frequency in the dictionary (over 5,600 examples) made it one of the first candidates for separation into its own parsing category. As the parser developed, it became obvious that other optional elements could be present without the structure changing sufficiently to need a different parsing strategy, and that these slightly variant definitions could be dealt with by a fundamentally similar approach. This method, which is discussed in more detail later in section 4.3.1, allowed the extension of a strategy which originally covered 5,626 definitions to allow it to deal with 10,494, or over a third of the total number.

Being able to identify patterns in this way is both efficient and rewarding, but two major problems became apparent once the first few obvious structures had been identified. Firstly, although there were significant patterns which were signalled directly by the initial word, such as the 'if/when' and 'a/an/the' patterns already discussed, many others were embedded slightly more deeply within the definition text and so were more difficult to detect in this initial investigation. Secondly, to ensure that the analysis covered all the definitions it was necessary to establish suitable controls. The methods used to overcome these two problems are dealt with in the following sections.

4.2.4.1 Identifying less obvious patterns

Once the definitions with more obvious structural patterns, particularly those dependent on initial words, had been eliminated, it became necessary to search more deeply to identify the remaining structures. This involved a cyclic process of string matching applied to phrases in the definitions beyond the initial words. As an example, the frequency list given in section 4.2.3 has 'you' as the fourth most common initial word, beginning 1,908 definitions. Unlike the words 'if', 'when', 'a', 'an' and 'the', which belong to relatively small closed sets of words in the definition sublanguage, 'you' is a relatively frequent realisation of a sublanguage component which is much more widely variable. Because of this, its presence as the initial word of a definition is less likely to guarantee a relatively restricted set of structural patterns. A sample taken from the definitions which begin with 'you' shows something of the range of possibilities:

> You address a judge in court as **your honour;** (p. 268)
> You can refer to a disorganized group of things of various kinds as **odds and ends;** (p. 384)

> You also say '**There you are'** or '**There you go'** when you are giving something to someone. (p. 588, phrases)
> You use **time** after numbers to say how often something happens. (p. 594, sense 5)
> You can acknowledge someone's thanks by saying '**You're welcome'.** (p. 641, phrases)

These do not obviously fit one specific pattern of definition structure, but subsequent words and phrases do tend to recur within these definitions and in others beginning with other words. As an example, the word 'use' in the definition of 'time' shown above occurs in a similar context in more than 1200 of the definitions beginning with 'you'. When similar contexts were explored for the remaining definitions, it was found that while the initial word 'you' was found in the majority of the definitions following this pattern, there were definitions with an exactly parallel structure beginning with the words 'people', 'some people', 'Americans' and 'communists', as in these examples:

> Americans use **after** to tell the time. (p. 11, sense 5)
> Communists use **bourgeois** when referring to the capitalist system and to the social class who own most of the wealth in that system. (p. 57, sense 2)
> Some people use **love** as an affectionate way of addressing someone; (p. 334, sense 6)
> People use **really** in questions when they want you to answer 'no'; (p. 462, sense 3)

This cyclic process of pattern analysis, extending further into the initial phrases of the definitions, also allowed those introduced by the less frequent initial words to be explored fully. In a similar way, elements such as 'use', 'refer to', 'say' etc. were identifiable as the basic components of these patterns, capable of extension through optional elements, such as 'can' and 'also', already detected in other structures.

Rather more difficulty was caused by patterns which were superficially similar to major structures already identified but which varied from them in ways which seemed relatively trivial but which had significant effects on the possibilities of parsing. As an example, within the definitions beginning with 'if' or 'when' described in section 4.2.3, a small number follow a similar pattern to the definition of 'just', sense 1:

> If you say that something has **just** happened, you mean that it happened a very short time ago. (p. 306, sense 1)

Although this pattern seems similar to the main 'if/when' sentence structure shown in 4.2.3, it contains a further element, in this case realised by the words 'say that' in the part before the headword, and 'you mean that' in the part

afterwards. This puts the whole definition of the word 'just' into a metalinguistic frame, in which the meaning of the word is being examined specifically as a phenomenon of spoken language. As already discussed in section 2.4.4.3, in the terms used by Hanks (1987, p. 135) all dictionary definitions deal with word use, but where this is made explicit within specific definitions the fact needs to be acknowledged. These definitions therefore needed to be considered as potential candidates for separate categorisation and for treatment by a more specific parsing strategy. The final number of definitions with a pattern sufficiently similar to be included in this separate category was nearly 600, a relatively small and inconspicuous group compared to the major types, but by a cyclic process of pattern construction, subdivision of definition files, and checking for anomalies it was possible to extract these definitions into a coherent and useful type.

A further aid to the identification of more subtle differences in the broad structural patterns was found in the grammar codes contained in the definitions. Once a pattern had been identified it was relatively easy to summarise the distribution of major grammatical categories within the group of definitions. This usually revealed an obviously dominant part of speech within the structural group, which could be used to assess the uniformity of distribution of the definition structure which had been identified. This analysis proved very useful in the assessment of the potential for using a single parsing strategy with apparently similar structures, as discussed below in 4.3.1. An example is provided by the definitions of 'kindly' and 'meteor':

> A **kindly** person is kind, caring, and sympathetic. (p. 309, sense 1)
> A **meteor** is a piece of rock or metal that burns very brightly when it enters the earth's atmosphere from space. (p. 351)

The similarities between these definitions are obvious: both begin with an indefinite article immediately preceding the headword, both use 'is' as the link to the explanation. However, because the first definition deals with an adjective and the second with a noun, any parse of the two definitions should treat the other elements of the two sentences taking their relationship to the headword into account.

4.2.4.2 *Control of uncategorised definitions*

To ensure fulfilment of the second requirement, the complete coverage of all definitions, a strict routine was followed in which definitions which con-

formed to a particular pattern were split off from the current working group into a file for testing, and those remaining were also extracted into their own complementary file. This meant that at any given time a complete set of files existed which contained all the definitions whose patterns had not yet been fully identified. At its simplest, this involved a repeated splitting of the file of uncategorised definitions, using one command to split off the next pattern type identified, and then using the inverted form of the same command to collect the remaining items into the next version of the uncategorised file. Constant line count reconciliations were performed to ensure that no definitions had been lost because of incorrect command entry, poor pattern specifications or other possible errors.

4.3 The construction of the taxonomy

The groups of definitions with apparently similar structures which were obtained from the cyclic analysis process described earlier in section 4.2.4.1 now needed to be checked for structural integrity. The ultimate objective of the exercise was the development of a coherent local grammar for the definitions and an associated set of automatic parsing algorithms, and the integrity of a category in the taxonomy depends on its capability of being parsed using a single strategy. The only way of assessing this capability was by the formulation of a parsing strategy for each taxonomic group followed by an exhaustive testing process, designed to allow the refinement of the parsing strategy to accommodate minor variations within the structural pattern, or the formulation of more appropriate groups. The detailed stages of this process are described below.

4.3.1 Assessment of single parsing strategy potential

The structural pattern described earlier in section 4.2.4 is represented at its simplest by definitions such as:

> A **destroyer** is a small warship with a lot of guns. (p. 145)
> A **loch** is a large area of water in Scotland. (p. 329)
> A **screwdriver** is a tool for fixing screws into place. (p. 502)

These can certainly all be dealt with by a single parsing strategy, which could analyse them into sections such as those shown below:

A
destroyer
is
a
small
warship
with a lot of guns.

A
loch
is
a
large
area of water
in Scotland.

A
screwdriver
is
a
tool
for fixing screws into place.

The nature of these components is discussed in detail in Chapter 6. For the moment, the most important item is the component realised by 'warship', 'area of water' and 'tool'. In the normal hierarchical system of lexical relations these items would be seen as the superordinates or hypernyms of each of the three headwords. In the text surrounding these three superordinates it will be seen that there are slight but important variations in structure. The headwords 'destroyer' and 'loch' both have a component between the second indefinite article and the superordinate, while 'screwdriver' does not: its superordinate 'tool' is given with no prior modification. This does not affect their ability to be analysed by a common parsing algorithm, but the algorithm must be designed to allow for the non-realisation of specific components of the definition.

As a further example of the possibilities for extending the application of this single parsing strategy, consider the following definitions:

> Your **bloodstream** is your blood as it flows around your body. (p. 52)
> A person's **contemporaries** are people who are approximately the same age as them, or who lived at approximately the same time as them. (p. 112, sense 2)
> A kangaroo's **pouch** is a pocket of skin on its stomach in which its baby grows. (p. 431, sense 2)
> A woman's **uterus** is her womb; (p. 624)

Superficially, these definitions no longer match the simple structure of the examples just dealt with, but they can be analysed in a roughly similar way:

Your
bloodstream
is
your
blood
as it flows around your body.

A person's
contemporaries
are
people
who are approximately the same age as them, or who lived at approximately the same time as them.

A kangaroo's
pouch
is
a
pocket of skin
on its stomach in which its baby grows.

A woman's
uterus
is
her
womb;

The major difference in this analysis is the nature of the first component. In the three earlier definitions it was realised by the indefinite article: in these it takes the form of a possessive pronoun (e.g. 'your') or a possessive phrase (e.g. 'a person's'). Relatively minor alterations are needed to the mechanics of the parsing program to allow these definitions to be dealt with by the same strategy as the others. This process of constant extension of the parsing strategies coupled with checks on the validity of the new structural categories created formed the basis of the refinement of the taxonomy.

4.3.2 Identification and elimination of problem items

During its development the taxonomy underwent a constant process of validation in which various problems were identified which affected the structural

integrity of particular categories. In each case the reason for the problem needed to be identified so that a decision could be made on the treatment of the definitions affected. In some cases these problems were caused by individual anomalies in the writing of specific definitions, which otherwise followed an established structural pattern. These did not necessarily affect the overall grammar of the definition sub-language, but could instead be regarded as less well-formed manifestations of it. As examples, consider the following definitions:

> In games such as football **full time** is the end of a match. (p. 225, sense 2)
> In Britain the **ground floor** of a building is the floor that is level with the ground outside. (p. 246)
> In American English a **subway** is an underground railway. (p. 565, sense 2)

As already explained in section 4.2.2.1, the initial phrases of these definitions, such as 'In games such as football', should have been removed and treated as usage notes during pre-processing. However, the pre-processing software relies on the presence of a comma at the end of the note preceding the definition text proper. These three definitions do not contain a comma in the appropriate position, so the usage note is still present at the start of the definition. The problem caused by this was capable of being overcome within the definition type analysis software, but it may be considered more appropriate to insert the comma before the definition texts are put through pre-processing. This would represent a modification of the dictionary to make it more suitable for machine processing, which requires greater consistency and explicitness of structure than may be needed by human users.

Similar principles could apply to an error found in the definition of 'eminently':

> **Eminently means** very, or to a great degree;

In this case the word 'means', a crucial element of definition structure, has been included as part of the headword and therefore hidden from the investigation of structural patterns. This error, which probably has little or no effect on the human user of the dictionary, would prevent the parser from dealing with the definition correctly and would need to be corrected before processing. These and other similar results provide useful feedback which allows problems in the production of the dictionary to be detected and rectified, as explained in more detail later in sections 7.3 and 7.7.

In other cases the problems encountered during the validation of the taxonomy revealed an invalid application of a structural pattern to variant definition types which needed to be allocated to their own categories. As an example, the basic pattern represented by the definition of sense 1 of 'flow' is used primarily to define verbs:

> If a liquid, gas, or electrical current **flows** somewhere, it moves steadily and continuously. (p. 212)

While the limits of this structural pattern were being investigated it became obvious that a similar pattern was being used to define other parts of speech. For example:

> When you take a **chance,** you try to do something although there is a risk of danger or failure. (p. 81, sense 2)
> If the weather is **fresh,** it is fairly cold and windy. (p. 222, sense 7)
> If something **ordinarily** happens, it usually happens. (p. 392)
> When something is scarce, your **ration** of it is the amount that you are allowed to have. (p. 459, sense 1)

Because of the different nature of the defining process in these and similar cases, their structures needed separate parsing strategies and so were allocated to their own categories within the taxonomy.

4.3.3 Combination of similar categories

As has already been seen in the consideration of the extension of single parsing strategies to cover apparently different structure groups in section 4.3.1, categories which began as separate entities within the taxonomy sometimes needed to be combined. The basic approach to the construction of the taxonomy, described in sections 4.2.3 and 4.2.4, uses the linear text pattern as its starting-point, and in some cases definitions which begin with different words exhibit marked similarities of structure which allow them to be dealt with by single parsing strategies. In these cases the separate groups have been combined within the taxonomy and then subjected to the usual validation processes.

An example of this process has already been used in section 4.3.1 to illustrate the process of extension of parsing strategies to cover similar types. The definitions which begin with a possessive pronoun or phrase, examples of which are shown in that section, are not restricted to a single initial word. An

analysis of the text before the headword in definitions of this type shows the following initial texts occurring more than once:

your	454
someone's	120
a person's	19
a woman's	16
a bird's	8
a country's	7
a man's	7
an animal's	7
a vehicle's	3
a car's	2
a performer's	2
the earth's	2
your sense of	2

This means that these definitions are most likely to be introduced by 'your', 'someone's', 'a', or 'an'. The common structural element for all these definitions except those beginning with 'your' is the possessive apostrophe, and this has been used in the analysis software as a means of identifying definitions belonging to this type. Once this further type of definition had been identified, it was found that, subject to differences in the grammatical structure of the initial part of the sentence, it was possible to parse them accurately using a very similar parsing strategy to that developed for the basic definition type. This use of the parser to test the structural integrity of the taxonomy formed an important feature of the methodology of this research. As is explained in more detail in the next section, the development of the taxonomy and of the grammar and its associated parser were carried out in parallel.

4.4 Development of the grammar and parser

Once some of the major categories of the taxonomy had been established it became possible to experiment with parsing strategies. In theory, a parser would be expected, as already described in section 3.2, to be constructed on the basis of a pre-existent grammar. In practice, the earliest versions of the parser were attempts to establish the optimum methods of analysis and, in so doing, to test hypothetical grammars against the characteristics of the members of the taxonomic categories, simultaneously testing the usefulness of the

taxonomy as a description of definition structures. In other words, instead of being neatly divided into separate specific stages for each of the taxonomy, the grammar and the parser, the development process constantly involved all three elements in an interconnected cycle of formulation, testing and refinement. The intermediate stages of the taxonomy, the grammar and the parser all worked as development tools for each other, allowing hypotheses to be tested thoroughly and to be refined accurately. The next two sections give examples of the operation of this process at different stages during the development of the sublanguage description.

4.4.1 Developing the grammar and parser in the early stages

As an example of the process described in the previous section, one of the earliest forms of the parser was based on six definition types. An extract from the coded input used at this stage of the development is shown below:

> a |churchyard |is an area of land around a church where dead people are buried.
> if you are |flabbergasted, |you are extremely surprised;
> if you |manhandle |someone, you treat them very roughly.
> something that is |plush |is smart, comfortable, and expensive.
> if you do something |thankfully, |you do it feeling happy and relieved that something is the case or that something has happened.
> |vastly |means very much or to a very large extent.

This very early version of the input file is based on an extract from the dictionary containing only definition texts, as described earlier in section 4.2.1. To make processing easier the text was reduced to lower case throughout. Rudimentary type allocation software, developed from the earliest stages of the taxonomy, produced an annotated version of the input definitions. The examples shown above produced the following output from this program:

> 1 *a |churchyard |is an area of land around a church where dead people are buried.
> 2 *if you are |flabbergasted, |you are extremely surprised;
> 3 *if you |manhandle |someone, you treat them very roughly.
> 4 *something that is |plush |is smart, comfortable, and expensive.
> 5 *if you do something |thankfully, |you do it feeling happy and relieved that something is the case or that something has happened.
> 6 *|vastly |means very much or to a very large extent.

The number in front of each definition sentence is the provisional definition type, and the asterisk is used as an easily identified separator. This type of analysis, coupled with a simple set of parsing routines, allowed the following parsed output to be produced from these six example definitions:

```
Operator : a
Headword : churchyard
Hinge  : is
Match  : an
Superordinate : area
Discriminator : of land around a church where dead
people are buried.

Operator : if
Cotext  : you are
Headword : flabbergasted
Match  : you are
Explanation : extremely surprised;

Operator : if
Cotext  : you
Headword : manhandle
Cotext2 : someone
Match  : you
Explanation : treat *them* very roughly

Cotext  : something
Operator : that
Hinge 1 : is
Headword : plush
Hinge 2 : is
Explanation : smart, comfortable, and expensive.

Operator : if
Cotext  : you do something
Headword : thankfully
Match  : you do it
Explanation : feeling happy and relieved that something
is the case or that something has happened.

Headword :vastly
Hinge  : means
Explanation :very much or to a very large extent.
```

Even the simple definition types dealt with by this very primitive stage of the taxonomy accounted for reasonably large numbers of definitions:

Type	Number
1	9404
2	580
3	4249
4	1826
5	161
6	575

Total 16,795 or 53.5% of the total.

From then on, the development of the taxonomy was based on a process of continual reassessment of unallocated definitions, coupled with experimental extensions of the existing grammar and parsing strategies and thorough testing of their effectiveness.

The gradual refinement of the broad principles of the grammar and its associated parser arose naturally from this development of the taxonomy, although the more detailed aspects were developed, to some extent, independently once the taxonomy had provided a basis for their specification. As an example, in the original parsing software used to produce the output reproduced above, type 1 definitions, those with the same structure as the first example above, the definition of 'churchyard', were analysed into components labelled Operator, Headword, Hinge, Match, Superordinate and Discriminator. The identification of the Operator, Headword, Hinge and Match elements were unproblematic, being based almost entirely on the position of the text within the overall data structure. As has already been described in section 4.2.2.4, the basic three-section structure of the records extracted from the dictionary database identifies the major structural divisions of the definition texts, and in almost all cases the second field contains the headword.

The main difficulty in this type of definition is the division of the definiens text following the Match element into Superordinate and Discriminator. The relatively small sample of 500 type 1 definitions used in the initial investigation of the taxonomy led to the identification of a small group of boundary words which could be used to mark the division between these two components. A file was constructed as the investigation proceeded, which eventually contained the words:

of
which
who
that
whose

where
such
for
with
in
at
on
of
from
made
used
near
especially
between
to
around
towards
about
caused

This file was then used in the parsing software as the basis for splitting the definiens text into the two components. The investigation used to establish this list proved to be a useful starting-point for the development of the much more complex list which was eventually produced to deal with type A1 definitions, the equivalent in the final taxonomy of the original type 1. The detailed investigation carried out in the later stages of development used a combination of word frequency analysis of the text in this part of the third field and an assessment of the use of frequent words highlighted by the analysis. The assessment was carried out by using the parsing software with different versions of the boundary-word list and checking the resulting split between Superordinate and Discriminator.

This development could be carried out independently of the development of other areas of the taxonomy because it only applied to those definitions already allocated to type 1 and did not affect the allocation process itself.

4.4.2 Checking the operation of the parser in the final stages

As an example of the type of test used to control the development of the parser, consider the problem described in the previous section. The development of the set of boundary words and rules used in the identification of the

superordinate element in type A1 definitions, the extended taxonomic group equivalent to the earlier type 1, was carried out using frequency list analysis on the unanalysed definition texts. Once all of the type A1 definitions were parsed in the later stages of development it was very easy to construct a frequency distribution of the elements identified by the parser as superordinates. The contents of this list were then used as a basic check on the accuracy of the parser's identification of the superordinate boundaries. The following extract from a list prepared from a late stage of the parser's development shows the superordinates appearing twenty times or more:

person	401
no superordinate[3]	347
someone	246
something	184
place	137
substance	100
people	78
part	70
device	66
same	66
animal	64
container	62
object	62
abbreviation	55
man	54
behaviour	52
building	52
things	52
used	49
liquid	46
area	45
group of people	44
period of time	44
plant	42
room	42
way	42
parts	41
bird	40
woman	40
ability	39
instrument	39
machine	39

tool	37
part of it	35
vehicle	35
amount of it	34
game	34
situation	34
area of land	29
system	29
thing	29
fact	28
food	28
book	27
event	27
money	27
time	27
amount	26
fruit	26
illness	25
amount of money	24
material	24
feeling	23
belief	22
disease	22
shop	22
statement	22
drink	21
vegetable	21
covering	20

Problems were immediately apparent with those definitions with no superordinate (347), and those with the superordinates 'same' (66) and 'used' (49). There were also possible parsing problems with the superordinates containing matching pronouns, such as 'part of it' (35) and 'amount of it' (34). The surrounding text in the definitions with these superordinates were checked and the parsing algorithms adjusted accordingly. The test was then repeated until it showed results which seemed more accurate and acceptable. Similar tests were carried out on the other components identified by the latest version of the parser, and the results were used to correct the algorithms and develop a revised version.

4.5 Summary

The relatively simple analysis techniques described in this chapter formed the basis of the development of the entire taxonomy, grammar and parser for the definition sentences. The process combined the rigorous examination of the data by the computer with thorough manual evaluation of the results, using the taxonomy, the grammar and the parser to check the integrity of each other at all development stages. The resulting taxonomy is described in Chapter 5, and the grammar and parser derived from it in Chapter 6.

Notes

1. As explained above.

2. The majority of the definition sentences which begin with 'in' have already been removed during preprocessing, as explained in section 4.2.2.1 above.

3. In these cases the data item which corresponds to the superordinate was empty in the output from the parsing software.

Chapter 5

The definition type taxonomy

Chapter 4 describes the approach adopted for the investigation of the definition sentences through the construction of a structural taxonomy, which formed the basis of the grammar and the parsing software. The taxonomy itself is outlined in section 5.1, and its relationship to the structural descriptions provided in Sinclair's original analysis of the definitions is explored in section 5.2. The development of the terminology of the model is described in section 5.3, while 5.4 contains a detailed account of the structural patterns typical of each of the definition types. The taxonomy's relationship with the grammar and the parser are discussed in detail in section 5.5.

5.1 An outline of the taxonomy

The results of the investigation described in Chapter 4 are set out in summary below. The original labels used for these types (in, for example, Barnbrook 1996 pp. 160–1) were allocated during the development of the taxonomy and reflected the order in which types were identified rather than any meaningful structural relationship between them. The revised type labels used below were first used in Barnbrook and Sinclair (2001). The individual definition types have been grouped into four major structural categories, within which they are listed in approximate order of similarity to each other and frequency.

For each individual definition type in the table below, the frequency with which it occurs in *CCSD* is given, followed by a typical example.

Group A

A1	10,494	An **issue** of a magazine or newspaper is a particular edition of it. (p. 301, sense 3)
A2	689	The earth's **crust** is its outer layer. (p. 127, sense 3)
A3	358	**Forgot** is the past tense of **forget**. (p. 218)
A4	2,212	A **secluded** place is quiet, private, and undisturbed. (p. 504)
A5	2,202	Something that is **hidden** is not easily noticed. (p. 263, sense 1)

A6	1,441	To **commit** money or resources to something means to use them for a particular purpose. (p. 101, sense 2)
A7	172	New people who are introduced into an organization and whose fresh ideas are likely to improve it are referred to as **new blood, fresh blood,** or **young blood**. (p. 52, phrases)
		Group B
B1	7,528	When a country **liberalizes** its laws or its attitudes, it makes them less strict and allows more freedom. (p. 322)
B2	1,813	If someone is **run-down,** they are tired or ill; (p. 491, sense 1)
B3	1,714	If you do something **in class,** you do it during a lesson in school. (p. 89, Phrases)
B4	14	You ask what has **got into** someone when they are behaving in an unexpected way; (p. 233, sense 3)
		Group C
C1	1,524	You can also say you **admire** something when you look with pleasure at it. (p. 8, sense 2)
C2	561	If you say to someone that something is **their own affair,** you mean that you do not want to want to know about or become involved in their activities. (p. 10, sense 4)
C3	224	You can refer to a change back to a former state as a **return** to that state. (p. 480, sense 10)
C4	76	When someone creates something that has never existed before, you can refer to this event as the **invention** of the thing. (p. 298, sense 3)
C5	362	**Equatorial** is used to describe places and conditions near or at the equator. (p. 182)
		Group D
D1	17	In **humid** places, the weather is hot and damp. (p. 272)

The illustrative examples given above for each of the types in the taxonomy show their basic structural characteristics. A full description of the distinguishing features of each type is given in section 5.4. This description uses a special terminology for the linguistic units making up the definition structures, and it is first necessary to establish the set of terms used and their precise significance within the definition language.

Unallocated

Six definitions could not be allocated to any of the types shown above. These are described in detail in section 5.4.5.

5.2 The terminology of the taxonomy

The terminology used to describe the functional components of the definition sentences developed during the construction of the taxonomy. The starting-point was the set of terms used by Sinclair (1991, Chapter 9) in his discussion of 'the capacity of language to talk about itself' (p. 123). The definitions which he uses as examples (p. 124) are divided into a first part, which contains the headword, or topic, and a second part, which contains the comment. These terms do not exactly coincide with the definiendum and definiens of conventional lexicography, already discussed in some detail in section 2.1.1, but there is a straightforward relationship between them. The first part *contains* the definiendum, and the second part the definiens, but in most definitions they both also contain other elements. The nature of these other elements and their implications for the definition grammar and its associated parser are considered in detail in section 5.3.

An application of the basic model provided by Sinclair (1991) to the definition types described in section 5.1 is given in the next section.

5.2.1 The original analysis and the taxonomy

Table 1 below shows Sinclair's original level of definition analysis, as shown on p. 125 of Sinclair (1991), applied to the example definitions used above to illustrate definition types other than A7, C3 and C4. The examples used to typify types A7, C3 and C4 cannot be analysed under the same columnar headings and are shown in table 2, immediately below table 1. The reversal of the normal sequence of the definition sentence shown in these examples, which means that the second part, containing the comment, precedes the first part, containing the topic, is an important feature of the full sentence definition. The concept of the lexicographic equation, shown, for example, in section 2.4.4.2, suggests that definition structure involves a conventional 'Left Hand Side' and 'Right Hand Side' equivalent to mathematical or chemical models. The layout of most dictionaries other than the Cobuild series actually forces this conventional arrangement through the physical separation of the definiendum from its definiens on the page. In definition types A7, C3 and C4, by contrast, the demands of the expression of meaning have been allowed to reverse the normal order of the equation. The reasons for this and the implications for the grammar and the parser are discussed in more detail in section

Table 1.

Type	First part				Second part		Chunks
	Operator	Co-text(1)	Topic	Co-text(2)	Operator	Comment	
A1		An	**issue**	of a magazine or newspaper	is	a particular edition of it.	
A2		The earth's	**crust**		is	its outer layer.	
A3			**Forgot**		is	the past tense of forget.	
A4		A	**secluded**	place	is	quiet, private, and undisturbed.	
A5		Something that is	**hidden**		is	not easily noticed.	
A6		To	**commit**	money or resources to something	means	to use them … for a particular purpose.	1 2
B1	When	a country	**liberalizes**	its laws or its attitudes,		it makes them less strict and allows more freedom.	
B2	If	someone is	**run-down,**			they are tired or ill;	
B3	If	you do something	**with**	someone else,		you do it together.	
B4		You ask what has	**got into**	someone	when	they are behaving in an unexpected way;	
C1		You can also say you	**admire**	something	when	you look with pleasure at it.	
C2	If	you say to someone that something is	**their own affair,**			you mean that you do not want to know about or become involved in their activities.	
C5			**Equatorial**		is	used to describe places and conditions … near or at the equator.	1 2
D1		In	**humid**	places, the weather	is	hot and damp.	

Table 2.

Type	Second part		First part			
	Operator	Comment	Operator	Co-text(1)	Topic	Co-text(2)
A7		New people who are introduced into an organization and whose fresh ideas are likely to improve it	are referred to as		**new blood, fresh blood, or young blood.**	
C3	You can refer to	a change back to a former state	as	a	**return**	to that state.
C4	When	someone creates something that has never existed before,	you can refer to this event as	the	**invention**	of the thing.

5.2.3.3. The other definition types can be fitted into the basic scheme outlined by Sinclair, but this only begins the process of analysis, and beyond this point the different structural types begin to need more specialised treatment to allow their texts to be analysed adequately.

5.2.2 Further analysis of the second part

The tables given above analyse the final part of the definition, the comment, into the chunks suggested by Sinclair (1991, p. 125). As is explained in section 5.3 below, the nature of this analysis is actually subject to different requirements for each definition type. There is, however, a general model running through this more detailed description of the definition components, which is derived from Sinclair's analysis of the second part (Sinclair, 1991, pp. 132–134). This divides the second part of each definition into operator, gloss and framework, the last of which matches the co-text in the first part of the definitions. This approach would produce the analyses shown in table 3 below for the definitions analysed in table 1.

There are some problems evident in the application of this analysis model to the definition examples, and these are discussed in the next section.

5.2.3 Problems with the analysis of the second part

While the original model proposed by Sinclair can be applied to the definitions shown in the previous section, there are some discrepancies. The main features of these problem areas are outlined in sections 5.2.3.1 to 5.2.3.3, and the alterations made to the basic model during the development of the taxonomy, the grammar and the parser are covered throughout section 5.3, where the terms used to describe the structural patterns found in the taxonomy are discussed in detail.

5.2.3.1 *Lack of equivalence between topic and gloss*

In all but three of these definitions the element labelled 'gloss' in this analysis refers directly to the headword or 'topic' of the first part. The exceptions are the type B4 definition of 'got into', the type D1 definition of 'humid' and the type C4 definition of 'invention':

Table 3.

Type	Operator	Framework	Gloss	Framework
A1	is	a	particular edition	of it.
A2	is	its	outer layer.	
A3	is		the past tense of **forget**.	
A4	is		quiet, private, and undisturbed.	
A5	is		not easily noticed.	
A6	means	to	use … for a particular purpose.	them …
A7			New people who are introduced into an organization and whose fresh ideas are likely to improve it	
B1		It	makes … less strict and allows more freedom.	them
B2		they are	tired or ill;	
B3		you do it	together.	
B4	when	they	are behaving in an unexpected way;	
C1	when	you	look with pleasure at	it.
C2		you mean that you	do not want to know about or become involved in	their activities.
C3		You can refer to a	change back to a former state	
C5	is		used to describe places and conditions near or at the equator.	
C4			When someone creates something that has never existed before,	
D1	is		hot and damp.	

> You ask what has **got into** someone when they are behaving in an unexpected way;
> In **humid** places, the weather is hot and damp.
> When someone creates something that has never existed before, you can refer to this event as the **invention** of the thing.

In the first two cases, the topic refers to the co-text in the first part and the gloss refers to its matching elements in the second part, but the two types of reference do not use the same syntax, so that the gloss cannot be used as a substitute for the topic. In the third case, the gloss 'When someone creates something that has never existed before' matches the words 'this event'. While this element is then equated directly with the topic 'invention', there is still a displacement of the relationship between the topic and its gloss. These fea-

tures of Cobuild definition strategies have been discussed in general terms in section 2.4.4.2, and their detailed implications for the grammar are dealt with in section 6.2.

5.2.3.2 *Embedded framework elements*

In the examples shown in table 3 above for definition types A6 and B1 matching co-text elements are embedded within the text of the gloss, so that the equivalence between topic and gloss appears to become:

> **commit** = use … for a particular purpose.
> **liberalizes** = makes … less strict and allows more freedom.

This interrupts the linear structure of the definitions, and is dealt with in the analysis process. It did not affect the construction of the taxonomy, largely because most of the type recognition is carried out on the earlier part of the text.

5.2.3.3 *Reversed sequence definitions*

The interpretation of the analyses of definition types A7, C3 and C4 is slightly complex, and it is important to remember that in these definition patterns the normal sequence is reversed. The original definition texts are:

> A7 New people who are introduced into an organization and whose fresh ideas are likely to improve it are referred to as **new blood, fresh blood,** or **young blood**.
> C3 You can refer to a change back to a former state as a **return** to that state.
> C4 When someone creates something that has never existed before, you can refer to this event as the **invention** of the thing.

The type A7 and C3 patterns are effectively rearranged versions of definition types A1 and C1 respectively, as shown in the examples below:

> A1 **Fantasy** refers to the activity of imagining things, or the things that you imagine. (p. 198, sense 2)
> C1 You can use **existence** to refer to someone's way of life. (p. 189, sense 2)

Type C4 in turn is an elaborated version of type C3, in which the entity which is being referred to or reported in some other way is a rather more complex piece of text introduced by 'if' or 'when'. The detailed descriptions of the definition components given in sections 5.3 and 6.7 reflect these relationships between the definition types.

5.3 The development of the definition analysis model

The terms explained in sections 5.3.1 to 5.3.9 have been developed from the original definition analysis model described in Sinclair (1991) which has already been discussed in detail in section 5.2, and the relationship of each component of the new model to its corresponding elements in the original is described within each of the sections. The range of definition structures revealed by the taxonomy shows that different types of headword need different definition structures, and that parallels between components which are specific to different types of definition structure may not always be complete or consistent. This demands a rather large set of terms, some of which overlap with standard linguistic labels. Any potential confusion arising from this state of affairs should be dispelled by the guidance on structural contexts given within the description of each of the components.

Chapter 6 describes the relationships between the components of the definition sentences in detail in its description of the definition sentence grammar. The outline given here is solely intended to make it possible to follow the descriptions of definition structures used in the taxonomy. The terminology is largely based on the grammar description produced for the ET–10/51 project (see section 7.6.2) and described in the project's Final Report (Sinclair, Hoelter & Peters, 1995).

5.3.1 Usage and other notes

All definition types can have embedded notes attached to them which may be placed before or after the main definition text. These should not affect the structure of the definition or its place in the taxonomy, since they are generally removed from the definition text during preprocessing and put into separate fields within the definition record. Because of this, and because they affect all definition types equally, they have not been included in the structural patterns and their own possible structures have not been considered as part of the description of the definition sentences. However, some minimal structural analysis was needed to develop the software which carries out the preprocessing described in section 4.2.2.1, and the basic characteristics of the notes have been established.

The preprocessing program recognises part of the initial text of the definition as a preceding usage note if the definition:

a) begins with 'in', 'at' or 'on' and
b) contains a comma in the text preceding the headword.

The comma usually marks the end of the note and the beginning of the definition text proper. The embedded note following the definition text is even more easily identified: the software checks for text following a full stop, semi-colon or colon, and treats it as a note. The effectiveness of this process is considered in detail in section 7.3.1.1.

5.3.2 Operator

In Sinclair's original analysis model the term 'operator' is used for the component of the definition text which forms the link between the two halves of the lexicographic equation. The term for this component has been changed to 'hinge' in the present study, and its characteristics and functions are discussed in section 5.3.5. For the purposes of this analysis the label 'operator' has been transferred to some elements which Sinclair's analysis includes as co-text. The reason for this change was the desire to distinguish between those elements of the headword's textual environment which provided syntactic information about its normal usage, and those which provided the corresponding lexical information. The operators are the components which provide purely syntactic information. This distinction relates to the typical syntactic and lexical properties of the word being defined, rather than the syntax or lexis of the definition sentence itself, since the hinge element is most likely to appear to have a purely syntactic function within the organization of the definition text. As an example, consider the definitions:

> In an army, the **cavalry** used to be the group of soldiers who rode horses. (p. 78)
> An **echelon** is a level of power or responsibility in an organization; (p. 170)
> **Piracy** was robbery carried out by pirates. (p. 419, sense 1)

In all of these definitions the presence or absence of an article before the headword denotes its normal syntactic behaviour, as described in the next paragraph, while the hinges 'used to be', 'is' and 'was' provide information relating to the currency of the lexical item being defined. The article is a typical example of an operator. Its presence or absence is particularly important in definitions of nouns. For example, the presence of an indefinite article before the headword in the standard form of noun definition normally implies that the word being defined is a countable noun, and sets up an expectation that

the article will be matched by a corresponding item in the other half of the definition. Where this does not happen, it will have significant implications for the description of meaning given by the definition. As an example, consider the definition of sense 4 of 'love' (p. 333):

> **Love** is a very strong feeling of affection or liking for someone or something.

It has not been possible to define the uncount noun 'love' using a corresponding uncount noun: instead the count noun 'feeling' has been used. The asymmetry of the article in the second part of the definition alerts the user to the difference in the properties of the two words in a totally consistent way, without the need for a full understanding of the explicit grammar notes.

Where articles perform as operators within a definition they are given a separate entry in its structural description. Where they are used in the text within some other component and do not fulfil this function of the definition language, they are, of course, simply contained within the grammatical unit of which they form part. The variability of the functions of a word in different contexts within definitions is a significant feature of the definition grammar. As has been explained in more detail in section 3.3.3.1, individual words are not generally regarded as the basic linguistic components of the definitions. Component boundaries are more often the basis of the analysis performed by the parser than the identification of complete components, and the basis of the pattern-matching performed by the parser is determined by the context within which it takes place.

The other main manifestation of the operator is the 'to' infinitive marker in definitions such as:

> To **liberate** a place means to free it from the control of another country. (p. 322, sense 2)

Again, the information provided by this component relates entirely to its normal syntactic environment rather than its lexical relationships.

5.3.3 Co-text

With the exception of the operators whose separation is described in the previous section, the general concept of co-text used in the descriptions of definition structures is that formulated by Sinclair (1991, p. 124): the words in the first part of each definition sentence other than the headword. Different definition types have different potential co-texts in varying positions. As

examples, type A1 definitions can have co-text before the headword, as in:

> A university or college **campus** is the area of land containing its main buildings. (p. 72)
> A theatre or dance **company** is a group of performers who work together. (p. 102, sense 2)
> A radio or television **series** is a set of related programmes with the same title. (p. 510, sense 2)

Definitions belonging to this same structural type may also have co-text following the definition, as in:

> An **approach** to a situation or problem is a way of thinking about it or of dealing with it. (p. 24, sense 5)
> A **consequence** of something is a result or effect of it. (p. 109, sense 1)
> The **pivot** in a situation is the most important thing around which everything else is based or arranged. (p. 420, sense 3)

To keep the distinction between these two possible co-texts clear, they were first numbered within the definition structures in order of occurrence. In the definition sentence grammar, described in Chapter 6, the functions of the co-texts within the definition are considered in detail.

In the most general terms, these functions vary with the nature of the headword: in type B1 definitions, which are used for verb headwords, their typical function is to provide the subjects and objects, direct and indirect, of the headword. The following examples show something of the range of possibilities:

> If you **beam** a signal or information to a place, you send it by means of radio waves. (p. 41, sense 3)
> When the police **breathalyze** a driver, they ask the driver to breathe into a special bag to see if he or she has drunk too much alcohol. (p. 61)
> If you **get** someone to do something, you ask or tell them to do it, and they do it. (p. 232, sense 6)
> If a blow or cold weather **numbs** a part of your body, you can no longer feel anything in it. (p. 381, sense 3)

In each of these definitions, co-text 1 ('you', 'the police', 'you' and 'a blow or cold weather') are the subjects of each of their verb headwords, while co-text 2 ('a signal or information', 'a driver', 'someone' and 'a part of your body') forms the object. The definitions of 'beam' and 'get' are slightly more complex: their meanings demand structures with an added adjunct or bound clause, and these extra elements in the definitions ('to a place' and 'to do something')

can be separately identified as co-text 3 and analysed accordingly within the definition texts.

5.3.4 Headword

These are relatively unproblematic elements within the definition text, although, as already explained in section 4.2.2.2, they can have a complex structure involving more than one headword element separated by text which is not printed in bold type in the dictionary. The preprocessing described in section 4.2.2.2 deals with this so that the headword can be treated as a single element.

5.3.5 Hinge

The basic two-part structure of dictionary definitions, in which the meaning of the definiendum is described by the definiens, requires the definition sentence to be constructed in two parts. The link between the two parts is called the 'hinge' in this description of the definition structures. As already described in section 5.3.2, this component was originally labelled the 'operator' in Sinclair's analysis model, but that term has been transferred in the description used in the project to specific elements of the co-text . The simplest form of the hinge, used for many of the definitions in Group A, is some form of the verb 'to be'. These examples are taken from type A1 definitions:

> **Anthropology** is the study of people, society, and culture. (p. 21)
> **Particulars** are facts or details; (p. 405, sense 5)
> **Warriors** were soldiers or experienced fighting men in former times; (p. 636)

In some definitions this is replaced by an equivalent phrase with subtly different relational implications:

> **Brushwood** consists of small branches and twigs that have broken off trees and bushes. (p. 65)
> **Mythology** refers to stories that have been made up in the past to explain natural events or to justify religious beliefs. (p. 368, sense 1)

Similar hinges are used for the main adjective definition type, type A4, although they relate to their adjective headwords in a slightly different way:

> A **busy** time is a time when you have a lot of things to do. (p. 69, sense 4)
> A **kindly** person is kind, caring, and sympathetic. (p309., sense 1)

> **Unsteady** objects are not held, fixed, or balanced securely. (p. 620, sense 3)

In all three examples, the subject of the verb 'is' or 'are' is the co-text of the adjective headword, rather than the headword itself, and this needs to be recognised in the grammar and parser.

Type A6 definitions use some form of the word 'means', sometimes within a phrase, as their hinge. The following examples show typical forms:

> To **convince** someone of something means to make them believe that it is true or that it exists. (p. 115)
> **Ecclesiastical** means belonging to or connected with the Christian Church. (p. 169)
> **Juicy** also means interesting or exciting, or containing scandal; (p. 306, sense 2)

Of the words which can realise the central hinge of Group A definitions, the variations of the verb 'to be' and the phrases based on 'consists of' produce definitions which deal with relations of genuine equivalence between the definiendum and the definiens. Hinges based on the word 'means' or phrases such as 'refers to', on the other hand, deal with purely linguistic relations between them and do not exploit the full structural and inferential possibilities of the definition syntax. Definitions containing these hinges are the closest equivalents to traditional dictionary definitions in the Cobuild dictionaries.

The third type A6 example definition given above, for sense 2 of 'juicy', shows a feature of many of the definition hinges: the addition of the word 'also' to relate the definition of a particular sense to those of previous senses. This is treated in the structural analysis as part of the hinge, along with other possible elaborations such as the use of the word 'can' in front of the normal hinge. These additional elements within the major functional components may, in some cases, need to be interpreted as part of the fine-tuning of the lexicographic equation. The word 'also' is, in fact, as discussed in section 3.5.2.2, a rare reference outside the definition sentence to another sentence within the same headword paragraph, and as such has no real effect on the meaning of the definition. The word 'can', on the other hand, has important implications for the probability of the usage being described in the definition. The second most common hinge type is found in the Group B definitions. These begin with 'if' or 'when' and this initial word forms their hinge. Type B1 is the most frequent definition type within this group, and three examples are given below:

> If you **overestimate** someone or something, you think that they are better, bigger, or more important than they really are. (p. 398)
> When you **reach** a place, you arrive there. (p. 460, sense 1)
> If your muscles or joints **stiffen,** they become difficult to bend or move. (p. 554, sense 2)

These definitions are constructed as conditional statements, in which the initial 'if' or 'when' forms the link between the two elements, although, unlike the hinge in Group A definitions, it is not in a central position in the sentence. This may become clearer if the definition of 'overestimate' is rewritten to change its word order:

> You **overestimate** someone or something if you think that they are better, bigger, or more important than they really are.

This may not be such an appropriate word order for the majority of these definitions, and the lexicographer has presumably chosen the normal arrangement to achieve optimum clarity. There is, in fact, another rather rare definition type, type B4, which uses this reverse order:

> You also **flick** something when you hit it sharply with your fingernail by pressing the fingernail against your thumb and suddenly releasing it. (p. 211, sense 4)
> Two places or objects **are linked** when there is a physical connection between them so that you can travel or communicate between them. (p. 327, sense 1)

More complex definition structures use more complex hinges. Type A5 definitions, used mainly for adjectives, have a branching structure which uses a hinge in two parts separated by other definition text elements. The verb 'to be' is normally used at least for the first part of this complex hinge:

> Something that is **abundant** is present in large quantities. (p. 3)
> Someone who is **impulsive** does things suddenly without thinking about them first. (p. 281)
> A place that is **off the beaten track** is in a quiet and isolated area. (p. 599, phrases)

This structure represents a rearrangement of the more common structure used in type A4 adjective definitions:

> **Incisive** speech or writing is clear and forceful. (p. 283)
> A **multinational** company has branches in many different countries. (p. 366, sense 1)
> The **winning** competitor, team, or entry in a competition is the one that has won. (p. 649, sense 1)

The selection of word order in these examples is linked to the nature of the co-text in the definition and the ability of the headword to be used as an attributive or a predicative adjective, or both. In the type A5 examples given above, the word 'is' in the first part of each definition corresponds to 'is', 'does' and 'is' respectively in the definition's second part. The fact that the hinges do not match in some definitions, such as the definition of 'impulsive' shown above, is crucial to the interpretation of the meaning of the headword as given in the dictionary. In the other two headwords the definitions, stripped of all matching elements, could be interpreted as the following lexicographic equations:

abundant = present in large quantities
off the beaten track = in a quiet and isolated area

For 'impulsive', this would need to be stated differently:

to be impulsive = to do things suddenly without thinking about them first

An increase in the complexity of the hinge similar to that shown above between type A4 and type A5 structures can take place between type A1 definitions, which use 'refers to' as a hinge, and type C3 definitions. In type A1 structures the phrase 'refers to' forms a simple, central hinge, as in:

The **accused** refers to the person or people being tried in a court for a crime. (p. 5)

A similar phrase is used in type C3 definitions, but the sequence of the definition is reversed and the hinge becomes more complex:

You can refer to a group of people with the same profession or interests as a **fraternity.** (p. 221, sense 2)
You can refer to books and magazines as reading **matter;** (p. 345, sense 4)
You can refer to working-class people, especially industrial workers, as the **proletariat;** (p. 443)

This version of the structure has a hinge with two separated parts, 'can refer to' and 'as', similar in some ways to that of the type A5 definitions.

5.3.6 Projection

Section 2.1.2 considers the nature of the metalanguage in full sentence definitions, and quotes Hanks' assertion that:

> Dictionaries are much concerned with accounting for what it is that an utterer may expect a hearer to believe.
>
> (Hanks, 1987, p. 135)

The same section also discusses the implicit nature of this process in most definition forms, and the fact that in some definitions it is made explicit, so that the definition becomes a direct comment on usage, or, in Sinclair's words:

> The statement may be about what people mean when they use a word or phrase, rather than what the word or phrase means.
>
> (Sinclair, 1991, p. 126)

Consider the following definitions:

> When you refer to the **aforementioned** person or subject, you mean the person or subject that has already been mentioned; (p. 11)
> If you describe a situation or event as **farcical,** you mean that it is completely ridiculous. (p. 198)
> If you say that something was not said **in so many words,** you mean that it was said indirectly, but that you are giving its real meaning. (p. 652, phrases)

This definition strategy, used for headwords whose meaning can only be conveyed through an explicit description of the circumstances of their use, involves a further definition component, identified by Sinclair (1991, pp. 126–7) as the 'Report' element of co-text 1. Applying his analysis to these definitions would produce the following descriptions of their first parts:

First Part					
OPERATOR	**CO-TEXT(1)**			**TOPIC**	**CO-TEXT(2)**
	REPORT	**'topic'**	**'operator'**	**'comment'**	**'topic'**
When	you refer to		the	**aforementioned**	person or subject,
If	you describe	a situation or event	as	**farcical,**	
If	you say that	Something was not said		**in so many words,**	

The presence of the extra element within co-text 1 in these definitions demanded that they be categorised separately from apparently similar structures and dealt with by a specific parsing strategy. Consider the following type C2 definitions:

> If you describe a place or event as **enchanted,** you mean that it seems as lovely or strange as something in a fairy story. (p. 176, sense 2)
> If you say that something is not done **lightly,** you mean that it is not done without serious thought. (p. 325)
> If you call someone a **savage,** you mean that they are cruel, violent, or uncivilized. (p. 497, sense 3)

Superficially, they have a very similar structure to type B3 definitions, such as:

> If you do something **under duress,** you are forced to do it. (p. 167)
> When a cat goes '**miaow',** it makes a short high-pitched sound. (p. 351)
> If someone or something is on a **short-list** for a job or prize, they are one of a small group chosen from a larger group. (p. 518, sense 1)

The distinguishing feature of the type C2 definitions is the reporting structure which provides a frame for the lexicographic equation which transforms it into a comment on usage rather than intrinsic meaning. In the case of sense 2 of 'enchanted', without this framing the lexicographic equation would be:

> enchanted = seems as lovely or strange as something in a fairy story

Taking the surrounding reporting structure into account this becomes:

> describe … as enchanted = mean that …seems as lovely or strange as something in a fairy story

Applying the same reduction to the type B3 definition of 'miaow' produces:

> goes '**miaow**' = makes a short high-pitched sound

While this contains an element from the first part other than the headword, it is still a direct statement of meaning rather than a comment on normal usage. For the purposes of the taxonomy, the grammar and the parser the report component and its matching elements in the second parts of these definitions are given the label 'projection', taken, as explained in section 6.4, from Halliday (1985, p. 196). The realisation of this element varies between the definition types in which it occurs (those that make up Group C in the taxonomy shown in section 5.1) but they typically include structures based around the reporting verbs 'say', 'refer to', 'describe', 'use' and so on.

5.3.7 Superordinates and discriminators

As already discussed in section 2.4.3, the basic strategy used for explaining meaning in the Cobuild dictionaries is the superordinate and discriminator

model, in which the headword is related to an appropriate level of superordinate and distinguished from its co-hyponyms by the most useful discriminators. This works particularly well for noun definitions, for example:

> An **alert** is a situation in which people prepare themselves for danger. (p. 14, sense 3)
> A **caterpillar** is a small, worm-like animal that eventually develops into a butterfly or moth. (p. 78)
> A **toaster** is a piece of electric equipment used to toast bread. (p. 596)

Sinclair (1991, p. 133) describes the form of the second parts of these sentences as a 'classic definition', and generalises it into the two element model:

superordinate restriction

The superordinates of 'alert' and 'caterpillar' are fairly clearly 'situation' and 'animal'. The restriction elements are 'in which people prepare themselves for danger' for 'alert' and 'small, worm-like' and 'that eventually develops into a butterfly or moth' for 'caterpillar'. As can be seen, restrictions can be used both before and after the superordinate. The superordinate for 'toaster' is perhaps not so clearly defined, but for reasons that are explained in more detail in section 6.6.2.2, it would probably be 'piece of electric equipment' with 'used to toast bread' as the discriminator.

This superordinate and restriction model can be extended to verb definitions, but in many cases an analysis of the second part of the definition into matching and non-matching elements is more significant. This feature of the definition texts is described in section 5.3.9.

5.3.8 Explanation

Where the second part of the definition cannot be usefully analysed into the superordinate and discriminator components described in the previous section it is labelled 'explanation' in the definition structure patterns. Further analysis of this component is described in Chapter 6.

5.3.9 Matching elements in the second part

Sinclair's original analysis of Cobuild definitions (Sinclair, 1991, pp. 132–134) divides the second part of the text into the framework, the parts of the text which 'recall words in the co-text, either by repetition or other types of

cohesion' (p. 132), and the gloss. This division has already been used in the analysis of the second part and the identification of the gloss in section 5.2.2. The nature of these matching items is of the utmost importance in analysing the definition text, since, as described in more detail in section 6.1, any part of the first part which is unmatched in the second part is likely to form part of the definiendum. For the purposes of the taxonomy these matching elements are of rather less importance, because the distinguishing features of the structural types are generally located within the first part of each definition. This is perhaps unsurprising, since the second part consists largely of elements which correspond to the items in the first part, even where they do not match them exactly. The unmatched portions of the definition sentence are, after all, the two sides of some form of lexicographic equation. The cohesion created by the elements in the second part which directly match those in the first part is also purely intra-sentential, as already discussed in section 3.5.2.2, rather than forming links between definition sentences and so contributing to the overall discourse structure.

5.4 The structural patterns of the taxonomy

The structural taxonomy produced from the analysis of recurring definition patterns has already been summarised in section 5.1 as a hierarchical arrangement of definition types. Sections 5.4.1 to 5.4.5 provide a detailed commentary which explains the groupings adopted in terms of their structural patterns.

5.4.1 Group A

Group A, made up of definitions with a hinge centrally placed between the left and right hand sides, includes seven definition types which cover 17,568 definitions or 55.94% of the total number. Within Group A, types A1, A2 and A3 use a simple central hinge, often part of the verb 'to be' or a related phrase such as 'consists of', 'involves' or 'refers to'. Types A1 and A2 are typically used to define nouns, while type A3 provides grammatical cross-references to other dictionary headwords. Types A4 and A5 are more typically used to define adjectives. Type A4 uses a similar range of hinges to those found in types A1, A2 and A3, while type A5 uses a more complex two-part hinge, already described in section 5.3.5 above. Type A6 uses a form of the verb 'mean' as a

hinge and is used for a wider range of word types. Type A7 uses a reversed form of the basic group A structure. The numbers of definitions falling within each type are shown below, together with the percentage of the total number of definitions represented by the type.

Type	Number	Percentage of total
A1	10,494	33.41 %
A2	689	2.19 %
A3	358	1.14 %
A4	2,212	7.04 %
A5	2,202	7.01 %
A6	1,441	4.59 %
A7	172	0.55 %

5.4.2 Group B

Group B includes 11,069 definitions or 35.24 % of the total. In their basic form they use a conditional statement structure with an initial hinge, realised by 'if' or 'when' and preceding the left hand side of the definition, and do not contain any form of projection. In the reversed form of this structure, exhibited by type B4, the hinge moves to a medial position. Type B1 is typically used to define verbs, while types B2 and B3 are typically used for adjectives and for a wider range of words respectively. The basic sentence structure is similar for all three types. Type B4 uses a reversed form of type B3.

Type	Number	Percentage of total
B1	7,528	23.97 %
B2	1,813	5.77 %
B3	1,714	5.46 %
B4	14	0.04 %

5.4. 3 Group C

Group C includes 2,747 definitions or 8.75% of the total. They all contain some form of projection which frames the definition in an explicit statement about normal usage. Four of the structures within this group, types C1 to C4, use active forms of projection (such as 'you can refer to…', while type C5 uses a passive form (such as 'is used'). A wide range of words is defined using these structures, which are effectively more explicit versions of the corresponding Group A structures.

Type	Number	Percentage of total
C1	1,524	4.85 %
C2	561	1.79 %
C3	224	0.71 %
C5	362	1.15 %
C4	76	0.24 %

5.4. 4 Group D

Group D includes only one type, D1, with 17 definitions or 0.05% of the total. Type D1 defines headwords which appear to be embedded within a structure at the beginning of the definition which would otherwise be treated as a usage note.

Type	Number	Percentage of total
D1	17	0.05 %

5.4.5 Unallocated definitions

As already explained in section 5.1, six definitions could not be allocated to the established structural categories. They are listed below, and their implications for the definition sentence description and for dictionary construction are discussed in sections 7.2 and 7.3.

> **Around** can be an adverb or preposition, and is often used instead of **round** as the second part of a phrasal verb. (p. 26, sense 1)
> **Eminently means** very, or to a great degree; (p. 175)
> Roads, race courses, and swimming pools are sometimes divided into **lanes.** (p. 313, sense 2)
> In a railway station or airport, you can pay to leave your luggage in a **left-luggage office;** (p. 319)
> You can also give your impression of something you have just read or heard about by talking about the way it **sounds.** (p. 537, sense 6)
> You can acknowledge someone's thanks by saying **'You're welcome'.** (p. 641)

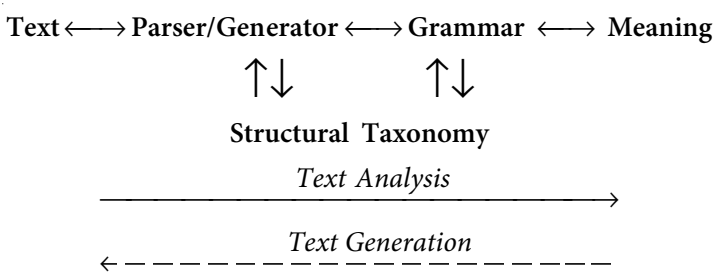
5.5 The relationship between the taxonomy and the grammar

The structural taxonomy described in this chapter is based on sequences of components common to a specific group of definition sentences. These components are generalised versions of the units which make up the sublanguage

grammar and which are used to describe the organisation of meaning within the definition sentences. Sentences made up of particular sequences of these components are gathered into groups within the taxonomy on the basis of their suitability for parsing by a single algorithm. This establishes the interconnectedness of the three elements of the model developed for the definition sentences. The details of this relationship are examined in section 5.5.1, and the special nature of the definition language model is considered in section 5.5.2.

5.5.1 The structural taxonomy, the parser and the grammar

The diagram below[1] outlines the relationships between the three elements of the definition sublanguage model.



The relationship between the parser and the grammar is obvious: as has already been discussed in the introduction to Chapter 3, the parser allows the grammar which governs the contents of a definition to be properly represented. In the analysis process shown above, the definition text is analysed by the parser and the meaning of the resulting analysis is obtained by reference to the appropriate part of the sublanguage grammar. The involvement of the structural taxonomy in this process is less obvious, but the selection of the appropriate part of the parsing software and the grammar associated with it depends on the position of the definition sentence within the structural taxonomy. In the process of text generation the semantic requirements of a proposed definition are fed through a selected part of the grammar and the associated parser algorithms are then used in reverse to generate the definition text. In this case the structural taxonomy would form the basis for selecting the most suitable definition type and its associated grammar and generator algorithms.

A similar relationship is evident in the earlier description in section 4.3 of the development of the structural taxonomy: the current versions of the parser and the grammar were constantly used to test the integrity of the definition groups, and the taxonomy is designed to allow the most efficient application of a single parsing strategy to a group of definition sentences. In its finished form, however, it is independent of them both and cannot be varied through their operations. Instead, the taxonomy forms the basis for any enhancement of the existing analysis or introduction of new definition types. Within the definition language as a whole, each definition type represents a subset of the sentences produced by the complete definition sentence grammar, with significant differences between their organisation and the interpretation of individual elements. The grammar and its associated parser can therefore not be used or understood without the structural taxonomy, which provides the basis for both their application and their development.

The main difference between the structural taxonomy and the other two elements is the level of detail of the information in each definition sentence which is used in constructing the taxonomy in the first place and in allocating individual definitions to a structural type at the start of the parsing process. Typically, the recognition program described in section 6.9 can perform this allocation using a relatively small amount of the overall sentence structure, often limited to the first part of the definition and the hinge. The parser and the grammar, on the other hand, both relate to complete sentences. This apparent difference of approach in fact owes more to the difference between categorisation and analysis than to any fundamental difference between the three elements of the language model. The structural taxonomy, the parser and the grammar all contain the same information. The differing modes of organisation of this information within each of these elements reflect their individual functions: the structural taxonomy uses fairly superficial similarities between sentences to categorise them into structural groups; the parser uses the restricted patterns of each individual group to analyse sentences into functional components; and the grammar provides the basis for the extraction of meaning from the analysis produced in this way.

5.5.2 The special nature of the definition language model

The consideration of the relationship between the taxonomy, the parser and the grammar reveals a major difference between the definition language

model and more conventional language descriptions. Chomsky (1965, pp. 16–17) dealing with the nature of deep and surface structure of sentences, states that:

> The central idea of transformational grammar is that they are, in general, distinct and that the surface structure is determined by repeated application of certain formal operations called "grammatical transformations" to objects of a more elementary sort.

The distinction between the more elementary objects referred to in this passage, often called 'kernel sentences', and the more elaborate sentences closer to the surface structure, does not seem to exist in the definition language grammar. The categorisation of individual sentences into the groups which make up the structural taxonomy creates a discrete classification: there is no continuum between the different types. As an example, consider the difference between type A1 definitions and type A4, represented by the following two examples:

> An **extravagance** is something that you spend money on but cannot really afford. (p. 193, sense 2: type A1)
> An **extravagant** person spends more money than they can afford or uses more of something than is reasonable. (p. 193, sense 1: type A4)

The structures of these two definitions seem remarkably similar, and in Chomsky's terms the sentences could be seen as products of slightly different transformations applied to the same kernel sentence and would be analysed in similar ways. For the structural taxonomy, the distinction between them lies in the nature of the element shown in bold type, the dictionary headword. If the word 'person' were being defined in the second sentence, rather than the word 'extravagant', it would be allocated to type A1 rather than type A4.

This fundamental distinction between the categories is independent of the general grammatical structure of the sentences and dependent only on the special features of the definition text within the dictionary. It means that transformations cannot be applied to base structures to produce different surface structures while leaving the deep structure unchanged. A change in surface structure is a change in deep structure, and there is no effective distinction between them.

5.6 Summary

It is possible to categorise the definitions contained in the sample and to produce a useful structural taxonomy on the basis of a simple analysis of definition sentence text patterns, with occasional reference to grammatical information where similar structures are applied to different types of head-word. The definition types revealed by this taxonomy have consistent structures in terms of the definition sentence grammar and are capable of automatic parsing using a limited set of algorithms developed for each definition type. Both the grammar and the parser are described in Chapter 6. Together, the structural taxonomy, the grammar and the parser form a model of the definition language capable of describing the definition sentences and allowing the extraction of semantic information from them.

Note

1. Suggested by John Sinclair.

Chapter 6

The definition language grammar and its parser

This chapter provides a detailed account of the grammar itself and the parser developed for it. It describes the functional components of the definition sentences, the structural combinations of those components and the variations in structure between the different definition types, together with an outline of the processing involved in the analysis of the definition sentences.

It is important to remember that the level of description provided by this grammar, and the analysis provided by the parser, both relate entirely to the function of the definition sentences as definitions, rather than as examples of English sentences in general. The rather generalised names used for some components in Chapters 4 and 5 have been made more specific in this account of the grammar, so that they can convey the part played by each element within individual definition types at a proper level of detail.

The grammar is described in sections 6.1 to 6.7, and the parser in sections 6.8 to 6.10.

6.1 The definiendum and the definiens in the definition sentences

Despite the variation and development evident in monolingual English dictionaries since their inception, already described in some detail in Chapter 2, two fundamental components can usefully be identified in all of them. These elements, described in section 2.1.1, are usually referred to as the definiendum, the linguistic unit which is to be defined, and the definiens, the words which perform the act of definition. Whatever the length and complexity of these two items, they remain the fundamental basis of dictionary structure. As has already been shown, most dictionaries other than the Cobuild range maintain a strict separation between them, usually showing the definiendum in bold type at the left edge of the column, and the definiens in normal type after it. The definiendum is rarely repeated in the case of multiple senses of the

same word, and any information which may be given relating to the normal context of the headword is presented in a highly abbreviated and encoded form. As an example, the entry for 'introduce' in *OALDCE* has the headword in bold type at the beginning as the basis of the definiendum. Sense 1 is then given as:

> ~ **sb** (**to sb**) make sb known formally to sb else by giving the person's name, or by giving each person's name to the other
>
> (p. 660)

In the case of the Cobuild dictionaries, of course, the definiendum and the definiens are both contained in the sentence making up the definition. The corresponding entry in *CCSD* is:

> If you **introduce** one person to another, you tell them each other's name, so that they can get to know each other. (p. 297, sense 1)

Here, the normal environment of the definiendum is included in a natural position in the sentence used to define it.

The basic definition structure put forward by Sinclair (1991), already described in section 5.2, divides the definition into a first and second part which contain the definiendum and the definiens respectively. An important feature of the first part, already referred to in sections 5.2.1 and 5.3.3, is that it may also contain other components, the co-text elements, which can give further information about the operation of the sense being defined. Consider the following definitions extracted from the first three noun senses of 'breast' in *OALDCE*:

> 1 either of the two parts of a woman's body that produce milk
> 2 (a) (rhet) upper front part of the human body
> (b) part of a garment covering this
> 3 part of an animal corresponding to the human breast, eaten as food
>
> (*OALDCE* p. 137)

In all these entries, the elements that provide information about restrictions on the operation of the sense are part of the text of the definiens. As an illustration of the Cobuild approach, consider senses 1 and 4 of 'breast' on p. 60:

> A woman's **breasts** are the two soft, round pieces of flesh on her chest that can produce milk to feed a baby.
> A bird's **breast** is the front part of its body.

In both of these entries the headword is a form of the word 'breast', but the first part of the definition also contains elements which specify the restrictions on the sense. Each definition is effectively stated to be dealing with a different linguistic manifestation of the word 'breast', and the co-text is being used to signal this from the start of the definition sentence. Only a woman's breasts, in the plural, are defined in sense 1 in terms of the production of milk to feed a baby; only a bird's breast is defined in sense 4 as the front part of its body.

The original analysis described by Sinclair (1991, pp. 124–125) would divide each of these definitions into two parts:

First part	Second part
A woman's breasts	are the two soft, round pieces of flesh on her chest that can produce milk to feed a baby.
A bird's breast	is the front part of its body.

This division leaves the link between the two halves, referred to by Sinclair as the 'operator', and in sections 5.3.2 and 5.3.5 as the 'hinge', within the second part of each definition. For the purposes of the grammar it is more useful to treat this as a separate element, and to divide the basic structure of each definition into three components. To avoid confusion with the original analysis, the First part, less any hinge element, is labelled 'L' (for left hand side), the Second part, also less any hinge element, is labelled 'R' (for right hand side), and the hinge element is labelled 'H'.[1] The analysis of these two definitions would then become:

L	H	R
A woman's breasts	are	the two soft, round pieces of flesh on her chest that can produce milk to feed a baby.
A bird's breast	is	the front part of its body.

In both definitions the co-text surrounding the definiendum in the first part is repeated in some form in the second part. Sinclair (1991, pp. 132–134) refers to this extra text within the second part as 'framework', and this has been discussed in sections 5.2.2 and 5.3.9. If the elements which match in this way are eliminated, the definiendum and its definiens can be isolated. The ex-

amples below show senses 1 and 4 of 'breast' stripped down in this way, with the hinge and all matching co-text elements removed:

Sense	Definiendum	Definiens
1	Breasts	the two soft, round pieces of flesh on … chest that can produce milk to feed a baby.
2	Breast	the front part of … body.

This produces a set of definitions much closer to the traditional format, but it ignores the effect of the hinge element and of the co-text in L and the matching elements in R. The characteristics of these elements are discussed in detail for individual definition types in sections 6.2 and 6.3, but it is worth considering their general implications here. The left and right sides of the definitions are made up as follows:

```
L = (C) Dm (C)
R = (M) Ds (M)
```

where:

D^m is the definiendum
(C) represents co-text elements, some of which are optional in some definition types
Ds is the definiens, and
(M) represents any framework elements matching co-text in L.

In some definition types the definiens can be further analysed into:

(d^r) S (d^r)

where:

S is a superordinate structure (possibly capable of further analysis) and
(d^r) represents optional discriminator structures

This is explored further in section 6.5.2.1.

The complete analysis of senses 1 and 4 of 'breast' would then become:

L		H	R		
C	Dm		Ds/M[2]		
			d^r	S	d^r
A woman's	breasts	are	the two soft, round	pieces of flesh	on her chest that can produce milk to feed a baby.
A bird's	breast	is	the front	part of its body	.

It is also important to remember that the headword does not always completely coincide with the definiendum. Consider the following definitions:

> If people are **agreed** about something, they have reached a decision about it. (p. 12, sense 3)
> When you bring a liquid to the **boil,** you heat it until it boils. (p. 54, sense 2)
> When you take a **chance,** you try to do something although there is a risk of danger or failure. (p. 81, sense 2)
> If you show **prejudice** in favour of someone, you treat them better than other people. (p. 435, sense 2)

The following table shows these definitions analysed into the three main structural units. Where co-text elements in L are matched in R both the original co-text and its matching component are shown in italics.

H	L	R
If	*people* are **agreed** *about something*,	*They* have reached a decision *about it.*
When	*you* bring *a liquid* to the **boil**,	You heat *it* until *it* boils.
When	*you* take a **chance**,	*You* try to do something although there is a risk of danger or failure.
If	*you* show **prejudice** in favour of *someone*,	*You* treat *them* better than other people.

The unmatched portions of L and R form the definiendum and definiens, and, as can be seen from the table below, the definiendum extends significantly beyond the headword shown in bold type in the dictionary:

D^m	D^s
are **agreed**	have reached a decision
bring… to the **boil**,	heat… until… boils.
take a chance,	try to do something although there is a risk of danger or failure.
show **prejudice** in favour of	treat… better than other people.

The process of checking each item in L for a matching element in R provides an extremely powerful basis for identifying the definiendum and definiens within definitions, and thus enables the full-sentence definition to be reconciled to some extent to more traditional methods. It is now necessary to consider the elements which are associated more specifically with the definition sentences. The hinge and its role in the lexicographic equation are discussed in the next section, and the text surrounding the definiendum in the left hand side of the definition is dealt with in section 6.3.

6.2 The hinge and the lexicographic equation

The *CCSD* definitions for senses 1 and 4 of 'breast' given in section 6.1 above can be used to illustrate one of the most important components of the definition sentences. The general form used by traditional dictionary definitions, stated using the notation introduced in the previous section, is:

```
Dm Ds
```

the definiendum followed immediately by its definiens. This form implies the equation between these two elements which has already been referred to in section 2.1.1:

```
Dm = Ds
```

The feature that distinguishes full sentence definitions from most traditional approaches is the fact that they contain both sides of this equation together with the equality operator itself. Within the grammar developed for the definition sentences this equality operator component is referred to as the 'hinge'. This element, already described briefly in sections 5.3.5 and 6.1, is of the utmost importance within the sentences. Apart from its significance as a basic

component of the grammar, it often provides the simplest practical means of recognising the division between the definition's first and second parts.

Both the position and the realisation of the hinge differ from one definition type to another. Type A2, the definition type used for senses 1 and 4 of 'breast' above, often uses an appropriate part of the verb 'to be' as its hinge, producing a straightforward linear rendering of the equation. Applying this to senses 1 and 4 of 'breast' gives:

> A woman's **breasts** = the two soft, round pieces of flesh on her chest that can produce milk to feed a baby.
> A bird's **breast** = the front part of its body.

The major definition strategy for verbs, found, for example, in type B1 definitions, uses a rather different approach. Consider sense 4 of the headword 'graduate':

> When a student **graduates,** he or she has successfully completed a degree course at a university or college and receives a certificate that shows this. (p. 242)

H may, at first, seem difficult to identify in this definition. On closer examination, however, the structure should become clear:

H	L		R	
	C	D^m	M	D^s
When	A student	**graduates,**	he or she	has successfully completed a degree course at a university or college and receives a certificate that shows this.

H is the initial word 'when', and the original linear structure of the equation form has been rearranged. It can be seen as a rewriting of the LHR form of the equation:

L	H	R
a student **graduates**	when	he or she has successfully completed a degree course at a university or college and receives a certificate that shows this.

This reordered version of the definition suggests a direct causal relationship between the process described in the definiendum and the process described

in the definiens, while the link between D^m and D^s in the original version of the definition seems more strictly linguistic. Type B4 definitions use the central hinge sequence, as in sense 9 of 'help':

> You shout '**Help!**' when you are in danger, in order to attract someone's attention.
>
> (p. 261)

There is, perhaps, a stronger causal relationship in these definitions, but this pattern is extremely rare. Another effect of the original sequence is that the hinge element is foregrounded. In other definitions of the same type H is realised by 'if' rather than 'when', and the choice provides important information about the nature of the definiendum.

The major difference, then, between these definition sentences and the forms used in other dictionaries, lies in the presentation of the linkage between the definiendum and the definiens. In most other dictionaries the relationship between them is implicit and hardly goes beyond simple equality. In the Cobuild range it is explicit and covers a far wider range of possibilities. The hinge is the first component of the full definition sentences which is peculiar to them. As has already been shown, both the words realising the hinge and their position in the definition can vary from one type of definition to another, but however it is realised, and whether it is actually present within the text or simply implied by it, it is a crucial component. It specifies the nature of the semantic relationship which links the definiendum to the definiens, a relationship which is often more complex than one of simple equality. A brief survey of the range of variation observed in the main definition types is given below.

6.2.1 Hinges in Group A definitions

In its simplest manifestation in the definitions which form Group A, the hinge occupies a central position between the definiendum and the definiens. In most cases the definiendum comes at the start of the definition and is followed by the definiens, but, as sense 4 of 'band' shows, this can be varied to suit the demands of the definition.

> **Another** can also be used to mean a different thing or person from the one just mentioned. (p. 20, sense 2)

A range of numbers or values within a system of measurement can also be referred to as a **band.** (p. 37, sense 4)
The **dial** of a clock or meter is the part where the time or a measurement is indicated. (p. 147, sense 1)
Experimental also means relating to scientific experiments. (p. 190, sense 2)
A **fearsome** thing is terrible or frightening; (p. 201)
Something that is **flat** is not sloping, curved, or pointed. (p. 210, sense 2)
People, jobs, or appearances that are **grand** seem important or socially superior. (p. 242, sense 3)
A **larch** is a tree with needle-shaped leaves. (p. 314)
A **misconceived** plan or method is the wrong one for a particular situation and is therefore not likely to succeed. (p. 356)
To **pitch** a tent means to erect it. (p. 419, sense 7)
-s and **-es** are added to nouns to form plurals. (p. 492, sense 1)

In most of these examples the hinge, though varying in form and implications, has a straightforward central position in a linear semantic equation and can be seen clearly as a component of the definition outside both the definiendum and the definiens. Some group A definitions deviate from this basic pattern, with important implications for the nature of the semantic information being provided. In the definitions of adjectives, for example, the most commonly encountered strategy is to use a pattern like that of sense 1 of 'abrupt':

An **abrupt** action is very sudden and often unpleasant. (p. 2)

At first sight this has the normal components described above, with a typical central hinge realised by 'is'. Now consider the definition of 'punishing':

A **punishing** experience makes you very weak or helpless. (p. 449)

It is not possible in this case to set up the usual equation:

A **punishing** experience = you very weak or helpless

with 'makes' as the hinge. In fact, no obvious candidate for the hinge is visible. On closer inspection, even the definition of 'abrupt' is more suspect than it seems. The equation:

An **abrupt** action = very sudden and often unpleasant

works no better than the equivalent statement for 'punishing', and the problem is the same. An element of the definiendum has not been repeated within the definiens, and without it the equation cannot work in the normal way. In

order to complete these equations, the missing element needs to be supplied by the reader of the definition. If the definitions were restated as:

> An **abrupt** action is one which is very sudden and often unpleasant

and

> A **punishing** experience is one which makes you very weak or helpless

they would produce completely viable equations. In both cases, what seemed likely to be the hinge for the definition now appears as part of D^s, and the hinge, like the repetition of the noun accompanying the adjective, is seen to be absent.

It is interesting to note that the corresponding definitions for these senses in the original *CCELD* are rather fuller:

> If an action, change, or ending is **abrupt**, it is sudden and perhaps surprising or unpleasant. (p. 5, sense 1)
> Something that is **punishing** makes you very weak or helpless. (p. 1165)

These original definition forms have been altered in *CCSD*, sometimes simply to save space, but sometimes to reflect the relative frequency of the attributive use of the adjective headword compared to its predicative use. The resulting structure also appears in *CCELD*, for example in the definition of 'disapproving':

> A **disapproving** action, expression, etc shows that you do not approve of something or someone.
>
> (*CCELD*, p. 397)

The use of structures like this, in which the hinge and other elements of the definition need to be supplied by the user, probably has little effect on the native speaker. The additional element in the restated versions of the two definitions shown above, 'is one which', adds no semantic information and probably contributes little to syntactic clarity. For a learner of the language, however, the effect may be more serious, and section 7.3.3 considers the implications of similar structural abbreviations.

The definition of sense 2 of 'flat' introduces a further complexity. The word 'is' appears twice, linking the two elements of the definition to the co-text 'something', but the co-text itself is not matched in the second part. It is, however, possible to expand the definition slightly so that a full match is provided:

> Something that is fl**at** is *something that is* not sloping, curved, or pointed.

The additional text shown in italics makes the sentence rather awkward and unnatural. It is implied in the original text, and its identification allows the elimination of matching elements to produce the lexicographic equation:

flat	=	not sloping, curved, or pointed

The equality operator in this equation is realised by the explicit hinge 'is' in the original definition sentence.

Sense 3 of 'grand' appears to follow the same pattern, but there is a crucial difference. The restatement process shown for 'flat' would produce the following definition:

People, jobs, or appearances that are **grand** are people, jobs, or appearances that seem important or socially superior.

The elimination of matching items leaves the equation:

are	=	seem important or
grand		socially superior

The lack of symmetry here is significant: to be grand is not to *be* important, but to *seem* important. In this case the hinge is implied rather than being present in the definition text.

6.2.2 Hinges in Group B definitions

In the following examples of definitions from group B, the equation uses the sequence already described in section 6.2, and has an initial hinge realised by 'if' or 'when':

> If you do something **on account of** something or someone, you do it because of that thing or person. (p. 5, phrases)
> When the weather is **fine,** it is sunny and not raining. (p. 206, sense 6)
> If someone or something **is geared** to a particular purpose, they are organized or designed to be suitable for it. (p. 230, sense 4)
> When criminals are sentenced to **life imprisonment,** they are sentenced to stay in prison for the rest of their lives or for a very long time. (p. 323)
> If a reaction is **muted,** it is not very strong. (p. 367, sense 2)
> If you say that you have found your **niche** in life, you mean that you have a job or position which is exactly right for you. (p. 376, sense 2)
> If a fact is made **public,** it becomes known to everyone rather than being kept secret. (p. 447, sense 8)

> When you **run,** you move quickly, leaving the ground during each stride. (p. 490, sense 1)

The examples above show some variation in the nature of the equations that they represent. For example, sense 1 of 'run' can be analysed into:

H	L			R
	C	D^m	M	D^s
When	you	**run,**	*you*	move quickly, leaving the ground during each stride.

Eliminating the hinge and matched co-text produces the equation:

run = move quickly, leaving the ground during each stride.

This is the typical semantic relationship, and similar considerations would apply to most of the other verbs which are defined using this strategy.

The definition of 'geared' is slightly more complex. A similar analysis to that used for 'run' would yield:

H	L			R		
	C	D^m	C	M	D^s	M
If	someone or something	**is geared**	to a particular purpose,	*they*	are organized or designed to be suitable	*for it.*

Both co-text elements in L, 'someone or something' and 'to a particular purpose', are matched in R, by 'they' and 'for it'. The use of the plural pronoun 'they' as a match for 'someone' is a feature of the definition language which is not universally accepted in Standard English, and it was specifically adopted by the compilers of the dictionaries to avoid the use of gender-specific singular pronouns. D^m and D^s both contain a further element — the verb 'to be' used to form the passive. The switch from singular to plural is caused, of course, by the use of 'they' already described. The small variation from the basic strategy has been adopted to highlight the normal usage of the verb headword, and again the lexicographic equation is straightforward:

is geared = are organized or designed to be suitable

The same analysis applied to the definition of 'life imprisonment' highlights a more problematic relationship:

H	L			R		
	C	C	D^m	M	M	D^s
When	criminals	are sentenced to	life imprisonment,	*they*	*are sentenced to*	stay in prison for the rest of *their* lives or for a very long time.

The major change in this definition compared to the previous two examples is that the headword is no longer the first verb in the sentence, but has shifted to a part of the adjunct to the verb. The phrase 'are sentenced to' in L is co-text, and is exactly matched in R. This generates the lexicographic equation:

life imprisonment	=	stay in prison for the rest of… lives or for a very long time.

This shows a further degree of complexity in this definition: the definition text appears to be no longer exactly substitutable for the headword element of the definition. In fact, the apparent matching of the word 'to' in the first and second parts hides a difference of meaning between the two instances of the word. In the first part it is a preposition, and in the second it is an infinitive marker. This difference in meaning extends back to the word 'sentenced', so that the equation becomes:

sentenced to life imprisonment	=	sentenced to stay in prison for the rest of… lives or for a very long time.

This raises questions about the limits of the definiendum in definitions which have similar structural properties, and the implications of these questions for the grammar and parser are explored in section 7.3.1.2.3.

6.2.3 Hinges in Group C definitions

In the following examples of definitions from group C, the hinges are rather more complex than in the two groups examined so far:

> People use **Your Excellency**, **His Excellency**, or **Excellency** to refer to or address important officials. (p. 187)
> You use **fabulous** to say how wonderful or impressive something is; (p. 194)

> You can refer to working-class people, especially industrial workers, as the **proletariat;** (p. 443)

There is no obvious hinge in these definitions, and the lexicographic equation is obscured by the complex relationship between their headwords and the other elements in the definition texts. Consider the definition of 'excellency', in which the group of headwords and its definiens are framed by a structure which comments directly on the usage of the headword:

People use...	to refer to or address...

To produce a form of lexicographic equation from this would need extensive restatement, which would collect the elements of this structure on the right hand side:

Your Excellency, His Excellency, or Excellency	=	something that people use to refer to or address important officials

This is rather like the form of the equations shown earlier for sense 2 of 'flat' and sense 3 of 'grand' in section 6.2.1, since some matching elements are implied rather than stated, and elements of the hinge structure remain in the equation, showing that they need to be taken into account as part of the relationship between the definiendum and its definiens.

6.3 The text surrounding the definiendum

The relationship between the definiendum and the other text in the left hand side of the definition has been dealt with at some length in section 6.1. It is now necessary to consider the other text elements within this part of the definition sentence. The first point to be made about these other elements is that they tend to be optional. The minimal L, obviously, consists only of the headword. Examples of such definitions are shown below:

> **Absolute** means total and complete. (p. 2, sense 1)
> **Abstinence** is the practice of not having something you enjoy, such as alcoholic drinks. (p. 2)
> **Costly** also describes things that take a lot of time or effort. (p. 118, sense 2)
> **Flying saucers** are round flat spacecraft from other planets, which some people say they have seen. (p. 213)
> **Lately** means recently. (p. 315)

> **Lentils** are dried seeds taken from a particular plant which are cooked and eaten. (p. 321)
> **Psychiatry** is the branch of medicine concerned with the treatment of mental illness. (p. 447)
> **Wild** is used to describe the weather or the sea when it is very stormy. (p. 647, sense 4)

Most, if not all, of these definitions read remarkably like the traditional lexicographic equation, with the addition of an explicit hinge, embedded in a full English sentence. In most of the definition sentences, however, even for words belonging to the same grammatical categories, other components are present within L. The following sections deal with the most common of them.

6.3.1 Operators

The previous section contained examples of definitions whose left hand sides contain only the headword. Roughly 4200 definitions have a similar pattern, and an analysis of their grammar codes shows that well over half — about 2300 — have headwords which are uncount, plural or mass nouns, while about another 300 are count nouns which tend to be used in the plural in the sense being explained. The grammar note, a feature shared with many traditional dictionaries, can provide information about normal usage, but unless the information is very straightforward the note is likely to become so complex as to be unhelpful to the average dictionary user. Consider the following definition examples and accompanying grammar notes, taken from different senses of ‘ material’ (p. 345):

> A **material** is a solid substance. COUNT N OR UNCOUNT N (sense 1)
> **Material** is cloth. MASS N (sense 2)
> **Materials** are the equipment or things that you need for a particular activity. PLURAL N (sense 3)

Without the need for detailed commentary, the form of the definition differentiates between these three possible manifestations of the headword and shows the normal usage for each sense. Hanks (1987, p. 117) refers to the advantages of this strategy in enabling non-native English speakers to grasp the distinction in usage between count and uncount nouns, especially where such a distinction does not exist in their own language. This component of the first part obviously needs to be treated as a separate element within the

grammar. As explained in detail earlier in section 5.3.2, the term used for it in the definition language grammar is 'operator'.

The set of articles forms an obvious part of the realisation of the operator, but they can also be realised by the word 'to' as an infinitive marker for verb headwords in type A6 definitions. The following examples show most of the possible realisations:

> To **accept** a difficult or unpleasant situation means to recognize that it cannot be changed. (p. 3, sense 4)
> A **doctor** is someone qualified in medicine who treats sick or injured people. (p. 159, sense 1)
> An **eagle** is a large bird that lives by eating small animals. (p. 168)
> The **mass media** are television, radio, and newspapers. (p. 344)

6.3.2 Co-text

The following definitions contain one element of co-text, italicised for ease of identification:

> **Appreciation** *of something* is recognition and enjoyment of its good qualities. (p. 23, sense 1)
> **Deep** *in an area* means a long way inside it. (p. 137, sense 3)
> **Fleshy** *leaves or stalks* are thick. (p. 211, sense 2)
> *Someone's* **life** is their state of being alive, or the period of time during which they are alive. (p. 323, sense 3)
> **Sheltered** *accommodation* is designed for old or handicapped people. (p. 516, sense 3)

The co-text in each of these definitions restricts the linguistic domain within which the sense operates by specifying its normal textual environment. Its detailed function varies between the examples but there is a general purpose. To understand the field of operation of the sense being defined, the user of the dictionary needs to be made aware of the nature and extent of any restrictions or tendencies affecting its normal usage. As an example, senses 1 and 2 of 'deep' have the following definitions:

> If something is **deep,** it extends a long way down from the surface.
> You use **deep** to talk about measurements. (p. 137, senses 1 and 2)

The main reason for the difference in meanings between these two senses and sense 3 is that the rather more specialised meaning described by sense 3

applies only or mainly in the context of the phrase, 'in an area' or other similar phrases.

This differentiation is also provided by more traditional dictionaries, but their definition structure provides less scope for setting the definiendum in its normal environment. As an example, consider sense 1 of 'appreciation' in *LDOCE*:

> understanding of the good qualities or worth of something
>
> (*LDOCE*, p. 41)

Although this contains almost the same elements as the Cobuild version, they are arranged differently. The words 'of something', set in the definiens in the *LDOCE* definition, are placed next to the definiendum 'appreciation' in Cobuild to show the typical text structures into which the headword normally fits. The traditional treatment used in *LDOCE* does not convey this typical environment so clearly. The matching co-text or framework element 'its' in the right hand side of the Cobuild definition is the exact equivalent of the *LDOCE* definiens element, but the use in the Cobuild version of anaphoric reference to the co-text in the left hand side produces a completely clear and symmetrical account of the meaning of this sense of 'appreciation'.

The *CCSD* definitions shown above have only one co-text element, but many have two or more. To allow multiple co-text elements to be identified satisfactorily for description and analysis they have been labelled in the parser output with a description of their function within the definition sentence which depends on the type to which they belong. This approach is rather different from the conventions used for the ET–10/51 project (see section 7.6.2), described in Barnbrook & Sinclair (1995), which uses a sequential numbering system. A further deviation from that convention is the replacement of the label 'co-text 0', used to mark general linguistic restrictions sometimes placed on a sense in an additional note preceding the definition text proper, by the label 'usage note'. As described in section 4.2.2.1, these notes were identified and isolated during pre-processing, before the separation of the definitions into their typed groups, and this element is therefore independent of definition type.

6.4 Projection

The definition for sense 1 of 'bitch' in *CCSD* (p. 49) is:

> If you call a woman a **bitch,** you mean that she behaves in a very unpleasant way;

There is a significant difference between this form of definition and that used by more traditional dictionaries. *LDOCE* (p. 93, entry 1 of bitch, sense 2) has:

> *derog* a woman, esp. when unkind or bad-tempered

and *OALDCE* (p. 109, sense 2(a)) has:

> *sl derog* spiteful woman

Both dictionaries, of course, also have examples of usage, and the abbreviated note at the beginning of each entry gives some indication of the normal context of this sense of the word. But if we rewrite these definitions using an appropriate full sentence strategy, we would probably get something like:

> A **bitch** is a woman, especially one who is unkind or bad-tempered.

and

> A **bitch** is a spiteful woman.

Neither of these is the real equivalent of the cited Cobuild definition. In order for them to become its equivalent the Cobuild definition would need to be rewritten as:

> A **bitch** is a woman who behaves in a very unpleasant way;

This has now lost an essential part of the original definition. The dictionary does not claim that there is an equality of the normal sort between the definiendum 'bitch' and this reconstituted definiens 'woman who behaves in a very unpleasant way'. Instead it claims an equality between something that you might say, and what you would mean by it. This explicitly metalinguistic element in the definition is not strictly part of the traditional definiendum and definiens. It is probably most usefully considered as a modification of the hinge, of the nature of the relationship between them. Because of its complexity, however, and because of the existence in many cases, as in the example quoted above, of a normal hinge in addition to the explicitly metalinguistic structure, it seems best to deal with it separately from the point of view of both terminology and analysis.

These metalinguistic structures can be considered in relation to the two major categories identified in Sinclair (1991, p. 126) in his examination of variation in co-text: those which are about the word itself, and those which are about what people mean when they use it. To some extent these correspond to neutral metalinguistic statements, at one end of the scale, and those which describe an inherently subjective use of the word in the second. As examples, consider the following definitions:

> If you say that a child or animal is **adorable,** you feel great affection for them. (p. 9)
> If you call someone a **fascist,** you mean that their opinions are very right-wing. (p. 199, sense 2)
> If you call a business a **goldmine,** you mean that it produces large profits. (p. 240)
> You can use **mug** to refer to a mug and its contents, or to the contents only. (p. 366, sense 2)
> You use **naked** to describe behaviour or strong emotions which are not hidden in any way. (p. 369, sense 3)

The definition for sense 2 of ‘ mug’ is neutral metalinguistic comment, while that for sense 2 of ‘fascist’ is almost entirely subjective. The other definitions perhaps lie somewhere between these two extremes. If the parser is to make this information available from the definitions, these metalinguistic structures need to be identified and properly distinguished from each other.

A limited range of phrases realises the metalinguistic structure, often using traditional reporting verbs such as ‘call’, ‘describe’, ‘say’, ‘refer’ and so on. Following Halliday, the term ‘projection’ was suggested by Sinclair during the ET–10/51 project (Barnbrook & Sinclair, 1995, p. 9). Halliday (1985, p. 196), in his consideration of logico-semantic relations between clauses, distinguishes two fundamental groups of relationships: expansion of the primary clause by the secondary, and projection of the secondary clause through the primary. This provides a usefully general description of the structures within these definitions.

6.5 The right hand side

The complexity and richness of the definiendum and its surrounding text, detailed above, is the hallmark of the Cobuild definition style. As Hanks points out (1987, p. 118):

> 'In general, then, the first part of each Cobuild definition shows the use, while the second part shows the meaning.'

This suggests that the right hand side, part of which corresponds to the definiens, should represent less of a departure from traditional lexicography. To some extent this is true, but there are elements within it which are influenced by the demands made on the first part and the methods adopted to satisfy them. Consider the following definitions:

> A **dyke** is a thick wall that prevents water flooding onto land from a river or from the sea. (p. 168)
> **Mathematics** is a subject which involves the study of numbers, quantities, or shapes. (p. 345)
> A **slander** is an untrue spoken statement about someone which is intended to damage their reputation. (p. 527, sense 1)

The second part of the definition in each case is almost pure traditional definiens. Comparing these examples with their corresponding definitions in other dictionaries, *LDOCE* has:

> a wall or bank built to keep back water and prevent flooding (p. 285, dike entry 1, sense 1)
> the science of numbers and of the structure and measurement of shapes, including algebra and geometry as well as arithmetic (p. 645)
> an intentional false spoken report, story, etc., which unfairly damages the good opinion held about a person by others (p. 987, entry 1, sense 1)

and *OALDCE* has:

> long wall of earth, etc (to keep back water and prevent flooding) (p. 335, dike sense 2)
> science of numbers, quantity and space, of which eg arithmetic, algebra, trigonometry and geometry are branches (p. 768)
> (offence of making a) false statement intended to damage sb's reputation (p. 1196)

While there are variations in the amount of information given, the structures of these definientia correspond quite closely to the second parts of the Cobuild definitions. *OALDCE*, interestingly, omits articles from the start of its definition even where the nouns used in them would typically take an article, while *LDOCE* and Cobuild definitions omit or include them in accordance with normal English usage. In the *CCSD* definitions of 'dyke' and 'slander', the operator 'a' in the definiendum is matched by a corresponding article in the

definiens. In the case of 'mathematics', however, there is no article in the definiendum, since 'mathematics' is an uncountable noun, but since it is being explained by the use of a countable noun, 'subject', there is a non-matching article in the second part. This may be useful in applying the parser output to natural language processing problems, but it also illustrates an important feature of the Cobuild definiens which is rarely present in traditional dictionaries.

The next section deals with the significance of matched and unmatched items in the right hand side of the definition, while section 6.5.2 deals considers the detailed analysis of the definiens.

6.5.1 Matched and unmatched items

The minimal form of the definiendum in the definition sentences is the marked headword. The traditional definiens is normally regarded as a potential substitute for the headword, at least in less complex definitions. Any additional components contained in the first part of the definition will need to be repeated in the definiens in some way, unless, regardless of marking, they actually form part of the definiendum. If items in the second part can be matched with those in the first part, it should be possible to analyse the implications of any such unmatched components.

The following definitions contain additional co-textual elements in the definiendum:

> A woman's **cleavage** is the space between her breasts. (p. 91, sense 1)
> Your **descent** is your family's origins. (p. 143, sense 2)
> If a company **launches** a new product, it starts to make it available to the public. (p. 315, sense 4)
> A **slab** of something is a thick, flat piece of it. (p. 527)

If the hinges and the matching elements in the second parts of the definitions are removed, this would leave the following equivalences between headwords and definientia:

cleavage	=	space between breasts
descent	=	family's origins
launches	=	starts to make available to the public
slab	=	thick flat piece

These look remarkably like the traditional definitions encountered in the other dictionaries, and the parsing needs of the second part of the definition perhaps now become clearer. Matching elements from the first part of the definition need to be identified, and the remaining components of the definiens need to be analysed according to their functions in the definition of meaning.

There is, however, one more consideration. As already noted above all items in the first part should be matched directly in the second part of the definition unless they form part of the definiendum. The following definitions have co-text elements in the first part which are not matched in the second part:

> If you behave with **aggression,** you behave angrily or violently towards someone. (p. 12)
> A tailor's **dummy** is a model of a person that is used to display clothes. (p. 166, sense 2)
> If you give someone a **lift,** you drive them in your car from one place to another. (p. 324, sense 3)

If the hinges and matching elements are removed from these definitions, they reduce to:

with aggression	=	angrily or violently towards someone
tailor's dummy	=	model of a person that is used to display clothes.
give a lift	=	drive in car from one place to another

There are certainly some problems with the rather telegraphic style of these newly stripped down definitions, but they are not unlike traditional lexicographic language. There is, perhaps, rather more of a problem with the first of these examples: the residual phrase 'towards someone' does not seem to fit as part of the definition of 'with aggression', and it may be that this matching process has highlighted a problem within this definition. Consider the rewritten version:

> If you behave with **aggression** towards someone, you behave angrily or violently towards them.

In this case the matching process would work perfectly, and it would look rather more like the standard form of similar definitions. This ability of the parsing process to identify potential problems or anomalies in the construction of definitions is dealt with in detail in section 7.7.1.

6.5.2 The analysis of the definiens

Once the matched items are stripped out, what we are left with from the second half can be thought of, as in the examples above, as the 'true' definiens, the text in the second part used to explain the meaning of the definiendum extracted from the first part. We now need to consider the components of this text, and the level of detail to which they need to be analysed. The definition of meaning in the dictionary is achieved in a variety of ways, depending on the complexity and individual requirements of the headword, but there is a fairly typical pattern which works for many of the more straightforward definition strategies. It can best be introduced by considering the typical noun definition form.

6.5.2.1 *Explaining the meanings of nouns*

Most of the nouns are explained using a variant of the form exemplified by:

> A **shadow** is a dark shape made when something prevents light from reaching a surface. (p. 513, sense 1)

Stripping away the hinge and matching article, the text which explains the meaning of sense 1 of 'shadow' is:

> dark shape made when something prevents light from reaching a surface

As has already been described in section 6.1, this can be broken down into:

(d^r) S (d^r)

This represents the lexical superordinate of the definiendum, with optional discriminators that specify the member of the superordinate class being dealt with. Using the label D^{r1} for the discriminator preceding the superordinate and D^{r2} for the following discriminator, a primary analysis of the definiens of 'shadow' would consist of:

D^{r1}	S	D^{r2}
dark	shape	made when something prevents light from reaching a surface

It might also be useful to be able to subdivide discriminators. Those which precede the superordinate will tend to have different characteristics from those that follow it, and will tend to be less complex. Following discriminators,

as can be seen in this example, and as described in Sinclair's original 'chunking' process (Sinclair, 1991, p. 124) might also be capable of recursive analysis into smaller sub-units. The main factors involved in this subdivision are considered in section 6.6.3.

6.5.2.2 *Verb definitions*

The concept of the superordinate and discriminator is also useful in the analysis of the definitions of verbs, although it is by no means the only strategy used for them in Cobuild definitions. The following definitions can easily be analysed on the basis used for nouns in the previous section:

> If someone **abducts** another person, they take the person away illegally. (p. 1)
> If someone or something **displeases** you, they make you dissatisfied, annoyed, or upset. (p. 154)
> If someone **lashes** you, they hit you with a whip. (p. 314, sense 4)
> If you **rush** something, you do it in a hurry. (p. 492, sense 3)

The definiens for each of these can be analysed into:

S	D^{r2}
take	away illegally
make	dissatisfied, annoyed or upset
hit	with a whip
do	in a hurry

The realisation of the discriminators is obviously rather different from the equivalent realisations for nouns, and there seems to be no equivalent of D^{r1}, but the model is useful for describing definitions which use this simple pattern. A potential problem, already visible in the selected examples, lies in the nature of the superordinates identified by this process. The words 'make' and 'do' have little real lexical content in these usages, and it might be thought preferable to group together the superordinate with the discriminator in these cases and use the whole unit as a phrasal synonym. However, if the analysis is carried out at the level of detail shown above, larger groupings could be recovered as desired from the parsed output.

6.5.2.3 *Adjectives*

One widely-used definition structure for adjectives is shown in the following examples:

> An **able** person is clever or good at doing something. (p. 1, sense 2)
> A **ferocious** animal, person, or action is fierce and violent. (p. 202)
> **Mild** weather is less cold than usual. (p. 353, sense 3)
> **Virtuous** behaviour is morally correct. (p. 631)

These could be analysed on the same basis as the noun pattern into:

D^{r1}	S	D^{r2}
	clever or good	at doing something
	fierce and violent	
less	cold	than usual
morally	correct	

These results may seem a little odd, especially in the case of 'mild', whose superordinate seems to be 'cold'. In fact, as has already been described in section 6.2.1 in an examination of the hinge, these definitions all have structures in which the hinge, together with the repetition of part of the co-text, is implied rather than actually realised in the second part of the definition. In terms of the restated structure described in 6.2.1, these definitions would be expanded to:

> An **able** person is (one who is) clever or good at doing something.
> A **ferocious** animal, person, or action is (one which is) fierce and violent.
> **Mild** weather is (weather which is) less cold than usual.
> **Virtuous** behaviour is (behaviour which is) morally correct.

An analysis of these expansions on the basis of the noun superordinate discriminator model would be more straightforward, but would not apply to the headword alone. It would apply to the combination of the headword and the co-text, as in a possible analysis for 'able':

S	D^{r2}
one	who is clever or good at doing something

This is obviously less useful, although it appears syntactically to be more correct. The essential problem with adjectives lies in the way in which they are used in English. By their nature, adjectives normally refer to nouns and differentiate them in some way from other examples of the same noun which do not have the same qualities. The example quoted for sense 2 of 'able' in *CCSD* is:

> *He was an unusually able detective*

In this sentence, 'able' differentiates between this example of a detective and others whose abilities are less well-developed. It is, in other words, a discriminator, and performs the same function in the definition text.

The problem with the analysis of the expanded definitions, then, is that the wrong elements of the definition text are being analysed. The expansion is a useful way of identifying the underlying structure of the definition, with its omission of the hinge and some co-text repetition, but for a functional analysis of the definition only the elements which explain the meaning of the headword should be considered. In some cases the superordinate and discriminator model will need to be replaced with something more suited to the adjective's linguistic behaviour.

6.5.2.4 *Other models of definition*

In the superordinate and discriminator model described in the preceding sections the definiens contains a set of words which can be isolated and, subject to minor inflectional changes, substituted for the definiendum. This meets the traditional lexicographical requirement of substitutability described by Hanks (1987, p. 119). As Hanks points out, however, this requirement stems from a formalism imposed on lexicography from philosophy, and may not produce the most useful definitions of the meanings of words. Some problems with substitutability are inherent in the adjective definitions referred to in section 6.5.2.3, but these can be removed by expanding them to create the hinge and co-text repetition which has been omitted. Other definition structures give rise to more intractable problems. In the definition of sense 1 of 'one-way':

> **One-way** streets are streets along which vehicles can drive in only one direction. (p. 388)

there is a perfectly good hinge, 'are', and the co-text 'streets' is faithfully repeated, but the expressions 'one-way' and 'along which vehicles can drive in

only one direction' are not substitutable for each other in the same position in a sentence. This is, of course, only a problem of English syntax, and the meaning for the human user should be clear from the definition. The syntactic problem may, however, not be trivial for a natural language processing application making use of the parsed output, and the parsed output could draw attention to the general problem involved in the definition of an adjective which forms a preceding discriminator by a phrase used as a following discriminator.

Schnelle (1995, section 2) suggests a fundamental change in the method of definition which, among other things, would remove the problems which appear to beset definitions like these. He proposes that, for the purposes of automatic analysis, all the explanations could be rearranged to convert them to the structure found in Group B of the taxonomy. Sense 3 of 'account', a type B3 definition, shows the basic pattern:

> If you have an **account** with a bank, you leave money with it and withdraw it when you need it. (p. 4, sense 3)

Schnelle argues that this form of definition, with its 'if… then' structure, operates according to 'the rules of sentential logic (propositional logic, predicate logic and their derivatives)' rather than the term logic which applies to definitions of the form:

> A **geranium** is a plant with small red, pink, or white flowers, often grown in houses. (p. 232)

The advantages of this transformation are based on the argument that 'sentential logic is much better understood than term logic', and therefore allows more straightforward analysis of interdependency between related definitions. In his description of the restructuring of definitions to fit the sentential logic format, he also briefly mentions the possibility of transforming 'some unorthodox explanations used in Cobuild' (Schnelle, 1995, section 2).

Applying this idea to the definition of sense 1 of 'one-way' would produce the 'if… then' form:

> If a street is **one-way**, vehicles can drive along it in only one direction.

Eliminating the hinge and matching items from this produces the equation:

> is one-way = vehicles can drive along… in only one direction

Many definitions already use this 'if… then' format for types of headwords which are more commonly defined using a Group A strategy. The definition for sense 1 of 'wry' provides an illustration:

> If someone has a **wry** expression, it shows that they find a bad or difficult situation slightly amusing or ironic. (p. 656)

In the second part of the definition, the subject of the clause forming the definiens has changed from 'someone' to 'a wry expression', and the adjective being explained, 'wry', is not simply paraphrased but described in terms of what an expression with that quality does. This strategy has presumably been used because alternatives did not work. Consider the alternative Group A format:

> A **wry** expression shows that someone finds a bad or difficult situation slightly amusing or ironic.

This does something like the same thing, but is probably not sufficiently explicit about the relationship between the expression and the person referred to as 'someone'. When we try to make the relationship explicit, as in:

> A **wry** expression on someone shows that they find a bad or difficult situation slightly amusing or ironic.

the definition becomes rather unnatural, and probably needs to be slightly expanded to:

> A **wry** expression on someone's face shows that they find a bad or difficult situation slightly amusing or ironic.

Similar considerations have led to other definitions being constructed in similarly asymmetrical ways. Their treatment in the grammar is rather limited. Instead of the decomposition of D^s into the D^{r1} S D^{r2} structure already described in sections 6.5.2.1 to 6.5.2.3, it is left intact and treated as a single structure called the 'explanation'. In the case of sense 1 of 'wry', this would be identified as the following text:

> *it* shows that *they* find a bad or difficult situation slightly amusing or ironic.

The italicised words 'it' and 'they' in this text are the framework elements which match the co-texts 'expression' and 'someone'. Eliminating them from the text allows the definiens proper to emerge from the right hand side of the definition sentence:

> shows that… find a bad or difficult situation slightly amusing or ironic.

The further analysis of these explanation structures is problematic, but it is a problem shared with more traditional lexicographic approaches, as shown by the following extracts from the *LDOCE* and *OALDCE* entries:

> (esp. of an expression on the face) showing a mixture of amusement and displeasure, dislike, or disbelief
>
> (*LDOCE*, p. 1222)

> **1** (of a person's face, features, etc.) twisted into an expression of disappointment, disgust or mockery: **2** ironically humorous; slightly mocking
>
> (*OALDCE*, p. 1482)

Both of these bring in the restricted application to a facial expression, and neither successfully produces a substitute for 'wry'. Adequate analysis of the explanation element in Cobuild definition sentences could perhaps be achieved more efficiently using a general language grammar, such as the one described in section 6.6.3.2.4.

6.6 Complex elements

The descriptions of the various elements of the definition sublanguage grammar already given in sections 6.1 to 6.5 do not deal with all of the complexities of structure that can arise within these elements. To some extent, these complexities are more properly dealt with under the description of the parser in sections 6.8 to 6.10 below, but it is useful to consider the range of variation within the main components as part of the definition language grammar.

6.6.1 Headwords

There are two common types of complexity within the headwords of definitions. The first is easily dealt with: most headwords are single words, as in:

> A **capricious** person often changes their mind unexpectedly.
>
> (p. 74)

This is not always the case. In some cases the basic lexical unit is a phrase rather than a word, and this must be recognised within the definition, as it is in the case of 'credit card':

> A **credit card** is a plastic card that you use to buy goods on credit or to borrow money.
>
> (p. 123)

This is a small complication, easily dealt with both theoretically and practically. More difficult are the definitions which deal with alternative lexical units, the most extreme example of which is shown by the phrasal definition given under sense 1 of 'bore':

> If something **bores** you **to tears, bores** you **to death,** or **bores** you **stiff,** it bores you very much indeed;
>
> (p. 56)

Definitions like this are given special treatment during the extraction process, documented earlier in section 4.2.2.2, and cause minor practical problems during the parsing process, described in section 6.10.2.2.1. From the point of view of the grammar, it is important to recognise that the co-text element 'you', common to all three alternatives, is embedded in the definienda, which can be reduced to:

> bores… to tears
> bores… to death
> bores… stiff

On the right hand side of the definition, the matching element 'you' is, of course, realised once only.

6.6.2 Superordinates

There are two main potential problem areas within the superordinate element of the definiens: the presence of alternatives and the treatment of superordinates containing the word 'of', which can be thought of as complex superordinates capable of further analysis.

6.6.2.1 *Alternative superordinates*

The superordinate can be made up from alternative elements, as for example in sense 1 of 'substance':

> A **substance** is a solid, powder, or liquid. (p. 565, sense 1)

This causes few, if any, problems: the entire set of alternative superordinates can be taken as a unit and subdivided as necessary using the commas and the word 'or'. The following definitions are rather more problematic:

> A **tower** is a tall, narrow building, or a tall part of a building such as a castle or church. (p. 599, sense 1)
> A **waterway** is a canal, river, or narrow channel of sea which boats can sail along. (p. 638)
> A **youth** is a boy or a young man, especially a teenager. (p. 658, sense 3)

The right hand sides of these definitions could be analysed as:

O	Dr^1	S	Dr^2
a or a	tall, narrow tall	building, part of a building	such as a castle or church.
a or	 narrow	canal, river, channel of sea	 which boats can sail along.
a or a	 young	boy man,	especially a teenager.

These textual complexities may cause difficulties for the parsing software, but these can be overcome. More problematic is the difficulty of establishing the scope of operation of the discriminators. The application of the D^{r1} element is generally straightforward, but it is difficult to be sure whether 'such as a castle or church' in the definition of 'tower' applies to both 'building' and 'part of a building'. The same is true of D^{r2} in the other two examples. This is a problem for the grammar and the parser, but is likely to cause more significant difficulties for the user of the dictionary. The embedding of the D^{r1} elements 'tall', 'narrow' and 'young' within the superordinate groups in the three examples could also cause confusion to the learner of the language, although they are relatively clear for the grammar.

6.6.2.2 *The complex superordinate*

The boundary between the superordinate and the D^{r2} element is generally quite clear, despite the relatively large number of words which can form this boundary (already mentioned in section 4.4.1, and dealt with in more detail later in section 7.3.3). During the early stages of the parser's development, described in section 4.4.1, the word 'of' was one of these boundary words. As

work progressed it became obvious that this was not necessarily appropriate. In the definitions below, the use of 'of' as a boundary would produce rather empty superordinates:

> An **academic** is a member of a university or college who teaches or does research. (p. 3, sense 3)
> The **admission** fee is the amount of money you pay to enter a place. (p. 8, sense 2)
> An **aerial** is a piece of wire that receives television or radio signals; (p. 10, sense 3)
> **Antics** are funny, silly or unusual ways of behaving. (p. 21)
> **Variety** is a type of entertainment including many different kinds of acts in the same show. (p. 626, sense 4)
> A **vigil** is a period of time when you remain quietly in a place, especially at night, for example because you are praying or are making a political protest. (p. 630)

The superordinates of these definitions would be 'member', 'amount', 'piece', 'ways', 'type' and 'period', none of which is sufficiently specific to be a useful superordinate. The phrases which are produced by ignoring the word 'of' seem more useful and informative:

> member of a university or college
> amount of money
> piece of wire
> ways of behaving
> type of entertainment
> period of time

The decision is not, however, completely straightforward. The analysis of the following definitions would probably be improved by treating 'of' as a boundary word:

> **Veneer** is a thin layer of wood or plastic which is used to improve the appearance of something. (p. 627, sense 2)
> A **waxwork** is a model of a famous person, made out of wax. (p. 638)
> **Windsurfing** i s the sport of riding on a windsurfer. (p. 648)
> **Woodworm** are the larvae of a particular type of beetle, which make holes in wood by feeding on it. (p. 651)

The distinction between the two sets of examples is not easily made using the pattern-matching techniques generally adopted for the parser. The grammar needs to account for both possible structural interpretations, and the resolution of the analysis of a specific definition may need to rely on the first element of the superordinate, such as 'member', 'amount', 'period', 'model', 'larvae' etc., together with the presence of 'of' and the nature of the following words.

The identification and interpretation of these words has already been considered in other areas of research. The first words mentioned above — 'member', 'amount', 'piece' etc. — belong to the class of words labelled 'subtechnical' vocabulary in general linguistics, and they seem to have much in common with the words which make up Winter's 'Vocabulary 3' (Winter, 1977, pp. 18–22). Winter contrasts the 'closed-system' Vocabulary 3 words with 'open-system' words in terms of their 'stages of reference':

> The open-system words refer to their items in the real world, which may be seen or unseen; Vocabulary 3 words refer to their open-system words in the utterance. These open-system words must be there; they can be explicit or implicit (e.g., deletions can be put back into the clause). The open-system words look directly at the world; Vocabulary 3 words look only at their open-system words. Each gets their meaning from what they refer to. Vocabulary 3 could perhaps be regarded as a natural metalanguage for the open-system words.
>
> (p. 88)

Winter's summary of these words as 'a natural metalanguage' coincides perfectly with the communicative purpose of the definition sublanguage.

The appropriate analysis of the superordinates described above depends on the identification of this metalinguistic vocabulary within the sublanguage and the use of context to disambiguate the structural effect of the word 'of'.

6.6.3 Discriminators

Both the D^{r1} and D^{r2} elements in definitions which follow the superordinate and discriminator model can consist of more than one logical unit. A full analysis of the definitions for natural language processing applications should be capable of extracting these units individually. The rather different considerations involved in achieving this analysis for the two types of discriminator are dealt with in the next two sections.

6.6.3.1 *Discriminators preceding the superordinate*

The following definitions contain more than one D^{r1} element:

> A **balloon** is a small, thin, rubber bag that you blow air into so that it becomes larger. (p. 37, sense 1)
> A **citrus fruit** is a juicy, sharp-tasting fruit such as an orange, lemon, or grapefruit. (p. 88)
> A **grimace** is a twisted, ugly expression on your face that shows you are displeased, disgusted, or in pain. (p. 245)

> **Jet** is a hard black stone that is used in jewellery. (p. 304, sense 4)
> A **kangaroo** is a large Australian animal which moves by jumping on its back legs. (p. 307)
> **Porridge** is a thick, sticky food made from oats cooked in water. (p. 429)
> **Rags** are old, torn clothes. (p. 456, sense 2)

In all the above examples the elements of D^{r1} form a simple list of shared properties combined in such a way that they all restrict their superordinates in the same way. In many of them these elements are separated by commas, but this is not an essential structural feature. The following examples show a slightly more complex organisation:

> A **gulf** is also a very large bay. (p. 248, sense 2)
> **Luxury** is very great comfort among beautiful and expensive surroundings. (p. 336, sense 1)
> A **pamphlet** is a very thin book with a paper cover, which gives information about something. (p. 402)

In these examples the element 'very' applies to the second D^{r1} element rather than to the superordinate, and needs to be treated differently. In a general grammatical model it could be called a 'submodifier' or something similar. The parser does not identify this component separately, but further analysis of the D^{r1} element to isolate this and similar items would be a straightforward process in the interpretation of parsed output for a specific natural language processing system.

6.6.3.2 *Discriminators following the superordinate*

The structure of the D^{r2} element is significantly more complex and correspondingly more difficult to analyse. The following examples of type A1 definitions illustrate the main problems:

> **Dawn** is the time of day when light first appears in the sky, before the sun rises. (p. 133, sense 1)
> A **fruit machine** is a machine used for gambling which pays out money when you get a particular pattern of symbols on a screen; (p. 224)
> A **light** is anything that produces light, especially an electric bulb. (p. 324, sense 2)
> **Socialism** is the belief that the state should own industries on behalf of the people and that everyone should be equal. (p. 534)

The discriminators following the superordinates in these definitions, the D^{r2} elements, are:

> when light first appears in the sky, before the sun rises
> used for gambling which pays out money when you get a particular pattern of symbols on a screen
> that produces light, especially an electric bulb
> that the state should own industries on behalf of the people and that everyone should be equal

As already referred to in section 6.5.2.1, Sinclair (1991, p. 124) provides a general description of the analysis of the second part of the definition sentence, which he refers to as the 'comment':

> Comments are sometimes divisible according to the surface syntax. This is called *chunking*; in this kind of sentence, successive chunks express gradually increasing depth of detail.

The application of this process to the D^{r2} elements of definitions involves three main considerations: the identification of chunk boundaries within the D^{r2} elements, dealt with in section 6.6.3.2.1, the assessment of their scope of reference within the definiens, dealt with in section 6.6.3.2.2, and the problem of conjuncts and disjuncts within D^{r2}, dealt with in section 6.6.3.2.3. Section 6.6.3.2.4 describes a general language grammar which could be useful in the description and interpretation of D^{r2} structures.

6.6.3.2.1 *Identification of chunk boundaries*

The subdivision of D^{r2} elements into chunks is based on similar considerations to those used in the original identification of the D^{r2} boundary. The words which are used to find the beginning of the D^{r2} element can also be used as chunk boundary markers, taking conjuncts and disjuncts into account as appropriate. Applying this principle to the examples shown in section 6.6.3.2 above would produce the following analysis:

Chunks			
1	**2**	**3**	**4**
when light first appears	in the sky,	before the sun rises	
used for gambling	which pays out money	when you get a particular pattern of symbols	on a screen;
that produces light,	especially an electric bulb		
that the state should own industries	on behalf of the people	and that everyone should be equal	

There are some obvious problems with this very simple analysis. In the first place, the scope of reference of chunk 2 of the first item in the table, 'in the sky', relates to chunk 1, 'when light first appears', whereas chunk 3 of the same item, 'before the sun rises', applies to the superordinate 'time of day'. This is discussed in detail in the next section. The second major problem concerns the extraction of information from the chunks. They have a wide range of possible structures which do not conform to the restricted patterns found in the other components of the definitions. While it is a relatively simple matter to identify the chunks on the basis of a limited number of boundary words, their interpretation is much more complex. Also, because the rules governing their structure are not specific to the definition sublanguage, this part of the analysis process could perhaps be dealt with more efficiently by a general language grammar. A potentially suitable grammar is considered in section 6.6.3.2.4.

6.6.3.2.2 *The scope of reference of the chunks*

The analysis given in the table below shows the scope of reference of each of the chunks identified in section 6.6.3.2.1 above (using 'S' for the superordinate and numbers for each of the chunks):

Chunks			
1	**2**	**3**	**4**
when light first appears (S)	in the sky, (1)	before the sun rises (S)	
used for gambling (S)	which pays out money (S)	when you get a particular pattern of symbols (2)	on a screen; (3)
that produces light, (S)	especially an electric bulb (S)		
that the state should own industries (S)	on behalf of the people (1)	and that everyone should be equal (S)	

This shows that there is significant nesting of chunks within the D^{r2} element. An extreme example of nesting is shown in sense 1 of 'telephone':

> The **telephone** is an electrical system used to talk to someone in another place by dialling a number on a piece of equipment and speaking into it. (p. 582)

The scope of reference of the chunks of D^{r2} can be shown as:

Chunk						
1	**2**	**3**	**4**	**5**	**6**	**7**
used to talk (**S**)	to someone (**1**)	in another place (**2**)	by dialling a number (**1**)	on a piece of equipment (**4**)	and speaking (**1**)	into it (**6**)

The automatic analysis of these structures is problematic, although a starting-point could be made by considering a higher level unit which consists of groups of chunks with the highest scope of reference, referring directly to the superordinate. In the table below these are collected together for the examples shown in section 6.6.3.2.1.

Multi-chunk Unit	
A	**B**
when light first appears in the sky,	before the sun rises.
used for gambling	which pays out money when you get a particular pattern of symbols on a screen;
that produces light,	especially an electric bulb.
that the state should own industries on behalf of the people	and that everyone should be equal.

If the boundary markers which delimit these multi-chunk units could be identified, this initial grouping could be used as the basis for a full analysis of scope of reference and chunk hierarchy. Again, this process might be more efficiently dealt with by a general language grammar, and this is discussed in more detail in section 6.6.3.2.4.

6.6.3.2.3 *Conjuncts and disjuncts*

A further problem in the analysis of D^{r2} is shown in the examples below:

> An **accent** is also a mark written above or below certain letters in some languages to show how they are pronounced. (p. 3, sense 2)
> **Accommodation** is a room or building to stay in, work in, or live in. (p. 4)
> The **country** is land away from towns and cities. (p. , sense 3)

> **Depression** is a mental state in which someone feels unhappy and has no energy or enthusiasm. (p. 143, sense 1)
> A wildlife **sanctuary** is a place where birds or animals are protected and allowed to live freely. (p. , sense 2)

Each of the conjuncts and disjuncts in the D^{r2} elements of these definitions creates a branched structure which needs to be analysed properly so that information can be extracted correctly. In sense 1 of 'accent' the structure can be shown in the following table:

written	above or below	certain letters in some languages to show how they are pronounced

The branch shown in the middle section of this structure effectively creates two separate chunks which are linked by the disjunct 'or':

> written above certain letters
> or
> written below certain letters

Each of these chunks can then be used with the following chunks to create two D^{r2} elements:

> written above certain letters in some languages to show how they are pronounced
> written below certain letters in some languages to show how they are pronounced

These expanded D^{r2} elements can be easily recovered from the structure shown in the table above.

The same approach can be used to deal with conjuncts. In sense 1 of 'depression' the structure becomes:

in which someone	feels unhappy and has no energy or enthusiasm

A slightly more complex problem is shown by sense 2 of 'sanctuary', but this can also be dealt with in the same way:

where	birds or animals	are	protected and allowed to live freely

In all these cases, the analysis can be performed by including the conjunct or disjunct as a component of the appropriate chunk of the D^{r2} element. In order to do this, its scope of reference must be properly assessed, and once again this is more likely to be achieved using a general language grammar, such as the one described in section 6.6.3.2.4.

6.6.3.2.4 *The use of a general language grammar for further analysis*
Brazil (1995) describes a 'grammar of communication' (p. 2) which deals with 'used speech as purpose-driven activity' (p. 21), and which thus contrasts with 'sentence-oriented grammars'. He sets out to show that Chomsky's contention (in Chomsky, 1957) that finite state grammars cannot account for the sentences of a natural language, does not apply to 'purpose-driven language' (pp. 20–21).

The grammar that Brazil proposes uses a concept which he calls 'communicative need', effectively the requirements of the participants in the interaction. Although his grammar sets out to deal specifically with speech, the well-defined communicative needs of the definition sentences should allow the same principles to be applied in the analysis of the more complex, less tightly structured parts of definition sentences, such as the following discriminator or the explanation.

Brazil's grammar is incremental (p. 39), and the 'telling increment' and 'asking increment' are both independent of the notion of the sentence. These increments, arranged in a chain which allows the participants to move from an 'initial state' to a 'target state', through an 'intermediate state' (pp. 47–48), can also be seen in the chunks of the D^{r2} elements described in section 6.6.3.2.1. Brazil refers to the basic units, similar to the chunks described above, as 'elements' (p. 47) and recognises the possibility of the elaboration of the basic three-element chain through the concept of extensions (p. 58). He also deals in detail with the further analysis of the basic units, still on the basis of the purpose of the communicative process, arriving at a complete, almost word by word analysis (e.g. pp. 215–218). The relative simplicity of this grammar, its linear nature and the fact that it is founded on communicative need rather than more abstract and formal linguistic concepts should make it

eminently suitable for use in the further analysis of the more complex definition components.

6.7 The grammar of the definition types: A formal summary

The table in section 6.7.2 provides a formal summary of the definition language grammar for each of the identified types. An explanation of the symbols and conventions used in the summary is given in section 6.7.1.

6.7.1 Explanation of symbols and conventions

Optional elements are shown in normal brackets, with a subscript '1' if they can only appear once in a definition. Matching elements have a subscript 'm'. If a definition can contain elements which have essentially similar functions but can occur in different positions with different realisations, they are distinguished by sequential superscript numbers. Alternative elements are separated by '|', with grouped items marked by square brackets.

Symbol	Meaning
A	Article
A^{d}	Adjunct
B	Binder e.g. 'that' in type A5 definitions
D^{r1}	Preceding discriminator
D^{r2}	Following discriminator
E	Explanation
H^{d}	Headword
H^{e}	Headword element
H^{i}	Hinge
I^{n}	Operator 'in' introducing type D1 definitions
L	Linker in type A3 definitions
M^{r}	Modifier, preceding a noun
N^{o}	Noun or noun phrase co-text
O^{b}	Object of a verb
P^{o}	Possessive pronoun or possessive noun phrase
P^{r}	Projection structure
P^{rs}	Projection subject
P^{rv}	Projection verb or verb phrase
P^{rc}	Projection complement
P^{rl}	Projection link
Q^{r}	Qualifier, following a noun
S	Superordinate
S^{b}	Subject of a verb
T^{o}	Operator 'to' in type A6 definitions
X	Cross-reference
V^{p}	Verb or verb phrase

6.7.2 Formal summary of the definition language grammar

Type	Formal Description
A1	$(A)_1$ (M^r) H^d (Q^r) H^i $(A_m)_1$ (D^{r1}) S (D^{r2})
A2	P^o (M^r) H^d (Q^r) H^i $(A_m\|P_m)_1$ (D^{r1}) S (D^{r2})
A3	H^d H^i $(A)_1$ E L X (N^2)
A4	$(A)_1$ (N^o) H^d N^o (H^i) E
A5	N^o $(B)_1$ $(H^i)_1$ H^d $(Q^r\|O^b)$ $(H^i{}_m)_1$ $[(D^{r1})$ S $(D^{r2})]\|E$
A6	$(T^o\|A)_1$ $(V^p)_1$ H^d $(Q^r\|O^b)$ (A^d) H^i $(T^o{}_m\|A_m)_1$ (D^{r1}) S (D^{r2})
A7	$(A)_1$ (D^{r1}) S (D^{r2}) H^i $(A_m)_1$ (M) H^d (Q^r)
B1	H^i S^b H^d (O^b) (A^d) $S^b{}_m$ E
B2	H^{i1} S^b $(H^{i2}\|H^e)_1$ H^d (A^d) $(S^b)_m$ $(H^{i3})_1$ E
B3	H^i S^b V^p (A) H^d (A^d) $(V^p)_m$ E
B4	S^b V^p (O^b) (A^d) H^d $A^d\|O^b$ H^i E
C1	P^{rs} P^{rv} (P^{rc}) (P^{rl}) H^d (A^d) $(P^r{}_m)$ E
C2	H^i P^{r1} S^b (P^{r2}) H^d (A^d) $P^r{}_m$ $(S^b{}_m)$ $(P^r{}_m)$ E
C3	P^{r1} $(A)_1$ E P^{r2} $(A_m)_1$ H^d (A^d)
C4	H^{i1} S^b $V^p\|H^{i2}$ (O^b) (P^{r1}) $(S^b{}_m)$ $(V^p{}_m\|H^{i3})$ H^d $(A^d\|Q^r)$
C5	$(A)_1$ H^d H^{i1} (A^d) (H^{i2}) E
D1	I^n $(A)_1$ H^d N^o (S^b) (H^i) E

6.8 An outline of the parsing process

The parsing process developed during this research operates in two main stages. The first stage uses the structural taxonomy as a basis for allocating individual definition sentences to appropriate parsing strategies, and these strategies are used in the second stage to implement the grammar. For ease of use the process is controlled by a short control program which passes the input file of definitions first to a recognition program, which appends a type marker to the input data, and then passes the marked data to a program which selects the appropriate parsing software. The sections below describe the main processing steps involved in these two stages. The recognition stage is applied to all definitions input and is dealt with in section 6.9. The second stage varies between definition types, and is described in outline in section 6.10.

6.9 The recognition of definition types

The recognition program uses the patterns of text in the definition sentences to allocate them to their definition types, occasionally resorting to the grammatical information contained in the record extracted from the dictionary database to make fine distinctions between structurally similar types. The input to the program is the preprocessed version of the extracted data described earlier in section 4.2.1, and the main features of this data are considered in section 6.9.1. Section 6.9.2 outlines the recognition process.

6.9.1 The definition record data structure

The most important part of the data for the recognition program, the definition text itself, is contained in the first three items of the data record. The table below shows the organisation of the definition text within these first three items for several different definition patterns.

Item 1	Item 2	Item 3
Text before headword	**Headword**	**Text after headword**
A	current account	is a bank account which you can take money out of at any time using your cheque book or cheque card;
	Impurities	are substances that are present in another substance, making it of a low quality.
If someone or something	wheels,	they move round in the shape of a circle or part of a circle.
People sometimes refer to a toilet as the	bathroom.	
In a	logical	argument or analysis, each statement is true if the statement before it is true.

The remaining six items contain the following data:

Item	4	5	6	7	8	9
Contents	Sense	Grammar	Definition	Headword	Usage notes	
					Following definition	Preceding definition

The internal organisation of the data records, described in section 4.2.1, allows the software to identify all of the items correctly even when some of them are empty.

6.9.2 The recognition process

To a large extent, the approach used in the recognition process[3] to allocate definition sentences to their structural types reflects the investigative approach used in the development of the taxonomy, described in detail in Chapter 4, to identify the original categories. As has already been described in Chapter 4, part of the development process included the combination of groups of definitions which appeared to have different text structures into types which represented a single grammatical structure category, and which were therefore capable of analysis using a single parsing strategy. This characteristic is also reflected in the recognition process. As an example, consider the following examples of type A2 definitions:

> Your **stepdaughter** is the daughter of your husband or wife by an earlier marriage. (p. 552)
> A person's **income** is the money that they earn or receive. (p. 283)

Both definitions begin with a possessive structure, but whereas the first uses a closed class determiner, the other uses a general morphological marker attached to an open class noun. The first group of type A2 definitions emerged early in the examination of initial words described in sections 4.2.3, but later investigation revealed the essential similarities with the second group, as shown in section 4.3.1. In the recognition process, the first group are identified early in the routine on the basis of an initial 'your', 'someone's' etc. The second group emerges later in processing, after other definition types have ben eliminated, on the basis of the inclusion of an apostrophe in the first data item.

This cumulative approach to the recognition process relies mainly on text patterns within data items 1 and 3, with occasional reference to the contents of item 2, the headword element. Data item 5, the grammar code, is also used at some points to differentiate between similar definition structures used in different ways for different parts of speech.

At various stages within the recognition process, definitions which fail to meet any of the criteria for the standard structural categories, are labelled as 'unallocated'. These anomalous definitions are discussed in greater detail in section 5.4.5, and their implications are explored in sections 7.2 and 7.3.

6.10 The second stage

Each definition type demands a different individual parsing strategy, but it is possible to generalise the overall approach used for the second stage of parsing. It is divided into two main subprocesses: analysis and display. In the initial analysis stage, described in the next section, the original definition sentences are split into their main functional components, as identified in the sublanguage grammar. In the display stage, described in section 6.10.2, the analysed definition record produced by the analysis process is arranged in the required output format. This separation originated in the practical considerations of software development, but it does have significant advantages, especially where a definition has complex components or embedded elements which need a more flexible output formatting approach. The analysis and display methods used for each of the individual definition types are illustrated with examples of analysed definition sentences in Appendices 1 and 2.

6.10.1 The initial analysis

The initial analysis stage works on two levels. The analysis into functional components, described in the next section, produces a subdivided version of the original definition text, with each component of the analysis allocated to a specific item within the output data record. The second level of analysis identifies some of the framework elements in the right hand sides of the definitions, already described in sections 5.2.3.2 and 6.1, which match elements of co-text in the left hand sides. Where these matching framework elements form easily separable components in their own right within the

sublanguage grammar they are dealt with in the first level of processing and allocated to individual data items. Where, on the other hand, they are embedded within other components, such as explanations or discriminators, they are identified by the second level of analysis and marked with an appropriate tag so that they can be treated correctly in the display stage. This process is described in detail in section 6.10.1.2.

6.10.1.1 *The first level — functional analysis*

The table in section 6.7.2 shows the formal representation of the definition types in the notation of the sublanguage grammar. The definition sentences are already divided into three sections during extraction and preprocessing, as shown in section 6.9.1. The functional analysis stage splits these three items into the components shown in section 6.7.2. The other data items contained in the definition record are unaffected by this analysis and pass unchanged to the display stage for use in the creation of the required output format. The table below shows the analysis performed on the definition text for each of the types, using the same notation as in the table in section 6.7.2. The output from the analysis stage, which is passed to the display stage, contains sixteen data items (except for types A6 and C3, which contain seventeen, and type C2 which contains eighteen). The first nine, (ten for types A6 and C3, eleven for type C2) items, shown in the table below, contain the results of the initial functional analysis, while the remaining seven consist of the six items described in section 6.9.1, together with the type marker added by the recognition software.

6.10.1.2 *The second level — identifying embedded framework elements*

Any element of co-text in the left hand side of a definition could potentially be matched in the right hand side. The analysis programs contain procedures which use the contents of the co-text elements to create lists of potential matching items, which are then searched for in the appropriate text elements of the right hand side. As an example, consider the definition:

> When the police **breathalyze** a driver, they ask the driver to breathe into a special bag to see if he or she has drunk too much alcohol. (p. 61)

The analysed version of this definition includes the following data items:

		Item											
Group:Type		**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**Type**
A:	**A1**	A	M^{r}	H^{d}	Q^{r}	H^{i}	A_{m}	D^{r1}	S	D^{r2}			**A1**
	A2	P^{o}	M^{r}	H^{d}	Q^{r}	H^{i}	$A_{m}\|P_{m}$	D^{r1}	S	D^{r2}			**A2**
	A3	H^{d}	H^{i}	A	E	L	X	N^{2}					**A3**
	A4	A	N^{o}	H^{d}	N^{o}	H^{i}	E						**A4**
	A5	N^{o}	B	H^{i}	H^{d}	$Q^{r}\|O^{b}$	H^{i}_{m}	D^{r1}	S	D^{r2}			**A5**
	A6	$T^{o}\|A$	V^{p}	H^{d}	$Q^{r}\|O^{b}$	A^{d}	H^{i}	$T^{o}_{m}\|A_{m}$	D^{r1}	S	D^{r2}		**A6**
	A7	A	D^{r1}	S	D^{r2}	H^{i}	A_{m}	M^{r}	H^{d}	Q^{r}			**A7**
B:	**B1**	H^{i}	S^{b}	H^{d}	O^{b}	A^{d}	S^{b}_{m}	E					**B1**
	B2	H^{i1}	S^{b}	$H^{i2}\|H^{e}$	H^{d}	A^{d}	S^{b}_{m}	H^{i3}	E				**B2**
	B3	H^{i}	S^{b}	V^{p}	$A\|O^{b}$	H^{d}	A^{d}	S^{b}_{m}	V^{p}_{m}	E			**B3**
	B4	S^{b}	V^{p}	O^{b}	A^{d}	H^{d}	$A^{d}\|O^{b}$	H^{i}	E				**B4**
C:	**C1**	P^{rs}	P^{rv}	P^{rc}	P^{rl}	H^{d}	A^{d}	P^{r}_{m}	E				**C1**
	C2	H^{i}	P^{rs}	P^{rv}	P^{rc}	P^{rl}	A	H^{d}	A^{d}	P^{rs}_{m}	P^{r}_{m}	E	**C2**
	C3	P^{rs}	P^{rv}	A	D^{r1}	S	D^{r2}	P^{r2}	A_{m}	H^{d}	A^{d}		**C3**
	C4	H^{i}	S^{b}	$V^{p}\|H^{i2}$	$O^{b}\|E$	P^{r1}	S^{b}_{m}	$V^{p}_{m}\|P^{r2}$	H^{d}	$A^{d}\|Q$			**C4**
	C5	A	H^{d}	H^{i1}	A^{d1}	A^{d2}	H^{i2}	E					**C5**
D:	**D1**	I^{n}	A	H^{d}	N^{o}	S^{b}	H^{i}	E					**D1**

Item Component						
1	2	3	4	5	6	7
H^i	S^b	H^d	O^b	A^d	$S_m{}^b$	E
When	the police	breathalyze	a driver,		they	ask @M2_the driver_M@ to breathe into a special bag to see if @M2_he or she_M@ has drunk too much alcohol.

The matching pronoun 'they' for the S^b co-text element 'the police' is allocated to its own data item, item 6, because it occupies a separate, well-defined position in the linear sequence of the definition. In contrast the elements 'the driver' and 'he or she' within data item 7 which match the O^b co-text, 'a driver', are identified by the boundary markers '@M2_' and '_M@'. These markers allow them to be treated correctly at the display stage even though they are embedded within the explanation element E which makes up data item 7. The number in the opening marker '@M2_' allows the display stage to identify the matched item correctly. The list of potential matching elements created for the co-text 'a driver' includes a range of pronouns and the word 'driver'. The inclusion of the article in the first match, and the amalgamation of 'he', 'she' and the connecting 'or' are achieved by a separate procedure after initial matching has been performed. The process of matching these elements is particularly useful, as has already been described in section 6.5.2.4, in definitions which follow Group B in using an explanation structure rather than the more easily analysed superordinate and discriminator model.

6.10.2 The display stage

The separation between the initial analysis described above and the process of formatting the analysed data for output has already been explained in section 6.10. Apart from the need to deal with complex or embedded elements correctly this separation also allows the final output format to be adjusted to suit the requirements of individual applications without disturbing the initial functional analysis. The following section explores different methods of presentation, and section 6.10.2.2 examines the further analysis carried out during this stage.

6.10.2.1 *Presentation of output*

The examples given below show some possible methods of presenting the analysed definition data. The first is a simple vertical list of data components:

```
breathalyze
VB      with OBJ
Hi      When
Sb      the police
Hd      breathalyze
Ob      a driver,
Sbm     they
E       ask
Obm     the driver
E       to breathe into a special bag to see if
Obm     he or she
E       has drunk too much alcohol.
```

In addition to the analysed definition text the output includes the headword and the grammar code. The vertical list format is relatively accessible for the human reader, and could also be used as a record structure for input to further computer processing.

An alternative approach, similar to the output of tagging programs for other forms of text, is to preserve the horizontal layout of the text, marking the boundaries of the components:

```
Hi_When_# Sb_the police_# Hd_breathalyze_# Ob_a
   driver,_# Sbm_they_# E_ask_# Obm_the driver_# E_to
   breathe into a special bag to see if_# Obm_he or
   she_# E_has drunk too much alcohol._#
```

This layout presents only the definition text, in a single line of information, in which each component is introduced by its standard notation followed by '_', and ended by the marker '_#'. The two presentations use slightly different versions of the display software and work from the same analysed data produced during the first stage. The range of possible presentation methods and formats is almost limitless, and some earlier examples (from the Chamberlain and ET/10–51 projects) are described in Barnbrook (1993) and Barnbrook & Sinclair (1995).

6.10.2.2 *Further analysis at the display stage*

There are two major areas of the original definition text which are not fully analysed during the initial functional analysis process: complex elements such as headwords and superordinates (dealt with in section 6.10.2.2.1) and embedded framework elements in the right hand side (dealt with in section 6.10.2.2.2).

6.10.2.2.1 *The analysis of complex elements*

Section 6.6.1 gives an example of a definition containing a complex headword:

> If something **bores** you **to tears, bores** you **to death,** or **bores** you **stiff,** it bores you very much indeed;
>
> (sense 1, p. 56)

In the initial analysis process the definition text is analysed into:

Item Component						
1	**2**	**3**	**4**	**5**	**6**	**7**
H^{i}	S^{b}	H^{d}	O^{b}	A^{d}	S^{b}_{m}	E
If	something	bores *you *to tears,* bores *you *to death, *or *bores *you *stiff,			it	bores @M2_you_M@ very much indeed;

The text allocated to item 3 contains several elements, including three versions of the headword, each with its own embedded co-text. During the display stage, this element is analysed into its constituent parts, so that the final output is:

```
bores
ADJ
Hi      If
Sb      something
Hd1     bores
Ob      you
Hd1     to tears,
Hd2     bores
Ob      you
Hd2     to death,
```

```
Or      or
Hd3     bores
Ob      you
Hd3     stiff,
Sbm     it
E       bores
Obm     you
E       very much indeed;
N2      an informal use.
```

Similar techniques are also used to separate alternatives within the superordinate and its discriminators, as is shown by sense 1 of 'door', which contains several sets of alternatives:

> A **door** is a swinging or sliding piece of wood, glass, or metal, which is used to open and close the entrance to a building, room, cupboard, or vehicle. (p. 160, sense 1)

The analysis shows how the alternatives are dealt with:

```
door    (1)
COUNT   N
A       A
Hd      door
Hi      is
Am      a
Dr1     swinging
Or      or
Dr1     sliding
S       piece of wood,
S       glass,
Or      or
S       metal,
Dr2     which is used to open and close the entrance to
        a building,
Dr2     room,
Dr2     cupboard,
Or      or
Dr2     vehicle.
```

The format is designed to bring out the branching structure created by the provision of alternatives at each stage. In further processing this structure could be used to produce alternative single definitions, such as:

> A **door** is a swinging piece of wood which is used to open and close the entrance to a building.
> A **door** is a sliding piece of glass which is used to open and close the entrance to a room.

None of these partial statements, of course, contains the full *CCSD* definition, which has been presented as a conveniently abbreviated list of all the possibilities expressed by the multiple alternatives.

6.10.2.2.2 *Dealing with embedded framework elements*

Section 6.10.1.2 describes the identification and tagging of embedded framework elements during the initial analysis stage. The display program contains a procedure which is capable of using these markers to label framework elements with the appropriate notation for the original co-text which it matches. This allows the marked text to be separated from its environment and labelled as necessary, while preserving the correct labels for the remainder of the text. All the decisions made by this procedure relate only to formatting: no actual analysis of the data is carried out.

6.11 Summary

The recognition software and the individual analysis and display routines for each definition type, which together form the parser, are capable of identifying the structural patterns which underlie the taxonomy described in Chapter 5 and of analysing the definition sentences into the functional components summarised earlier in this chapter in section 6.7. The adequacy of the analysis and the implications of any anomalies found, together with possible applications of the taxonomy, the grammar and the parser are discussed in Chapter 7.

Notes

1. The enhancements to the original analysis shown in this section, and the notation used for it, were suggested by Professor J.M.Sinclair.
2. Embedded matching elements are in italic type in both tables
3. A full description of the recognition process is given in Barnbrook (1995)

Chapter 7

Evaluation and applications

The taxonomy, grammar and parser described in the preceding chapters are given a critical evaluation in this chapter. Their implications for the construction of dictionaries and other sources of definitions are explored, together with present and potential future applications. Section 7.1 outlines the evaluation process, 7.2 the implications of the evaluation for the definition language description, and 7.3 the general implications for dictionary design and construction. Sections 7.5 to 7.8 outline possible applications.

7.1 Stages of the evaluation process

The evaluation of the definition taxonomy, grammar and parsing software falls naturally under three main headings:

a) continuous testing, error correction and enhancement during the development of the language description model and its associated software
b) formal testing to demonstrate the adequate operation of the final version of the software
c) assessment of the implications of the results of stages a) and b)

The first and second stages formed part of the development process itself and have already been described. The third stage is described in sections 7.2 and 7.3.

7.2 Implications of the results for the definition sentence description

The construction of the taxonomy and the use of the grammar and parser developed from it provided a useful opportunity to check the appropriateness and robustness of the language description model which they represent. The implications of the results of the development and testing processes for the taxonomy are considered in the next section, and their implications for the grammar and parser in section 7.2.2.

7.2.1 Implications for the taxonomy

During the course of the development of the taxonomy it became apparent that a very small number of definitions did not fit the criteria for any of the definition types, but equally did not constitute a coherent type in their own right. The six definition sentences involved have already been described in section 5.4.5, and their implications for the taxonomy are now considered individually:

> **Around** an be an adverb or preposition, and is often used instead of **round** as the second part of a phrasal verb. (p. 26)

The problem with this definition lies in its complexity. In terms of the taxonomy it mixes two types of definition, type A1 and type C5. If these elements were separated, two definitions would be produced:

> **Around** can be an adverb or preposition. (type A1), and
> **Around** is often used instead of **round** as the second part of a phrasal verb. (type C5)

In terms of the language description model, this hybridisation of two identified types seems to confirm the taxonomy's general appropriateness. The practical problems involved in the analysis of the original complex definition sentence can easily be circumvented by the separation described above, which could be performed automatically.

> **Eminently means** very, or to a great degree; (p. 175)

This is a simple typographical error in the positioning of the headword boundary markers. Again, its detection by the recognition software reflects the robustness and accuracy of the taxonomy.

> Roads, race courses, and swimming pools are sometimes divided into **lanes.** (p. 313, sense 2)
> In a railway station or airport, you can pay to leave your luggage in a **left-luggage office;** (p. 319)

Both these sentences give information about their headwords, but the structure used does not correspond to any form of definition recognised by the taxonomy. It is arguable, in fact, that they are not strictly definitions in any of the wide range of senses of that word encountered in the dictionary, but are rather illustrative sentences. In the case of 'lanes' this interpretation is reinforced by the second sentence found in the definition text:

> These are parallel strips separated from each other by lines or ropes.

This text has been treated by the preprocessing program as a following usage note because of its separation from the main definition text and its lack of a headword marker.

The original definition texts could perhaps be turned into type A1 structures by altering the sequence of words:

> **Lanes** are things that roads, race courses, and swimming pools are sometimes divided into.
> A **left-luggage office** is a place in a railway station or airport where you can pay to leave your luggage;

These new wordings perhaps seem rather clumsy and provide little or no genuine extra information. The uninformative superordinate 'things' in the first definition has had to be generated to make the definition complete, and constitutes a default option. The slightly more specific superordinate 'place' and its associated discriminator boundary 'where' in the second are both derived from the preposition 'in' in the original sentence. Given this lack of genuinely new information, it is possible that this form of rewriting could be automated to simplify computer analysis, and it might be a useful way of regularizing deviant patterns, although the information extracted from such quasi-definitions may not be as useful as that derived from the more normal forms. In the case of 'lanes', of course, a rewriting of the second sentence to give it a proper definition structure could achieve rather more. The definition text could then become:

> **Lanes** are parallel strips separated from each other by lines or ropes.

This could be followed by the note:

> Roads, race courses, and swimming pools are sometimes divided into lanes.

This simple reordering would produce a normal type A1 definition with a following usage note. Again, these anomalies reflect problems in the composition of the definition sentences rather than weaknesses in the taxonomy.

> You can also give your impression of something you have just read or heard about by talking about the way it **sounds.** (p. 537, sense 6)
> You can acknowledge someone's thanks by saying **'You're welcome'.** (p. 641)

Both the remaining two definitions are effectively reversed versions of type B4, exemplified by the definition of 'encore':

> An audience shouts **'Encore!'** at the end of a concert when they want the performer to perform an extra item. (p. 176)

A simple rearrangement of the text would convert them to this form with no real loss of information:

> You can also talk about the way something you have just read or heard about **sounds** when you want to give your impression of it.
> You can say **'You're welcome'** when you want to acknowledge someone's thanks.

These forms could certainly now be parsed using the type B4 algorithm, but there is a certain clumsiness about the wording from the point of view of a human reader, which is no doubt what led to the original choice of form. It is possible that this rewriting could also be performed automatically.

Overall, then, this very small number of deviant structures found in the sample of definition sentences contained in *CCSD* has no serious implications for the usefulness or successful operation of the parser or the adequacy of the description of the definition language provided by the taxonomy and grammar. In fact, the nature of these deviations serves to confirm the basic accuracy of the model which has been developed to describe the definition sentences.

7.2.2 Implications for the grammar and parser

The implications of the results for the integrity of the grammar or the effectiveness of the parser were taken into account during the development process, so that all problems encountered during the various stages of testing have already been dealt with. There are still, however, implications for the application and detailed interpretation of the description provided by the grammar and the output produced by the parser, and these have already been described in detail in Chapter 6.

7.3 Implications of the results for the design and compilation of dictionaries

As well as providing a useful review of the language description model developed for the definition sentences, the development and testing processes also revealed problems and potential areas of improvement in the design and compilation of the dictionary. Errors which had not been detected during the

production of the dictionary and problems in the application of compilation policies were both highlighted by the process. While the items described below relate directly to the dictionary selected for use as a source of sample definitions, this by-product of the development of the grammar and parser could be used to provide automated quality control during the construction of dictionaries in general. Possible applications of this aspect of the software are explored in detail in section 7.7.1.

7.3.1 Text anomalies

The detailed examination of the definition sentences demanded by the development and testing of the taxonomy, the grammar and the parsing software revealed some anomalous characteristics of the text which had not been detected by the testing procedures adopted during the compilation of the dictionary. This is not a criticism of those procedures. It is likely that only the type of investigation demanded by the thorough analysis of the definition language which was carried out for this project would be capable of revealing these problems. The following three sections describe the main anomalies revealed during the development of the software.

7.3.1.1 *Register notes*

Because the development of the definition sentence taxonomy depends on the existence of consistent patterns in structures formed using the same strategy, any anomalies that affected the recognition of those structures were quickly highlighted. For example, as already described in section 4.2.2.1, it was found at an early stage of the analysis that definitions beginning with 'in' often had usage notes at the start which, rather than being separately coded as register notes with the mark-up code [RN], had been included as part of the definition text under the code [DT]. On further investigation, many of these turned out to be similar to the definition of 'attorney':

> [DT]In the United States, an [HH]attorney [DC]is a lawyer.

This includes the register note 'In the United States' in the definition text, beginning at [DT]. In the definition of sense 2 of 'agency', on the other hand, the entry is:

> [RN]In the United States,
> [DT]an [HH]agency [DC]is an administrative organization run by a government.

This is clearly a more useful treatment, and should be applied consistently throughout the dictionary.

As parsing strategies developed, some definitions caused problems because of the presence of extraneous material at the end of the text. As an example, this is the definition text for 'bogged down' from the original dictionary file:

> [DT]If you are [HH]bogged down [DC]in something, it prevents you from making progress or getting something done; an informal use.

This is a very similar problem to that described earlier in this previous section. Once again, the treatment of the register note 'an informal use' contrasts with the normal treatment, which is shown in the dictionary entry for 'abate':

> [DT]When something [HH]abates, [DC]it becomes much less strong or widespread;
> [RN]a formal use.

Again, this is clearly the more useful treatment, and register notes which have not been dealt with in this way reduce the usefulness of the dictionary as a computer readable database. It is important to stress that there is no effect on the printed text in any of these cases.

Another anomaly affecting register notes, which did affect the printed form of the dictionary, was discovered as a direct result of the close investigation of the embedded initial register note described above. The normal form of an explanation containing a register note, regardless of the mark-up codes used, is shown in the explanation of 'backbencher':

> In Britain, a **backbencher** is an MP who does not hold an official position in the government or its opposition. (p. 35)

The comma after the register note was used, because of the inconsistency described above in marking these notes, as a basis for splitting them from the explanation during preprocessing. As the definitions were parsed, it became apparent that three of them had not been preprocessed properly, and that type recognition and parsing had been impaired simply because the commas were missing:

> In games such as football **full time** is the end of a match. (p. 225, sense 2)
> In Britain the **ground floor** of a building is the floor that is level with the ground outside. (p. 246)
> In American English a **subway** is an underground railway. (p. 565, sense 2)

As explained more fully in section 7.7.1, the parser could easily be adapted for use in checking the dictionary text for inconsistencies such as these.

7.3.1.2 *Headword boundaries*

The development of the definition language model drew attention to some apparent inconsistencies in the fixing of headword boundaries. Some of these, described in the next section, were obviously typographical errors, while others, described in section 7.3.1.2.2, raise more complex questions about the presentation of information in the dictionary entries to the human user and the implications for computer processing of the information.

7.3.1.2.1 *Typographical errors*

Because the parser uses the headword markers, interpreted in the printed version of the dictionary as bold type codes, as boundaries for the headword element of each definition text, anomalies in the positioning of these markers were quickly highlighted by routine testing carried out during the development of the parser. The two examples revealed during testing show the nature of the problem. The printed form of the definitions of 'eminently' and 'telegraph' sense 1 in the dictionary are:

> **Eminently means** very, or to a great degree; (p. 175)
> The **telegraph is** a system of sending messages over long distances by means of electrical or radio signals. (p. 582, sense 1)

These errors slipped past the proof-reading stages during dictionary preparation, but were detected by the type recognition software in the case of 'eminently' and by problems caused for the parser in the case of 'telegraph'. The source of the problem is the same in both cases: incorrect positioning of the headword mark-up codes, as can be seen from the original dictionary entries:

> [DT][HH]Eminently means [DC]very, or to a great degree;
> [RN]a formal use.
> [DT]The [HH]telegraph is [DC]a system of sending messages over long distances by means of electrical or radio signals.

In both cases, the [DC] marker (showing where the headword finishes and definition text continues) should be placed one word to the left, so that 'means' and 'is' are outside the headword boundary.

7.3.1.2.2 *Headword markers in cross-reference definitions*

During the development of the parsing algorithms, discrepancies between two definitions, both effectively cross-references to other dictionary headwords, raised questions of inconsistency of the use of bold type. The printed form of the definition text is given below:

> A **bathtub** is the same as a bath; (p. 40)
> **Hypnotism** is the same as **hypnosis**. (p. 274)

While the definition of 'bathtub' could be parsed using the type A1 algorithm, the definition of 'hypnotism' was initially problematic because of the cross-reference format used in the text, which contains two areas of bold type.

At first sight there appears to be an inconsistency here in the dictionary's treatment of the two headwords. On closer examination, it was found that three definitions followed exactly the same pattern as 'hypnotism':

> **Humanity** is the same as **mankind**. (p. 272, sense 1)
> **Hypnotism** is the same as **hypnosis**. (p. 274)
> **Racialism** is the same as **racism**. (p. 456)

A similar pattern is also used for the more obviously grammatical cross-references, such as:

> **Dried** is the past tense and past participle of **dry**. (p. 164, sense 1)
> **Media** is a plural of **medium**. (p. 347, sense 2)
> **SW** is a written abbreviation for 'south-west'. (p. 572)

Even within these items there is a slight anomaly in the method used for quoting the cross-referenced headword — bold type for 'dry' and 'medium', single quotes for 'south-west' — and this may in itself confuse human users, but there is an approximate consistency.

The pattern used for 'bathtub' was found in another 65 definitions altogether, including the following examples:

> A **budgie** is the same as a budgerigar; (p. 65)
> **Gasoline** is the same as petrol; (p. 229)
> A **telly** is the same as a television; (p. 583)

One possible reason for the difference of treatment was found. In all of these cases, the equivalence of the two words is qualified by a register or usage note. In the three definitions shown above, the notes are:

budgie	an informal use.
gasoline	an American use.
telly	an informal use.

In the three examples of grammatical cross-reference which use the same pattern as 'hypnotism', the equivalence seems to be unqualified, independent of the normal conditions of use. If this is the reason for the difference of treatment, the human user is not made fully aware of the implications of the methods adopted, and there may be a need to make this presentation more obvious and more consistent.

7.3.1.2.3 *The extent of the headword*

It has already been suggested in section 6.2.2 that for some definitions the definiendum and the headword marked in bold type in the dictionary are not necessarily identical. Consider the following type B3 definitions:

> If you do something with **aplomb,** you do it with great confidence. (p. 22)
> If you are allowed **entry** into a country or place, you are allowed to go in it. (p. 181, sense 5)
> If you are provided with **lodging,** you are provided with a place to stay for a period of time. (p. 330, sense 1)
> If you get **satisfaction** from someone, you get money or an apology from them because of some harm or injustice which has been done to you. (p. 496, sense 2)

In each case there is an important element in addition to the marked headword in the first part of the definition sentence which is repeated in the second part. Because of this repetition, the lexicographic equation can be stated in terms of the marked headword alone, as in:

aplomb	=	great confidence
entry	=	to go in
lodging	=	a place to stay for a period of time
satisfaction	=	money or an apology… because of some harm or injustice which has been done

In the following four definitions, however, also taken from type B3, the repetition is less complete:

If you are an **admirer** of someone, you like and respect them or their work. (p. 8, sense 2)
If you are a **champion** of a cause or principle, you support or defend it. (p. 81, sense 2)
If you have a **passion** for something, you like it very much. (p. 406, sense 2)
When a vehicle does a **U-turn,** it turns through a half circle and faces or moves in the opposite direction. (p. 625, sense 1)

The lexicographic equations produced from these definitions reflect the limited repetition:

are an **admirer** of = like and respect
are a **champion** of = support or defend
have a **passion** for = like… very much
does a **U-turn** = turns through a half circle and faces or moves in the opposite direction

The left hand sides of these equations include text items which are not included in the bold-type headword but which seem to form part of the definienda. These elements are automatically identified by the parser, which analyses them as headword elements rather than as part of the hinge structure. It may be more helpful if the entire definienda shown in these equations were set in bold type to make this identification easier for the human dictionary user.

7.3.2 Selection of definition strategies

The general structural groups described in section 5.1 seem to be associated with dominant word classes. A simple analysis of the headwords contained in group B, for example, whose definitions begin with 'if' or 'when' (types B1, B2 and B3), produces the following frequency list of grammatical classes:

verb	6614
adjective	1470
no grammar code	1273
noun	801
phrase	423
adverb	260
preposition	190
other	24
Total:	11055

Many of the headwords shown in the above table under 'no grammar code' or 'phrase' are also verbs, and this single word class accounts for more than two thirds of all the definitions which use group B strategies. They are generally defined using type B1 definitions, exemplified by the definition of sense 2 of 'pin':

> If you **pin** something somewhere, you fasten it there with a pin, a drawing pin, or a safety pin. (p. 418, sense 2)

Adjectives come a poor second, representing around 13% of the total. All of these use the type B2 strategy exemplified by sense 2 of 'meaningless':

> If your work or life is **meaningless,** you feel that it has no purpose and is not worthwhile. (p. 347, sense 2)

This strategy seems to be used (in preference to the more common type A4 strategy for adjectives) when the adjective is predominantly used predicatively rather than attributively. The typical type A4 definition of sense 2 of 'maiden' demonstrates this:

> The **maiden** voyage or flight of a ship or aeroplane is the first official journey that it makes. (p. 338, sense 2)

This seems a valid reason for adopting an alternative strategy, but nouns seem to present a more complex situation. Here are the explanation texts for a few of the 801 nouns explained using the type B3 strategy:

> If you gain **access** to a building or other place, you succeed in getting into it; (p. 3, sense 1)
> If you make an **assumption,** you suppose that something is true, sometimes wrongly. (p. 29, sense 1)
> When you take a **breath,** you breathe in. (p. 61, sense 2)
> If you have **change** for a note or a large coin, you have the same amount of money in smaller notes or coins. (p. 82, sense 11)
> If a street is a **dead end,** there is no way out at one end of it. (p. 133, sense 1)
> If you make an **effort** to do something, you try hard to do it. (p. 171, sense 1)
> When you get **feedback,** you get comments about something that you have done or made. (p. 201)
> When something is done with **ferocity,** it is done in a fierce and violent way. (p. 202)

The reason for the adoption of this strategy should now be much clearer. These nouns can only be described effectively in the contexts of verbs, as their direct objects (e.g. 'breath') or complements (e.g. 'dead end'), or in some

adverbial use (e.g. 'ferocity'). As with the predicative adjectives, the definition strategy is dictated by the need to incorporate the verb.

7.3.3 Consistency of definition wording

The need to identify the individual realisations of grammatical elements within the definition types focused attention on some aspects of the detailed wording of definitions and their implications for human users of the dictionary and for computational analysis. As an example, in the type A1 definitions which have discriminator text following the superordinate the beginning of the discriminator is marked in a variety of ways. This problem has already been referred to in section 4.4.1. As explained there, one of the main sets of possible introductory words is the set of relative pronouns, 'who', 'which', 'that' and so on. This, together with the set of prepositions, looked in the early stages of analysis as though it would form a reasonably complete description of the possible boundary markers, making the analysis into superordinate and following discriminator relatively straightforward. As the development of the parser proceeded, it became obvious that a policy decision had been taken during the compilation of the dictionary which made the text less consistent and less easy to parse. Consider the definitions below:

> **Abuse** is rude and unkind things that people say when they are angry. (p. 3, sense 1)
> An **affectation** is an attitude or type of behaviour that is not genuine, but which is intended to impress other people. (p. 10)
> A **consignment** of goods is a load that is being delivered to a place or person. (p. 110)
> A **ghetto** is a part of a city which is inhabited by many people of a particular nationality, colour, religion, or class. (p. 234)
> A **motorboat** is a boat that is driven by a small engine. (p. 363)

In each case, the word 'that' or 'which' introduces the following discriminator and forms a clear and straightforward boundary. Now consider the following similar definitions:

> **Dungarees** are trousers attached to a piece of cloth which covers your chest and has straps going over your shoulders. (p. 167)
> **Dutch** is the language spoken by people who live in the Netherlands. (p. 167, sense 2)
> A **ferret** is a small, fierce animal used for hunting rabbits and rats. (p. 202)

A **motel** is a hotel intended for people who are travelling by car. (p. 362)
A **prism** is an object made of clear glass with straight sides. (p. 440)

In each case the expected introduction to the discriminator is missing, because the full relative clause structure has been abbreviated: 'attached', for example, in place of 'that are attached'. There are no problems here in terms of the use of natural features of the English language, since in most cases these will be perfectly acceptable alternatives, but the lack of consistency in treatment may cause problems for the non-native speakers who form the target audience for the dictionary. It certainly caused problems in the design of the parser, since it greatly extended the set of potential boundary markers. The number of possible markers would have been around 65, if only relative pronouns, prepositions etc. had been used, as against the final list which includes over 200 words, and this made the task of exhaustively cataloguing them problematic. The problem is dealt with in the parsing software by using a list of possible discriminator boundary words, together with rules based on regular past and present participle formation, a list of irregular forms and an exclusion list to make the rules work more accurately.

The resulting set of possible boundary words includes items which, in a conventional general grammar of English, would be categorised as:

prepositions (e.g. about, into, through)
irregular past participles (e.g. dug, sewn, told)
adverbs (e.g. almost, easily, especially)
present participles (e.g. containing, extending, preventing)
adjectives (e.g. close, lower, qualified)
personal pronouns (e.g. he, it, they)

in addition to the normal relative pronouns. The problem seems to have been overcome for the parsing software, but it might be worth investigating the effect on the human user and considering whether it would make the dictionary easier to use if the definition pattern were simplified by the use of the limited set of relative pronouns, prepositions and so on to introduce all following discriminator phrases.

It is interesting to compare the second abbreviated set of definitions with the corresponding entries in *CCELD*. These are:

Dungarees are trousers that are attached to a piece of cloth which covers your chest and which has straps going over your shoulders. (p. 440)
Dutch is the language that is spoken in the Netherlands. (p. 441, sense 2)

> A **ferret** is a small, white, fierce animal related to the weasel, which is kept by people for hunting rabbits and rats. (p. 527)
> A **motel** is a hotel intended for people who are travelling by car, which has space to park cars near the rooms. (p. 940)
> A **prism** is a solid transparent object made of glass or plastic, which has many straight sides and angles. (p. 1141)

This shows a greater use of the relative pronoun, including the use of 'which' to introduce additional information in the definitions of 'motel' and 'prism', which omit the relative pronoun at the main discriminator boundary. A policy of abbreviation has obviously been imposed in the compilation of *CCSD*, but to some extent this is an extension of an option already exploited in the main dictionary.

7.4 Overall evaluation

The problems which have been revealed by the development of the definition language model could certainly affect the extraction of information from definition sentences for use in natural language processing systems, but their overall usefulness as a source of detailed linguistic information is still significant. The analysis of the definitions provided by the parser is generally accurate and sufficiently detailed. It must be remembered that the dictionary definitions used as a sample are designed entirely for human use, and that this would imply significant limitations on their usefulness for computational analysis. In fact, despite the problems described in this chapter, they lend themselves to detailed analysis using relatively simple pattern-matching techniques. As explained in the following sections, there are many applications of the parser, including some using the contents of the sample dictionary, which could contribute significantly to the exploration and processing of natural language.

7.5 Overview of applications

The main purpose of this research was the exploration of the language of the definition sentences, including the extraction of linguistic information for use in natural language processing. During the development of the taxonomy and the grammar and parser other possible applications became apparent, and the

main areas of potential are explored below. Section 7.6 deals with ways in which the use of the dictionary as a linguistic database can be facilitated and enhanced, while section 7.7 outlines potential uses in the construction and improvement of dictionaries. Section 7.8 describes possible extensions to the scope of the taxonomy, grammar and parser which would increase their general usefulness.

7.6 The dictionary as database

Monolingual English dictionaries usually contain information for each headword in addition to the definition or explanation of its meanings. At various times and in various dictionaries this has included information on pronunciation, syntactic characteristics, etymology, spelling and usage, often combined with illustrative quotations. In some cases the information given covers the past history of these features of the word as well as its current features. The selection of the information to be included and the way it is encoded in the entries are obviously crucial elements of dictionary design, but almost any modern dictionary will be constructed in such a way that the elements of the entries for each headword form a fairly consistently structured database. This will often allow a computer readable dictionary text to be accessed readily for linguistic applications even if it has not been designed specifically for this purpose. As we have seen, the Cobuild range is no exception to this general tendency.

In common with other learner's dictionaries the Cobuild range only contain those elements of this information that seem relevant to learners of current English — the forms of the headword lemma, its pronunciation, its syntactic details, details of lexical relations, a definition of its meaning, details of any unusual usage restrictions and examples of use taken from the corpus from which the dictionaries were constructed. These other items of information are generally given in a more traditionally encoded form which allows them to be fairly readily accessed by the computer without the need to write specialised parsing software.

In the case of *CCSD*, the mark-up codes allow access to these individual elements of each entry. Part of the *CCSD* dictionary database entry for 'drink', the printed form of which is shown in section 1.2, is shown below:

[EB]
[LB]
[HW]drink
[PR]/dr*!i!nk/,
[IF]drinks, drinking, drank [PR]/dr!a!nk/, [IF]drunk [PR]/dr*%u!nk/.
[LE]
[MB]
[MM]1
[GR]VB [GS]with or without [GC]OBJ
[DT]When you [HH]drink [DC]a liquid, you take it into your mouth and swallow it.
[XB]
[XX]We sat drinking coffee.
[XX]He drank eagerly.
[XE]
[ME]

This extract shows the main features of the mark-up system, similar in its essentials to those used by later editions of the Cobuild range. It delineates the beginning of the entire entry ([EB]), the information relevant to the whole entry (from [LB] to [LE]) and the information relating to each sense (from [MB] to [ME]). Within the headword information, the headword itself, ([HW]), its pronunciation ([PR]) and inflected forms ([IF]) are all separately accessible. Within the sense information, the sense number ([MM]), grammar code ([GR]), definition text ([DT]) and examples ([XB] to [XE]) can be isolated. There is some further analysis available within the texts of the grammar code and, of course, of the definition. The use of simple string-searching routines through standard utilities or awk programs would enable all of these pieces of data to be extracted and manipulated without further processing of the dictionary. Section 7.6.1 describes the enhancements to this process that can be achieved using the analyses provided by the parser.

7.6.1 Improving the navigation of the database

Facilities for accessing dictionary entries on the basis of different pieces of information are already well established. Dictionaries released on CD-ROM or through web-based interfaces, such as the *OED*, are usually indexed on several different pieces of information to allow searching to be carried out on most of the fields within an entry. This makes such things as cross-reference

between words extremely easy and efficient, but it can also be a powerful language investigation tool when combined with an interrogation language or macro system. In the case of the *OED*, it is possible to construct fairly sophisticated searches which can extract, for example, all headwords with a particular language included in their etymology whose first quotation date in the dictionary lies within a specified range. The results of the search can also be output to a text file for further processing and manipulation.

Facilities like these are extremely valuable, but they still limit the user's access to those items of data which were specifically identified by the mark-up system when the dictionary was compiled. The main benefit arising from a dictionary whose definitions can be automatically analysed is the potential for the use of the whole text as an element of database structure without prior explicit indexing. The information contained in each entry for a word can, of course, be accessed using the word itself as an index in any computer readable dictionary, but processing from that point on depends on the human user. If the definitions can be parsed the computer will have access to all the information contained, explicitly or implicitly, within the definition text, organised on the basis of the function of the information and not merely its form.

As an example, it would be useful if the dictionary database could be accessed by cross-references between words which share linguistic characteristics, including those not normally considered for indexing as individual pieces of information. For example, if you were considering the definition of sense 2 of 'girlfriend' in *CCSD*:

> A woman's **girlfriend** is a female friend. (p. 234)

you might feel a need to know what senses of other headwords had the same restrictive possessive element, 'a woman's'. Once the definitions have been parsed, software can easily be produced to select the definitions which contain the possessive. A simple application of such software to the parsed definitions within type A2 produces the following list of headwords and senses:

> admirers 1
> bonnet 2
> bosom 1
> breasts 1
> bust 5
> cleavage 1
> dowry
> girlfriend 2

husband
maiden name
negligee
ovaries
period 3
suit 2
suitor
uterus
vagina
womb

This list is, of course, only one possible arrangement of the data, extracted from the parsed output. Once the parsed definitions containing this possessive have been identified the system could access the complete original dictionary text for these entries. At such a simple level as this it would, of course, be possible to use standard string search utilities to produce similar results, although these would throw up all definitions containing the same sequence of characters regardless of their position or function within the sentence. The original database structure of the dictionary does not distinguish such elements of the definition entries, and one of the main benefits arising from the availability of parsed definitions lies in the extent to which analyses and searches such as these can be carried out on the basis of this kind of information, despite the fact that it has not been explicitly considered when the dictionary was set up.

The example above listed senses in the dictionary where the possessive element was realised by the phrase 'a woman's'. The parser can take this exploration of the dictionary further. For example, it can identify the superordinate of 'woman' from the word's own definition:

A **woman** is an adult female human being. (p. 651)

When parsed this has the superordinate 'being'. Headwords which share this superordinate can be regarded as the co-hyponyms of 'woman', and these can easily be found using the parsed definitions. A simple search for type A1 definitions with this superordinate produces the following list of senses:

child 1
foetus
man 1
spirit 3
woman

If the search carried out for 'a woman's' as possessive element in type A2 definitions is performed in a similar way for each of these co-hyponyms, it produces the following list of senses:

child's
  playmates
man's
  beard
  buddy 1
  moustache
  penis
  suit 1
  testicles
  wife
man's or boy's
  girlfriend 1

Because the structures of definitions vary from one type to another, these searches have been carried out within the same definition type, in this case A2. As an example of a similar possibility within another type, the definition for sense 2 of 'bung' is:

> If you **bung** something somewhere, you put it there in a quick and careless way; (p. 67)

A learner may be interested in other verbs which have the same object and adjunct elements — 'something somewhere' — to explore the words used in English for moving things around. Searching the parsed definitions for these elements yields the following list of senses:

chuck
dash 3
deposit 1
dump 2
ease 5
fit 5
fix 1
fling 1
fly 5
hang 1
hoist 1
jab 1
jam 2
lay 2

nail 2
pin 2
pitch 2
place 12
plant 6
pop 6
position 3
ram 2
secrete 2
set 2
shift 1
shovel 3
sling 1
slip 4
smack 2
sneak 2
stand 5
stick 7
strap 2
stuff 2
thrust 1
tip 3
toss 1
trundle 2
tuck 2
wedge 2

This provides scope either for guided browsing by learners exploring the linguistic restrictions of groups of related words, or for the development of dynamically focused searching and matching algorithms for natural language processing applications. It is unlikely that the above list could have been compiled exhaustively even by experienced language teachers.

The difference between this process and the use of information already coded into a dictionary relating to superordinates, synonyms, antonyms etc. is fundamental. A completely parsed dictionary would allow lexical relations and any other features of words which are implied by the definition text to be identified, even though they may not have been explicitly considered by the lexicographer, and even though they may not be known to native users of the language on a conscious level. It also allows the level of detail and the whole nature of the analysis to be adjusted through adjustments to the parsing software. Each form of analysis produced by different versions of the parsing

software would be capable of using all of the information contained in the dictionary text, with no limitations imposed by the lexicographer at the compilation stage beyond those inherent in the wording chosen.

This ability to interrogate the language of definitions fully is of crucial importance for the relevance of the dictionary as a source of information for natural language processing systems. As already described in section 7.3.2, although there were strongly preferred strategies for each grammatical class of headword sense, where this did not seem to work for a particular item lexicographers have chosen other approaches. The flexibility of defining approach inherent in this process allows the lexicographer to use linguistic intuition to override formulaic constructions where this seems more appropriate. This in turn implies that the process of construction of definitions may bring in features of the language which do not represent conscious decisions made by the lexicographer purely on the basis of the policies of dictionary compilation, but which are incorporated because they produce the definition sentence which seems most useful. If this is the case, the analysis carried out by the parsing software may be capable of revealing important features of the language of which native speakers and even the lexicographers themselves are not consciously aware, thus enhancing the richness and accuracy of the linguistic data available from the definition.

7.6.2 Conversion to database format

As described in the previous section, almost any modern dictionary is a form of database, in which at least limited elements of the entries can be accessed using the coding system. However, the demands of formal language processing systems may not be met efficiently or adequately by a format chosen originally for human readability. Sinclair (1994) describes Schnelle's suggestion in 1989 that the repetitive shapes of the Cobuild definitions, varying from one kind of meaning to another, should be capable of conversion into a logical form. This led ultimately to the development of a research project (project ET–10/51, part of the Eurotra programme, already mentioned in Chapters 5 and 6) in which a version of the definition parser was developed to carry out one of the stages of this conversion process.

The project is described in detail in its final report (Sinclair, Hoelter & Peters, 1995). The contribution made by the University of Birmingham group was the development of software capable of producing an analysis of the

definitions relating to a test vocabulary of nearly 400 words. The analysed output was passed to the other partners in the project, working in teams based at the Sprachwissenschaftliches Institut at Ruhr-Universität-Bochum and the Istituto di Linguistica Computazionale del C.N.R. at Universitá di Pisa. Both of these teams then developed software, based on different principles, which converted the Birmingham output into formal type-feature structures or attribute-value matrices. Both achieved the aims of the project, showing the tractability of the definition sentences, written in natural language, in the creation of formal linguistic descriptions. The level of detail of the information extracted from the parsed definitions demonstrates the potential of the parser and software developed from it for more complex projects leading, among other things, to the creation of lexica for natural language processing directly from human-readable dictionaries, as discussed in the next section.

7.6.3 The acquisition of computer lexica

The background to the use of machine readable dictionaries in the acquisition of lexica for NLP systems has already been discussed in section 1.2. Boguraev & Briscoe (1989) deal in detail in their introduction with the need for such lexica and with the advantages and disadvantages of the use of machine readable dictionaries in general as a basis for the construction of them and of the particular advantages of *LDOCE*. This dictionary is described in the chapter (p. 2) as 'uniquely suitable for computational lexicography', i.e. the derivation of lexica for computational linguistic processing. It is worth examining the detailed claims made for *LDOCE* to assess their implications for the suitability of full-sentence definitions such as those used in the Cobuild dictionaries for the same purposes, especially considering the extra information made available by the parser. The description of the information contained in *LDOCE* and its organisation is given on pp. 13–21.

After the general account of the type of information shared by *LDOCE* with most other similar dictionaries (pp. 13–14) Boguraev & Briscoe (pp. 14–17) highlight two major features as its specific advantages: the restricted defining vocabulary and the provision of detailed semantic and syntactic information via the 'subject' and 'box' codes. This provides explicit information in the machine-readable form of the dictionary, encoded in a form that makes it easy to access the data. This information, covering such areas as general context of use, details of subject and object preference for verbs and so

on, has obviously been assembled by the lexicographers and represents their conscious estimation of the headword's linguistic features. Similar information can be extracted at varying levels of detail from the parsed versions of full sentence definitions, although it is not necessarily available in the same consistent form for all headwords. The advantage of the type of information provided from the use of the parser is that it is not based on the conscious linguistic knowledge of the lexicographer or expressed as part of a preconceived and limited data structure. If a definition sentence needs to contain a specific piece of information, it will be incorporated by the lexicographer to satisfy the headword's semantic and syntactic demands, evidenced by the corpus data and realised partly through the lexicographer's unconscious knowledge of the language.

In the case of an explicitly coded dictionary, such as *LDOCE*, the decisions made before the dictionary's compilation as to what constitutes a general semantic area or the level of syntactic information to be explicitly encoded limit the possibilities of future information extraction. A survey of co-hyponyms, using techniques similar to those described in section 7.7.4.2, could provide a more useful indication of the semantic area or areas within which a headword operates. Information derived directly from the dictionary's definition texts in this way describes the linguistic features more naturally, fitting them into the context of the language itself, rather than an inflexible semantic taxonomy constructed intuitively without a full analysis of the language. In the definition sentences, the context provided for each headword does not simply fulfil an explanatory role: it also provides an acceptable lexico-grammatical context for the headword. The analysis performed by the parser can then make available both the explicitly encoded elements and the information implicit in the definition sentence.

7.6.4 Disambiguation

Because the parsable dictionary can provide access to all the linguistic information contained in the definitions, it could help to make one of the major problems of natural language processing, the disambiguation of words in context, much more tractable. Where alternative meanings of words exist, the definition sentences do not simply provide an explanation of the sense of each of them; they also provide the most relevant context for each sense. This could be used as the starting point for a dynamic comparison process which would

identify any similar contextual features in the text being processed which tend to make one sense more likely than another. In the following invented example:

> I need to go to the bank because I've got no cash.

the word 'bank', looked up in *CCSD*, would give the following set of definitions:

> A **bank** is a place where you can keep your money in an account. (p. 381, sense 1)
> You use **bank** to refer to a store of something. (sense 2)
> A **bank** is also the raised ground along the edge of a river or lake. (sense 3)
> A **bank** of something is a long, high row or mass of it. (sense 4)
> If you **bank on** something happening, you rely on it happening.

At first sight, there is not enough information here to allow a sense to be selected automatically. If we assume a simple matching system working on the words in the target sentence and trying to find them repeated in suitable place in the definition text, there is no match for the sentence's word 'cash' in any of the definitions. However, if we now look up the word 'cash' in *CCSD*, we get:

> **Cash** is money in the form of notes and coins rather than cheques. (p. 76, sense 1)
> If you **cash** a cheque, you exchange it at a bank for the amount of money that it is worth. (sense 2)
> If you **cash in** on a situation, you use it to gain an advantage for yourself;

The parsed versions of these definitions are:

```
cash    (1)
UNCOUNT N
Hd      Cash
Hi      is
S       money
Dr2     in the form of notes and coins rather than
        cheques.

cash    (2)
VB with OBJ
Hi      If
Sb      you
Hd      cash
Ob      a cheque,
Sbm     you
E       exchange
```

```
O^b_m      it
E          at a bank for the amount of money that
O^b_m      it
E          is worth.

cash in
PHR VB
H^i        If
S^b        you
H^d        cash in
A^d        on a situation,
Sb_m       you
E          use
A^d_m      it
E          to gain an advantage for yourself;
N^2        an informal use.
```

The part of speech represented by 'cash' in the target sentence may not be known at this stage, but both sense 1 and sense 2 have 'money' as elements in their definitions, and sense 1 actually has it as the superordinate of 'cash'. The replacement of 'cash' by 'money' in the sentence, to give:

> I need to go to the bank because I've got no money.

makes it much more likely that the most appropriate sense of 'bank' could now be selected.

The routing software that would be needed to determine search strategies and evaluate results would involve complex decision processes. Successful disambiguation may also need more information than is contained solely within the definitions, and would probably draw on the grammar information, the usage notes and the examples as further evidence. However, the availability of parsed definitions should make it possible to develop a system capable of making accurate choices from the alternative senses.

7.7 Dictionary construction

The exploration of the nature of the definition sentences has provided a basis for a comprehensive critique of the definition process itself, a process at the heart of lexicography. The specific issues arising within *CCSD*, dealt with earlier in 7.3, can be extended to form a critical analysis of the construction

process of dictionaries in general. This section details some of the practical ways in which this could be achieved.

7.7.1 Dictionary refinement — the taxonomy and parser as quality control tools

The need to rewrite some of the dictionary explanations to make them more amenable to automatic parsing has already been discussed in section 7.2.1, but this rewriting would be purely for the benefit of the parser and does not reflect any dissatisfaction with the dictionary as a human tool. However, this research has inevitably involved an evaluation of some of the decisions taken during the writing of the definitions and the effect of these decisions on the usefulness of the dictionary. This has happened partly because the construction of the parser has forced a close and systematic investigation of the structure of the definition sentences, and partly because by its operation the parser has made the functional components of the definitions available for automatic processing and comparison, so that any anomalies in them quickly become apparent.

As described in section 7.2, during the research work carried out to develop the parser various anomalies and errors came to light. Some of these were structural peculiarities, highlighted by the grouping of definitions with similar patterns into taxonomic classes or by the failure of an interim parsing strategy to deal with all members of a definition type properly, some were typographical errors revealed almost accidentally because of the close attention required for the construction of the taxonomy or the development of parsing strategies. The examples already given in section 7.3.1 show the range of types of inconsistency that can be brought to light even by an investigation that has no direct bearing on the integrity of the dictionary. These errors had not been detected by the careful checking that would have been carried out manually and with the assistance of standard computer utilities during the production of the dictionary, but were brought to light because the taxonomy or parser software was effectively reading explanations and considering their structures in detail. This could obviously be exploited as a form of quality control during the compilation process.

In addition to these checks on the structural consistency of explanations, which happened as a by-product of parser development, there are forms of quality control which can be carried out using the information made available

by the taxonomy and the parser, so that they can be made the basis of a set of quality control tools which could be used in the compilation of future dictionaries. This should provide a more efficient and more rigorous check than any manual form of proof-reading, and may reveal aspects of dictionary construction which would be impossible to investigate by any other means. Some detailed examples of this possible approach are given in the sections below.

7.7.1.1 *Using the taxonomy to check explanation strategy selection*

The production of the taxonomy grouped explanations into similar patterns regardless of the nature of the headword. As already described in section 7.3.2, when the grammar codes of the headwords dealt with by the various explanation strategies were analysed, patterns emerged which showed the dominant word classes for each general structure. Where headwords belonging to other word classes were also found, usually in much smaller numbers, it was possible to check them to see why that strategy had been chosen and whether it was the most appropriate. This would be a useful tool within the final stages of editing to identify any non-standard decisions made by the lexicographers and to check their validity.

7.7.1.2 *Using the parser to check relationships between definitions*

Definition sentences often define one word or phrase in terms of a more general one, the superordinate or its equivalent, and the potential applications of this feature in the production of a thesaurus are discussed in 7.7.4. If the dictionary is to be an effective source of information for its users, the links between definitions should be complete. This means, among other things, that all the words used to define a particular sense of a headword should themselves be properly defined elsewhere in the dictionary, so that users can decode the definitions that use them even if they are not already familiar with the entire defining vocabulary. It also means that where a superordinate is used in an explanation, it should be properly linked to its own superordinate in such a way that the user can move usefully upwards through the lexical hierarchy.

As an example, consider the definition of 'imperfection':

> An **imperfection** is a fault or weakness. (p. 279)

The explanations of the non-phrasal senses of 'fault', on p. 200 in *CCSD*, are:

> If a bad situation is your **fault,** you caused it or are responsible for it. (sense 1)
> A **fault** in something is a weakness or imperfection in it. (sense 2)
> If you say that you cannot **fault** someone, you mean that they are doing something so well that you cannot criticize them for it. (sense 3)
> A **fault** is also a large crack in the earth's surface; (sense 4)

Disambiguation would obviously be necessary before an assessment could be made, and this should lead to the selection of sense 2. Unfortunately, this generates a completely circular explanation which covers both 'imperfection', the word being examined, and 'weakness', the alternative synonym to 'fault'. An examination of the definition of 'weakness' shows that the relevant meaning is not defined:

> If you have a **weakness** for something, you like it very much, although this is perhaps surprising or undesirable. (p. 640)

Examples of the use of 'weakness' in a similar sense to that in the definition are found under the definition of 'weak', which is cross-referenced from 'weakness', but there is no direct definition of that sense of the word itself.

The parser would aid the automatic exploration of links like these, so that any gaps or inconsistencies between definitions could be identified and remedied.

7.7.2 *Dictionary translation*

The Cobuild range of learner's dictionaries, in common with others of the same type, are monolingual English dictionaries and are aimed, therefore, at the more advanced learner of English. Less advanced learners would use bilingual dictionaries which typically provide translation equivalents between two languages but do not provide the detailed linguistic information of the monolingual dictionary. To fill the gap between these two types of learner's dictionary, Cobuild has experimented with the translation of its monolingual English dictionaries into special hybrid forms of bilingual dictionary, called 'bridge bilinguals'. These use the learners' mother tongue to define the English headwords, using the same definition style as the monolingual versions. The normal Cobuild definition components are replaced by their equivalents in the language used for the dictionary, and the English headword is incorporated in the appropriate position within the sentence. As an example of the approach, the definition of 'map' in the *Bridge Bilingual Portuguese* version of *CCSD* (Sinclair et. al., 1995), is:

> **A map** é um desenho de uma área que mostra como ela seria se fosse vista do alto, às vezes incluindo informações especiais. (p. 343)

The original English definition in *CCSD* is:

> A **map** is a drawing of an area as it would appear if you saw it from above, sometimes with special information on it. (p. 342)

In this case, and in the case of many of the headwords, the translation is straightforward and involves little or no rearrangement of the original English definition text. In other cases, for example the noun definitions which use a possessive co-text preceding the headword (type A2 in the taxonomy), significant changes of structure have been needed and have been applied to the definition sentences to produce the most appropriate wording for the individual headword. For example, the original English definitions of 'beak', 'moustache' and 'negligee' are:

> A bird's **beak** is the hard curved or pointed part of its mouth. (p. 41)
> A man's **moustache** is the hair that grows on his upper lip. (p. 364)
> A woman's **negligee** is a dressing gown made of very thin material. (p. 373)

In the *Bridge Bilingual* these become:

> **A beak** é a parte dura, curva ou pontaiguda da boca de um pássaro. (p. 42)
> O pêlo que nasce acima do lábio superior de um homem é **his moustache**. (p. 365)
> **A negligee** é um roupão feminino feito de tecido muito leve. (p. 374)

In each of these definitions the possessive co-text has caused a problem for the translators and this problem has been solved in different ways. For the headwords 'beak' and 'moustache' the co-text has been relocated in a similar form — 'de um pássaro' and 'de um homem': for 'negligee' it has been changed to the adjective 'feminino'. For 'beak' and 'negligee' the original sequence of the definition has been preserved: for 'moustache' it has been reversed.

A similar process can be seen at work in the Slovenian versions of the definitions used in the bilingual Slovenian Bridge Dictionary (Polonaštern., 2000). The type A4 definition of 'secluded' can be used as an example. In *CCSD* it is:

> A **secluded** place is quiet, private, and undisturbed. (p. 504)

In Slovenian this structure does not work, and a relative clause structure is needed:

Kraj, ki je **secluded,** je miren, zaseben in nas tam nihče ne moti.
(Polonaštern, 2000, p. 674)

Appendix 3, compiled by Simon Krek, shows the application of the definition analysis model shown in sections 5.2.1 and 5.2.2 above to the partial translation of examples of the definition types into Slovenian.

The preparation for these translation processes demanded a thorough knowledge of the basic definition patterns present in the original English form of the dictionary. The analysis of recurring patterns carried out during the construction of the taxonomy was used as part of the material for training the teams involved in the translation process, and this enabled them to identify problems like these which were likely to occur in translation, to assess their significance, and to decide what action should be taken to deal with them. Languages for which dictionaries have been produced using Cobuild dictionaries as their basis include, in addition to Brazilian Portuguese and Slovenian, Czech, Danish and Finnish.

The principles of construction of the bridge bilingual English dictionary could be used to create bilingual dictionaries for other pairs of languages, using the original English dictionary text as a form of interlanguage key to align the translations into the other languages. In such a process, the definition parser would provide a basis for structural analysis of the definitions to aid their alignment and to make it possible to exploit the information contained in them in a computer assisted translation system which could be much more powerful and effective than existing approaches.

This process should, in the initial stages, be more or less automatic, though careful post-editing would be necessary to identify and remove any anomalies and to ensure that presentational and stylistic decisions are made properly.

7.7.3 Automatic lexicography

The possibility of producing dictionary entries automatically from corpora arises directly from the use of natural language for the construction of dictionary definitions and the development of the taxonomy and parser. As Barnbrook & Sinclair (1995, pp. 16–17) point out, the structures used to form Cobuild definitions are also structures that could be used to define words in non-dictionary texts, so that corpora could be searched for sentences which had these structures and so were potential definitions. Once

these sentences had been selected they could be parsed, investigated to assess their suitability and, if appropriate, used to provide the first stage in a process of genuinely automatic lexicography. As part of this process, the parsing routines developed during this study are currently being amended to allow them to identify definition sentences in unmarked text, and to analyse them without the headword identification and grammatical information contained in the dictionary entries.

7.7.4 The automatic thesaurus

This is an obvious practical application of the extended scope for identification and exploration of lexical relations offered by a parsable dictionary. If superordinates, synonyms, co-hyponyms, antonyms and so on can be idenstified by an analysis of the components of a definition, the information normally available from a thesaurus could be generated automatically. This could then be used to produce a draft text for a printed thesaurus, which may need some manual refinement, but would still reduce the amount of human effort needed, and should be more comprehensive than any manually produced version, or to generate a database which could be used as a computer-readable thesaurus. It would also be possible for the dictionary itself to be used as both dictionary and thesaurus, although the need for some manual reorganisation of entries and questions of processing speed may make the former option more attractive.

The three forms of analysis suggested below are simply starting-points for the process of developing software capable of constructing at least a framework for an automatic thesaurus.

7.7.4.1 *Identification of synonyms within definitions*

The most direct way of identifying potential synonyms is to extract from the parsed output headwords with no discriminators attached to their superordinates. A simple program run against the parsed output from type A1 definitions extracted nearly 200 definitions in which the superordinate was a single word and the two discriminator elements were blank. An extract is shown below:

cookie	biscuit
co-op	co-operative
corn 2	maize
cosmos	universe

course 4	route
cover 12	outside
creed 2	religion
dame 1	woman
den	home
dialogue 2	conversation
diaper	nappy
difficulty 1	problem
disagreement 2	argument
discord	disagreement
discotheque	disco
door 2	doorway
drapes 3	curtains
dynamite	explosive

On further investigation, these potential synonyms fall into several distinct classes. Words like 'co-op' and 'discotheque' relate different forms of words to their more common forms and act as cross-references within the dictionary. The synonymy between the words 'diaper' and 'nappy' and 'drapes' and 'curtains' is restricted by their usage notes: 'an American use'. The senses of 'course' and 'cover' that are synonymous with 'route' and 'outside' are restricted by the co-text 2 in each definition: 'of a ship or aircraft' and 'of a book or a magazine'. 'dynamite' is actually a hyponym of 'explosive', but this is signalled within the full definition:

> **Dynamite** is an explosive. (p. 168)

The switch from an uncountable noun with no article in the definiendum to a count noun with article in the definiens suggests 'dynamite' as an example of an explosive rather than as its synonym. In some cases, however, such as 'corn', 'dialogue', 'difficulty' and 'door', the selected items seem to be actual synonyms of each other, subject to the ambiguity of the headwords themselves and the extra information provided in other senses to enable disambiguation to be properly performed by the user. For example, consider all three senses of 'difficulty':

> A **difficulty** is a problem. (1)
> If you have **difficulty** doing something, you are not able to do it easily. (2)
> If you are **in difficulty** or **in difficulties**, you are having a lot of problems. (3)

Only sense 1, the count noun use of 'difficulty' in isolation, has the synonym 'problem'. All of the information needed to distinguish these different types of

synonyms within definitions can be retrieved automatically from the dictionary either through other elements of the parsed definition or through existing coded information.

7.7.4.2 *Investigation of co-hyponyms*

Once the superordinates used in definitions are identified in the parsed output they can be used to group words into sets of co-hyponyms to investigate their lexical relations with each other. As an example of the process, consider this extract from a list of superordinates, produced in order of frequency of occurrence from the parsed type A1 definitions, over 10,000 altogether. This small extract from the list shows the frequencies of occurrence of those superordinates found in 100 or more definitions:

person	401
someone	246
something	184
place	137
substance	100

As would be expected, these most frequent superordinates are very general, and only restrict their hyponyms in the most fundamental ways. In these cases, 'person' and 'someone' restrict the field to single human protagonists, 'something' and 'substance' to inanimate objects and materials, and 'place' to the set of possible locations. The hyponyms of any of these superordinates or groups of superordinates could quickly be assembled so that further analysis could be carried out on their discriminators to assess which co-hyponyms are likely to be synonymous with each other, which are antonymous, and which are subsidiary superordinates in their own right.

To make this processing totally automatic may require recursive processing which consults the dictionary definitions for the discriminator elements to determine their lexical relations and to discover the nature of the discrimination being made. In simple cases, however, it might be possible to identify likely synonyms on the basis of the similarity of their discriminator elements. Such an exploration of the co-text or discriminators of co-hyponyms could be a very valuable source of information for the thesaurus, since where identical or similar distinguishing features were found in more than one definition this would strongly suggest that their headwords were at least near synonyms.

The list below shows a sample taken from the 647 headwords which have 'someone' or 'person' as their superordinates:

Headword (sense)	Discriminator 1	Superordinate	Discriminator 2
juvenile delinquent	young	person	who is guilty of committing crimes, especially vandalism or violence.
keeper		person	who takes care of the animals in a zoo.
killer (1)		person	who has killed someone.
labourer		person	who does a job which involves a lot of hard physical work.
Latin American (2)		someone	who lives in or comes from south or Central America.
lawyer		person	who is qualified to advise people about the law and represent them in court.
layman		person	who is not qualified or experienced in a particular subject or activity.
leader (1)		person	who is in charge of it.
leader (2)		person	who is winning at a particular time.

If the second discriminators of this group of definitions are collected and counted, the first 12 lines of the resulting frequency list, sorted in frequency order, are:

in charge of it.	6
who has been elected to represent people in a country's parliament.	2
who is qualified to treat sick or injured animals;	2
who writes plays.	2
who wrote it.	2
at a beach or swimming pool whose job is to rescue people who are in danger of drowning.	1
believed to be chosen by God to say the things that God wants to tell people.	1
between thirteen and nineteen years of age.	1
chosen to make decisions on behalf of a group of people, especially at a meeting.	1
employed by a company at a senior level.	1
employed by a government to find out the secrets of other governments.	1
employed by a hotel, theatre, or cinema to open doors and help customers.	1

This does not, at first sight, look promising. Only 'in charge of it' is repeated more than twice, and this depends on the co-text that 'it' matches. It is, however, important to remember that a large part of the lexicographer's skill lies in the ability to differentiate finely between similar lexical items, and that there are very few genuinely complete synonyms in the language. Despite this,

it would still be valuable to be able to estimate the nearness and the nature of lexical relations between different headwords. A very crude but fairly effective way of investigating this area is suggested by the discriminator frequency list above. The last three items quoted begin with 'employed by a'. The following words in the discriminator vary, but these headwords all relate to employees of one organisation or another, and this might be a useful type of thing to know about other headwords. If we simply sort the file containing the headword, superordinate and discriminator information already shown above on the field containing the post-discriminator, those which begin with similar phrases will be forced together. This produces some interesting groups, among them a more complete collection of the employees seen above:

executive (1)	someone	employed by a company at a senior level.
secret agent	person	employed by a government to find out the secrets of other governments.
commissionaire	person	employed by a hotel, theatre, or cinema to open doors and help customers.
buyer (2)	someone	employed by a large store to decide what goods will be bought from manufacturers to be sold in the store.
home help	person	employed by a local government authority to help sick or old people with their housework.
courier (1)	someone	employed by a travel company to look after holidaymakers.
worker (1)	person	employed in an industry or business who has no responsibility for managing it.
housekeeper	person	employed to cook and clean a house for its owner.
gamekeeper	person	employed to look after game animals and birds on someone's land.

The application of the same technique also produced the following group of people from different countries:

African (2)	person	who comes from Africa.
Australian (2)	person	who comes from Australia.
Chinese (2)	person	who comes from China.
European (2)	person	who comes from Europe.
German (2)	person	who comes from Germany.
Briton	person	who comes from Great Britain.
Greek (2)	person	who comes from Greece.
Asian (2)	person	who comes from India, Pakistan, or some other part of Asia.

Indian (2)	person	who comes from India.
Italian (2)	person	who comes from Italy.
Japanese (2)	person	who comes from Japan.
Scot (1)	person	who comes from Scotland.
Russian (2)	person	who comes from the Soviet Union.
American (2)	person	who comes from the United States of America.

It can also be used to produce this group of believers in various things:

purist	person	who believes in absolute correctness.
Jew	person	who believes in and practises the religion of Judaism.
feminist	person	who believes in and supports feminism.
capitalist (2)	someone	who believes in and supports the principles of capitalism.
worshipper	someone	who believes in and worships a god.
democrat	person	who believes in democracy.
Hindu	person	who believes in Hinduism.
Muslim	person	who believes in Islam and lives according to its rules.
Christian	person	who believes in Jesus Christ and follows his teachings.
Sikh	person	who believes in the Indian religion of Sikhism.

Even without an investigation of the meanings of the discriminators, then, a simple examination of their exact wording can provide some useful information on lexical relations. This process could be considerably refined if the discriminators which convey more than one piece of differential information were split into their logical components. As an example, in the group given above which begins 'who comes from...', the discriminators could be split into:

who comes from	Africa.
who comes from	Australia.
who comes from	China.
who comes from	Europe.
who comes from	Germany.
who comes from	Great Britain.
who comes from	Greece.
who comes from	India, Pakistan, or some other part of Asia.
who comes from	India.
who comes from	Italy.
who comes from	Japan.
who comes from	Scotland.

who comes from the Soviet Union.
who comes from the United States of America.

This gives two levels of discrimination. There is a general group of headwords with the superordinate 'person' for which the lexicographers have chosen 'who comes from' as the introduction to the following discriminator, and the words in this group specify the country of origin. In the group of employees shown above a more complex possibility emerges. Consider the discriminators of the first five items in the list:

employed by a company at a senior level.
employed by a government to find out the secrets of other governments.
employed by a hotel, theatre, or cinema to open doors and help customers.
employed by a large store to decide what goods will be bought from manufacturers to be sold in the store.
employed by a local government authority to help sick or old people with their housework.

The pattern shown in four of these examples enables the discriminator to be split into two major elements represented by:

employed by **the employer** to **carry out the duties of the employment**

The elements in this pattern which are shown in bold type are the variable items in these discriminators, and where this pattern exists the fixed elements could easily be used as a framework to identify them for further lexical and semantic analysis. Indeed, a further development of the parser could attempt to split all discriminators into similar logical units to allow this comparison and summarisation to be performed automatically.

7.8 Possible extensions

The taxonomy, grammar and parser described in this study have been developed on the basis of the set of definition sentences provided by the *Student's Dictionary*. While there is no reason to believe that this does not constitute a representative sample of definition sentences in general, it would be useful to extend the study to cover definitions from other sources. The following sections describe the main possibilities.

7.8.1 Other dictionaries in the Cobuild range

CCSD is the smallest of the original set of Cobuild dictionaries, and the version used for this study was the first edition, published in 1990. The main dictionary in the series, the *Collins Cobuild English Language Dictionary*, is currently in its third edition (2001), and revisions of the *Student's* and other dictionaries in the range have also been produced. It would be useful to apply the principles of the taxonomy, grammar and parser to these other editions, both to verify their applicability to a larger sample and to gain access to the wider linguistic information available from these other sources.

As a preliminary step, the recognition and parsing software is currently being successfully adapted for use with the second edition of *CCELD* (1995). The adaptation is necessary largely because of differences in the encoding system used in the dictionary text files.

7.8.2 Other forms of dictionary definition

The analysis described throughout this study has had as its focus English definition sentences in general, as exemplified by the specific set of definitions contained in *CCSD*. While the information contained in the more conventional, non-sentence form of dictionary definition is less full and therefore less informative, it would be possible to adapt the grammar and its recognition and parsing software to carry out a similar analysis on these texts. As an example, consider the definition of sense (**a**) of **discus** in *OALDCE*:

> [C] heavy disc thrown in athletic contests

The parser could be adapted to provide the following analysis:

```
discus
C
Dr1      heavy
S        disc
Dr2      thrown in athletic contests
```

The headword and the grammar information ('C' for 'Countable noun') have been put into the same positions as in the Cobuild sentence analyses, and the definition text itself has been allocated the same functional labels as used for the sentence parser. The information available from this form of definition could then be used in a similar way to that provided by the analysis of full definition sentences.

7.8.3 Non-dictionary definitions

As has been made clear throughout this study, the recognition and parsing software developed for the definitions has made extensive use of the special characteristics of the dictionary text encoding system. In particular, the identification of the headword within the sentences has been used as a basic structural subdivision. In definition sentences occurring naturally in free text this identification would obviously not be available. As already described in section 7.7.3, the software is currently being developed to allow it to recognise and analyse definition sentences without this special mark-up and without the associated grammatical information provided elsewhere in the dictionary entry.

Initial results from this enhancement suggest that broad discrimination between definition sentences and non-definition sentences is fairly straightforward, the main problems relating to more subtle distinctions between related definition types. For example, consider the following two definition sentences from *CCSD*, stripped of their structural marking:

(**a**) A current account is a bank account which you can take money out of at any time using your cheque book or cheque card;
(**b**) A secluded place is quiet, private, and undisturbed.

In both cases the position of the hinge 'is' would suggest a group A definition, but in the absence of the emboldened headwords ('account' in (**a**) and 'secluded' in (**b**)) further analysis would be necessary to identify (**a**) as type A1 and (**b**) as type A4. On the basis of current findings this further analysis would not represent a significant complication in the enhancement of the software. This enhancement would be particularly useful in technical texts, where terms are defined on their first appearance in the text. The automatic extraction and analysis of term definitions would be a very powerful tool in information retrieval from such texts, as suggested in Pearson (1998, p. 209).

7.9 Summary of potential applications

The applications described in this chapter represent examples of the main areas in which the parser could contribute to natural language processing and lexicography. The set of examples is not exhaustive, and it would be difficult to set limits to the range of possible application areas. The central nature of the

information which can be provided by a dictionary, especially one which, as in the Cobuild range, uses the natural features of the language as the basis of its own description, gives the definition language model and its parser a potentially fundamental role within all areas of the study and manipulation of natural language.

7.10 Conclusion

The taxonomy, grammar and parser developed in this study provide both a description of the nature of the definition sentences which allows us to explore the process of definition itself, and an ability to analyse and extract the linguistic information contained in the sentences. The various forms of the lexicographic equation together with the more indirect metalinguistic description of usage and intention contained within the definition structure taxonomy provide a comprehensive survey of the ways in which the meanings of linguistic units can be expressed in dictionaries. The analysis of these various forms of definition made possible by the parser allow a complete and flexible extraction of the individual elements of the definition text without the limitations imposed by explicit encoding at the dictionary compilation stage.

This initial study is based on the sample of definitions from *CCSD* , and current developments include the extension of the parsing software to cover later and fuller editions of the Cobuild dictionaries, adjustments to the software to allow it to deal with unmarked definition sentences within the text of corpora and the development of a thesaurus produced using the parser from dictionary entries.

Appendix 1

Examples of initial analysis of definitions

For each of the definition types identified in the taxonomy, an example is shown below of the initial functional analysis produced by the parsing software. The conventions outlined in section 6.10.1.1 have been used in these tables.

Group A

	1	2	3	4	5	6	7	8	9
A1	A		current account		is	a	bank	account	which you can take money out of at any time using your cheque book or cheque card;
A2		A bird's	plumage		is		all @M1_its_M@	feathers.	
A3	Does	is	the	third person singular of the present tense	of	do.			
A4	An		abrasive	person	is	unkind and rude.			
A5	Someone	who	is	fraught		is	very	worried or anxious.	
A6	To		anaesthetize	someone	means	to		make @M2_them_M@ unconscious	by giving @M2_them_M@ an anaesthetic.
A7	The	wild	parts of some hot countries		are referred to as	the	.	bush	

Group B

	1	2	3	4	5	6	7	8	9
B1	If	you	confirm	something,		you	say that @M2_it_M@ is true.		
B2	If	you	are	content	with something,	you	are	satisfied @M2_with it._M@	
B3	If	there	is	a	reaction	against something,			@M3_it_M@ becomes unpopular.
B4	You	do	something	in a	careless	way	when	@M1_you_M@ are relaxed or confident.	

Group C

	1	2	3	4	5	6	7	8	9
C1	You	describe	something such as a quality	as	enviable			when someone else has it and @M1_you_M@ wish that @M1_you_M@ had it yourself.	

	1	2	3	4	5	6	7	8	9	10	11
C2	If	you	describe	something	as		amateurish,		you	mean	@M2_it_M@ is not skilfully made or done.
C3	You	can refer to	a		change	back to a former state	as	a	return	@M3_to that state_M@	

	1	2	3	4	5	6	7	8	9
C4	If	you	get	a lot of questions or complaints about something,	@M1_you_M@ can say that	@M1_you_M@	@M2_are getting_M@ a	barrage	@M3_of them._M@
C5		Mini-	is added	to nouns		to form	other nouns that refer to a smaller version of something.		

Group D

	1	2	3	4	5	6	7	8	9
D1	In	a	pressurized	container or area,	the pressure inside	is	different from @M3_the pressure_M@ outside.		

Appendix 2

Examples of final parsed output

The final output for each of the examples shown in the tables in Appendix 1 is shown below. The conventions already adopted for the description of the grammar in section 6.7 and of the functional analysis in section 6.10.1.1 above have been used in the output.

Group A

Type A1

```
current account
COUNT N
A       A
Hd      current account
Hi      is
Am      a
Dr1     bank
S       account
Dr2     which you can take money out of at any time
        using your cheque book
Or      or
Dr2     cheque card;
N2      a British use.
```

Type A2

```
plumage
UNCOUNT N
Mr      A bird's
Hd      plumage
Hi      is
Dr1     all
Mrm     its
S       feathers.
```

Type A3

```
Does
Hd      Does
Hi      is
A       the
E       third person singular
L       of
```

```
E       the present tense
L       of
X       do.
```

Type A4

```
abrasive (1)
ADJ
A       An
Hd      abrasive
No      person
Hi      is
E       unkind and rude.
```

Type A5

```
fraught (2)
ADJ
No      Someone
B       who
Hi      is
Hd      fraught
Him     is
Dr1     very
S       worried
Or      or
S       anxious.
```

Type A6

```
anaesthetize
VB with OBJ
To      To
Hd      anaesthetize
Ob      someone
Hi      means
Tom     to
S       make
Obm     them
S       unconscious
Dr2     by giving
Obm     them
Dr2     an anaesthetic.
```

Type A7

```
bush. (2)
SING N
A       The
Dr1     wild
S       parts of some hot countries
```

H^{i} are referred to as
A_{m} the
H^{d} bush.

Group B

Type B1

confirm (2)
REPORT VB
H^{i} If
S^{b} you
H^{d} confirm
O^{b} something,
S^{b}_{m} you
E say that
O^{b}_{m} it
E is true.

Type B2

content (6)
PRED ADJ
H^{i} If
S^{b} you
H^{i2} are
H^{d} content
A^{d} with something,
S^{b}_{m} you
H^{i2}_{m} are
E satisfied
A^{d}_{m} with it.
N^{2} If you are *content *to do something, you do it willingly.

Type B3

reaction (3)
COUNT N with ‘against’
H^{i} If
S^{b} there
H^{e} is
A a
H^{d} reaction
A^{d} against something,
A^{d}_{m} it
E becomes unpopular.

Type B4

careless (2)
ADJ

S^b You
V^p do
O^b something
A^d in a
H^d careless
A^d way
H^i when
S^b_m you
E are relaxed or confident.

Group C

Type C1

enviable
ADJ
P^{rs} You
P^{rv} describe
P^{rc} something such as a quality
P^{rl} as
H^d enviable
E when someone else has
P^{rc}_m it
E and
P^{rs}_m you
E wish that
P^{rs}_m you
E had
P^{rc}_m it
P^{rs}_m yourself.

Type C2

amateurish
ADJ
H^i If
P^{rs} you
P^{rv} describe
P^{rc} something as
H^d amateurish,
P^{rs}_m you
P^{rv}_m mean
P^{rc}_m it
E is not skilfully made or done.

Type C3

return (10)
SING N with PREP ‘to’
P^{rs} You

P^{rv}	can refer to
A	a
S	change
D^{r2}	back to a former state
P^{r2}	as
A_m	a
H^d	return
$D^{r2}{}_m$	to that state.

Type C4

barrage
COUNT N with SUPP

H^i	If
S^b	you
V^p	get
O^b	a lot of questions or complaints about something,
$S^b{}_m$	you
P^{r1}	can say that
$S^b{}_m$	you
$V^p{}_m$	are getting
A_m	a
H^d	barrage
$O^b{}_m$	of them.

Type C5

Mini-
PREFIX

H^d	Mini-
H^{i1}	is added
A^{d1}	to nouns
H^{i2}	to form
E	other nouns that refer to a smaller version of something.
N^2	For example, a mini-computer is a computer which is smaller than a normal computer.

Group D

Type D1

pressurized
ADJ

I^n	In
A	a
H^d	pressurized
N^o	container or area,

```
S^b      the pressure inside
H^i      is
E        different from
S^b_m    the pressure
E        outside.
```

Appendix 3 (by Simon Krek)

Partial translations of CCSD deWnitions for the Anglesko-slovenski slovar BRIDGE

These examples of the definition types have been analysed using the approach shown in sections 5.2.1 and 5.2.2, and show the relationship between the structures used in the English and Slovenian versions of the definitions.

	Type	First part				Second part	
		Operator	Co-text(1)	Topic	Co-text(2)	Operator	Comment
E	A1		An	**issue**	of a magazine or newspaper	is	a particular edition of it
S	A1		An	**issue**	of a magazine or newspaper	je	določena izdaja neke revije ali časopisa.
E	A2		The earth's	**crust**		is	its outer layer.
S	A2		The earth's	**crust**		je	zunanja plast Zemlje
E	A3			**Forgot**		is	the past tense of **forget**.
S	A3			**Forgot**		je	preteklik glagola **to forget**
E	A4		A	**secluded**	place	is	quiet, private, and undisturbed.
S	A4		Kraj, ki je	**secluded**		je	miren, zaseben in nas tam nihče ne moti.
E	A5		Something that is	**hidden**		is	not easily noticed.
S	A5		Kadar je nekaj	**hidden**			tistega ne opazimo zlahka.
E	A5/2		Something that is	**abominable**		is	very unpleasant or very bad.
S	A5/2		Kar je	**abominable**		je	zelo neprijetno ali slabo.
E	A6		To	**commit**	money or resources to something	means	to use them for a particular purpose.

	Type	First part				Second part	
		Operator	**Co-text(1)**	**Topic**	**Co-text(2)**	**Operator**	**Comment**
S	**A6**		To	**commit**	money or resources to something	pomeni	uporabiti denar ali sredstva v neki poseben namen.
E	**B1**	When	a country	**liberalizes**	its laws or its attitudes,		it makes them less strict and allows more freedom.
S	**B1**	Kadar	neka država	**liberalizes**	its laws or its attitudes		svoje zakone ali ravnanje naredi manj strogo in dovoli več svobode.
E	**B2**	If	someone is	**run-down,**			they are tired or ill
S	**B2**		Kdor je	**run-down**		je	utrujen ali bolan;
E	**B2/2**	If	someone is	**ailing**			they are ill and not getting better.
S	**B2/2**	Kadar	je človek	**ailing**		je	bolan in mu ne gre na bolje.
E	**B3**	If	you do something	**with**	someone else,		you do it together.
S	**B3**	Kadar	nekaj storimo	**with**	someone else		tisto storimo še z nekom.
E	**B4**		You ask what has	**got into**	someone	when	they are behaving in an unexpected way
S	**B4**			**What has got into you?**	vprašamo,	kadar	se nekdo vede nepriakovano;
E	**C1**		You can also say you	**admire**	something	when	you look with pleasure at it.
S	**C1**		Kadar človek	**admires**	something,		na tisto stvar gleda z zadovoljstvom.
E	**C2**	If	you say to someone that something is	**their own affair,**			you mean that you do not want to know about or become involved in their activities.
S	**C2**	Če	nekomu rečemo, da je nekaj	**their own affair,**			hočemo povedati, da o tistem nočemo nič vedeti ali se nočemo vpletati v njegovo dejavnost.

	Type	First part				Second part	
		Operator	Co-text(1)	Topic	Co-text(2)	Operator	Comment
E	C5			**Equatorial**		is	used to describe places and conditions near or at the equator.
S	C5		Z besedo	**equatorial**			opišemo kraje in razmere blizu ekvatorja ali na njem.
E	D1		In	**humid**	places, the weather	is	hot and damp.
S	D1		V kraju, ki je	**humid,**		je	vroče in vlažno.

	Type	Second part		First part			
E		Operator	Comment	Operator	Co-text(1)	Topic	Co-text (2)
E	A7		New people who are introduced into an organization and whose fresh ideas are likely to improve it	are referred to as		**new blood, fresh blood, or young blood.**	
S	A7	so	novinci, ki so sprejeti v neko organizacijo, da bi jo njihove sveže zamisli izboljšale.			**New blood, fresh blood** ali **young blood**	
E	C3	You can refer to	a change back to a former state	as	a	**return**	to that state.
S	C3	je	sprememba okoliščin v prejšnje stanje.		**A**	**return**	**to a former state**
E	C4	When	someone creates something that has never existed before,	you can refer to this event as	the	**invention**	of the thing.
S	C4	je	stvaritev nečesa, česar prej ni bilo.		The	invention	of something

COMMENT:

	Type	Operator	Framework	Gloss	Framework
E	A1	is	a	particular edition	of it.
S	A1	je		določena izdaja	neke revije ali časopisa.
E	A2	is	its	outer layer.	
S	A2	je		zunanja plast	Zemlje.
E	A3	is		the past tense of **forget**.	
S	A3	je		preteklik glagola to **forget**	
E	A4	is		quiet, private, and undisturbed.	
S	A4	je		miren, zaseben in nas tam nihče ne moti.	
E	A5	is		not easily noticed.	
S	A5		tistega	ne opazimo zlahka.	
E	A5/2	is		very unpleasant or very bad.	
S	A5/2	je		zelo neprijetno ali slabo.	
E	A6	means	to	use → for a particular purpose.	them
S	A6	pomeni		uporabiti → v neki poseben namen.	denar ali sredstva
	A7			New people who are introduced into an organization and whose fresh ideas are likely to improve it	
S	A7			novinci, ki so sprejeti v neko organizacijo, da bi jo njihove sveže zamisli izboljšale	
E	B1		It	makes → less strict and allows more freedom.	them
S	B1		svoje zakone ali ravnanje	naredi manj strogo in dovoli več svobode.	
E	B2		they are	tired or ill;	
S	B2		je	utrujen ali bolan;	
E	B3		you do it	together.	
S	B3		tisto storimo	še z nekom.	
E	B4	when	they	are behaving in an unexpected way;	
S	B4	kadar		se nekdo vede nepričakovano;	
E	C1	when	you	look with pleasure at	it.
S	C1	kadar		na → gleda z zadovoljstvom	tisto stvar
E	C2		you mean that you	do not want to know about or become involved in	their activities.
S	C2		hočemo povedati, da	o tistem nočemo nič vedeti ali se nočemo vpletati v	njegovo dejavnost.

	Type	Operator	Framework	Gloss	Framework
E	C3		You can refer to a	change back to a former state	
S	C3		je	sprememba okoliščin v prejšnje stanje.	
E	C5	is		used to describe places and conditions near or at the equator.	
S	C5			opišemo kraje in razmere blizu ekvatorja ali na njem.	
E	C4			When someone creates something that has never existed before,	
S	C4		je	stvaritev nečesa, česar prej ni bilo.	
E	D1	is		hot and damp.	
S	D1	je		vroče in vlažno.	

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